

ANNEE : 3ème année / 3rd year - ECTS

SEMESTRE : 1er semestre / 1st semester - 30 ECTS

PARCOURS : Parcours Hors GI - ECTS

UE : Semestre hors GI sans ECTS / Semester outside GI with no ECTS - 0 ECTS

[EC : Semestre sans activité pédagogique / Semester with no pedagogical activity - - 0 ECTS](#)

[EC : Césure / Gap year - - 0 ECTS](#)

UE : Semestre hors GI avec 30 ECTS / Semester outside GI with 30 ECTS - 30 ECTS

[EC : Echange académique S1 / Academic exchange S1 - - 30 ECTS](#)

PARCOURS : Parcours standard / Standard track - 30 ECTS

UE : Pilotage de la production / Production management - 7 ECTS

[EC : Systèmes automatisés de production / Automated production systems - APS - 3 ECTS](#)

[EC : Gestion industrielle / Industrial Management - GIN - 4 ECTS](#)

UE : Humanités et activités physiques et sportives / Humanities and sports activities - 6 ECTS

[EC : Français Langue Etrangère \(FLE\) Toutes compétences - Remédiation \(S1\) - L-FLE-TTC-REMEDIATION-S1 - 2 ECTS](#)

[EC : Japonais \(3 S1\) / Japanese - L-JAP-3-S1 - 2 ECTS](#)

[EC : FLE option Sciences humaines et sociales Cycle Ingénieur Cultures Comparées B2-C1 \(S1\) - L-FLE-SHSCIVB2C1-S1 - 2 ECTS](#)

[EC : Chinois \(3 S1\) / Chinese - L-CHI-3-S1 - 2 ECTS](#)

[EC : Anglais, mise à niveau \(3 S1\) / English foundations - L-ANG-A2-3-S1 - 2 ECTS](#)

[EC : Italien \(3 S1\) / Italian - L-ITA-3-S1 - 2 ECTS](#)

[EC : Portugais \(3 S1\) / Portuguese - L-POR-3-S1 - 2 ECTS](#)

[EC : Arabe \(3 S1\) / Arabic - L-ARA-3-S1 - 2 ECTS](#)

[EC : Anglais \(3 S1\) / English - L-ANG-3-S1 - 2 ECTS](#)

[EC : Français Langue Etrangère \(FLE\) Toutes compétences B1 à C1 \(S1\) - L-FLE-TTCB1B2C1-S1 - 2 ECTS](#)

[EC : Russe \(3 S1\) / Russian - L-RUS-3-S1 - 2 ECTS](#)

[EC : Allemand \(3 S1\) / German - L-ALL-3-S1 - 2 ECTS](#)

[EC : Tandem et Intercompréhension en Langues Etrangères \(3 S1\) / Tandem and intercomprehension - L-TAN-3-S1 - 2 ECTS](#)

[EC : Espagnol \(3 S1\) / Spanish - L-ESP-3-S1 - 2 ECTS](#)

[EC : Théâtre Sciences humaines et Communication / Theater Social Sciences and Communication - COM - 1 ECTS](#)

[EC : Semaine de rentrée / Welcome week - - 0 ECTS](#)

[EC : EPS3-S1 - Education Physique et Sportive / Physical Education - EPS3-S1 - 1 ECTS](#)

UE : Informatique et mathématiques décisionnelles / Computer Science and Decision Mathematics - 11 ECTS

[EC : Probabilités, statistiques, plans d'expériences / Probability, statistics, design of experiments - PSX - 4 ECTS](#)

[EC : Algorithmique, programmation et modélisation en UML / Algorithms, programming and modeling in UML - APM - 4 ECTS](#)

[EC : Recherche Opérationnelle et optimisation / Operational research and optimization - ROO - 3 ECTS](#)

UE : Conception et fabrication de produits et systèmes industriels / Design and manufacturing of industrial products and systems - 6 ECTS

[EC : Procédés de fabrication, industrialisation / Manufacturing processes, industrialization - PFI - 3 ECTS](#)

[EC : Résistance des matériaux / Strength of materials - RDM - 3 ECTS](#)

SEMESTRE : 2ème semestre / 2nd semester - 30 ECTS

PARCOURS : Parcours standard / Standard track - 30 ECTS

UE : Gestion de la chaîne logistique / Supply chain management - 5 ECTS

[EC : Gestion de flux / Flow management - GFL - 2 ECTS](#)

[EC : Introduction à la Supply Chain / Introduction to Supply Chain - ISC - 3 ECTS](#)

UE : Conception de produits et systèmes industriels / Design of industrial products and systems - 6 ECTS

[EC : Gestion d'un projet de conception de machine / Management of a machine design project - GPC - 4 ECTS](#)

[EC : Penser Système et Cycle de vie / Lifecycle and system thinking - 2 ECTS](#)

UE : Pilotage des systèmes industriels / Control of industrial systems - 7 ECTS

[EC : Commande des systèmes dynamiques / Control of dynamic systems - CSD - 4 ECTS](#)

[EC : Conception, dimensionnement et analyse de performance d'un système de pilotage / Design, sizing and performance analysis of a control system - CDA - 3 ECTS](#)

UE : Humanités et activités physiques et sportives / Humanities and sports activities - 7 ECTS

[EC : Théâtre Sciences humaines et Communication / Theater Social Sciences and Communication - COM - 1 ECTS](#)

[EC : Français Langue Etrangère \(FLE\) Toutes compétences B1 à C1 \(S2\) - L-FLE-TTCB1B2C1-S2 - 2 ECTS](#)

[EC : FLE option Sciences humaines et sociales Cycle Ingénieur Cultures Comparées B2-C1 \(S2\) - L-FLE-SHSCIVB2C1-S2 - 2 ECTS](#)

[EC : Français Langue Etrangère \(FLE\) Toutes compétences - Remédiation \(S2\) - L-FLE-TTC-REMEDIATION-S2 - 2 ECTS](#)

[EC : Portugais \(3 S2\) / Portuguese - L-POR-3-S2 - 2 ECTS](#)

[EC : Tandem et Intercompréhension en Langues Etrangères \(3 S2\) / Tandem and intercomprehension - L-TAN-3-S2 - 2 ECTS](#)

[EC : Anglais, mise à niveau \(3 S2\) / English foundations - L-ANG-A2-3-S2 - 2 ECTS](#)

[EC : Arabe \(3 S2\) / Arabic - L-ARA-3-S2 - 2 ECTS](#)

[EC : Allemand \(3 S2\) / German - L-ALL-3-S2 - 2 ECTS](#)

[EC : Russe \(3 S2\) / Russian - L-RUS-3-S2 - 2 ECTS](#)

[EC : Japonais \(3 S2\) / Japanese - L-JAP-3-S2 - 2 ECTS](#)

[EC : Chinois \(3 S2\) / Chinese - L-CHI-3-S2 - 2 ECTS](#)

[EC : Espagnol \(3 S2\) / Spanish - L-ESP-3-S2 - 2 ECTS](#)

[EC : Anglais \(3 S2\) / English - L-ANG-3-S2 - 2 ECTS](#)

[EC : Italien \(3 S2\) / Italian - L-ITA-3-S2 - 2 ECTS](#)

[EC : EPS3-S2 - Education Physique et Sportive / Physical Education - EPS3-S2 - 1 ECTS](#)

[EC : Introduction à la Recherche / Introduction to Research - - 1 ECTS](#)

UE : Conception et gestion des systèmes d'informations / Design and management of information systems - 5 ECTS

[EC : Conception de bases de données et architecture des systèmes d'informations / Database design and information systems architecture - CBD - 3 ECTS](#)

[EC : Résolution informatique d'un problème / Computational problem solving - RIP - 2 ECTS](#)

PARCOURS : Parcours hors GI - ECTS

UE : Semestre hors GI sans ECTS / Semester outside GI with no ECTS - 0 ECTS

[EC : Semestre sans activité pédagogique / Semester with no pedagogical activity - - 0 ECTS](#)

[EC : Césure / Gap year - - 0 ECTS](#)

ANNEE : 4ème année / 4th year - ECTS

SEMESTRE : 1er semestre / 1st semester - 30 ECTS

PARCOURS : Parcours DD KAIST IN / KAIST DD track - 30 ECTS

UE : Gestion et exploitation des données de l'entreprise / Management and exploitation of business data - 4 ECTS

[EC : Business Intelligence / Business Intelligence - BIN - 2 ECTS](#)

[EC : Introduction à la science des données / Introduction to data science - ISD - 2 ECTS](#)

UE : Projet labo KAIST - 10 ECTS

[EC : Projet labo pour étudiant d'échange / Research project for exchange student - LABECH15 - 15 ECTS](#)

UE : Amélioration continue et innovation - 5 ECTS

[EC : Ordonnancement / Scheduling - ORD - 2 ECTS](#)

[EC : Qualité / Quality - QLT - 1 ECTS](#)

[EC : Ecologie industrielle et économie circulaire / Industrial ecology and circular economy - EIE - 2 ECTS](#)

UE : Humanités et activités physiques et sportives / Humanities and sports - 5 ECTS

[EC : EPS4-S1 - Education Physique et Sportive / Physical Education - EPS4-S1 - 1 ECTS](#)

[EC : Chinois \(4 S1\) / Chinese - L-CHI-4-S1 - 2 ECTS](#)

[EC : Espagnol \(4 S1\) / Spanish - L-ESP-4-S1 - 2 ECTS](#)

[EC : Anglais \(4 S1\) / English - L-ANG-4-S1 - 2 ECTS](#)

[EC : Arabe \(4 S1\) / Arabic - L-ARA-4-S1 - 2 ECTS](#)

[EC : Allemand \(4 S1\) / German - L-ALL-4-S1 - 2 ECTS](#)

[EC : Japonais \(4 S1\) / Japanese - L-JAP-4-S1 - 2 ECTS](#)

[EC : FLE option Sciences humaines et sociales Cycle Ingénieur Cultures Comparées B2-C1 \(S1\) - L-FLE-SHSCIVB2C1-S1 - 2 ECTS](#)

[EC : Français Langue Etrangère \(FLE\) Toutes compétences - Remédiation \(S1\) - L-FLE-TTC-REMEDIATION-S1 - 2 ECTS](#)

[EC : Français Langue Etrangère \(FLE\) Toutes compétences B1 à C1 \(S1\) - L-FLE-TTCB1B2C1-S1 - 2 ECTS](#)

[EC : Russe \(4 S1\) / Russian - L-RUS-4-S1 - 2 ECTS](#)

[EC : Anglais, mise à niveau \(4 S1\) / English foundations - L-ANG-A2-4-S1 - 2 ECTS](#)

[EC : Italien \(4 S1\) / Italian - L-ITA-4-S1 - 2 ECTS](#)

[EC : Tandem et Intercompréhension en Langues Etrangères \(4 S1\) / Tandem and intercomprehension - L-TAN-4-S1 - 2 ECTS](#)

[EC : Portugais \(4 S1\) / Portuguese - L-POR-4-S1 - 2 ECTS](#)

UE : Projets / Projects - 4 ECTS

[EC : Projet collectif S1 / Collective project S1 - PCO - 4 ECTS](#)

UE : Options 4GI / 4th year options - 2 ECTS

[EC : Optimisation exacte et approchée / Exact and approached optimization - OEA - 2 ECTS](#)

[EC : Initiation à la recherche / Introduction to scientific research - INR - 2 ECTS](#)

[EC : Data-driven decision making / Data-driven decision making - DDD - 2 ECTS](#)

[EC : Sûreté de fonctionnement / Dependability - SDF - 2 ECTS](#)

PARCOURS : Hors GI / Outside GI - ECTS

UE : Semestre Hors GI avec 30 ECTS / - 30 ECTS

[EC : Echange académique S1 / Academic exchange S1 - ECH - 30 ECTS](#)

UE : Semestre hors GI sans ECTS / Semester outside GI with no ECTS - 0 ECTS

[EC : Césure / Gap year - - 0 ECTS](#)

[EC : Semestre sans activité pédagogique / Semester with no pedagogical activity - - 0 ECTS](#)

PARCOURS : Parcours standard / Standard track - 30 ECTS

UE : Gestion et exploitation des données de l'entreprise / Management and exploitation of business data - 7 ECTS

[EC : Introduction à la science des données / Introduction to data science - ISD - 2 ECTS](#)

[EC : Business Intelligence / Business Intelligence - BIN - 2 ECTS](#)

[EC : Systèmes d'informations d'entreprise / Business Information Systems - SIE - 3 ECTS](#)

UE : Projets / Projects - 4 ECTS

[EC : Projet collectif S1 / Collective project S1 - PCO - 4 ECTS](#)

UE : Humanités et activités physiques et sportives / Humanities and sports - 7 ECTS

[EC : Semaine de rentrée / Welcome week - - 0 ECTS](#)

[EC : Chinois \(4 S1\) / Chinese - L-CHI-4-S1 - 2 ECTS](#)

[EC : Italien \(4 S1\) / Italian - L-ITA-4-S1 - 2 ECTS](#)

[EC : Portugais \(4 S1\) / Portuguese - L-POR-4-S1 - 2 ECTS](#)

[EC : Anglais, mise à niveau \(4 S1\) / English foundations - L-ANG-A2-4-S1 - 2 ECTS](#)

[EC : Arabe \(4 S1\) / Arabic - L-ARA-4-S1 - 2 ECTS](#)

[EC : Anglais \(4 S1\) / English - L-ANG-4-S1 - 2 ECTS](#)

[EC : Russe \(4 S1\) / Russian - L-RUS-4-S1 - 2 ECTS](#)

[EC : Japonais \(4 S1\) / Japanese - L-JAP-4-S1 - 2 ECTS](#)

[EC : FLE option Sciences humaines et sociales Cycle Ingénieur Cultures Comparées B2-C1 \(S1\) - L-FLE-SHSCIVB2C1-S1 - 2 ECTS](#)

[EC : Français Langue Etrangère \(FLE\) Toutes compétences - Remédiation \(S1\) - L-FLE-TTC-REMEDIATION-S1 - 2 ECTS](#)

[EC : Espagnol \(4 S1\) / Spanish - L-ESP-4-S1 - 2 ECTS](#)

[EC : Français Langue Etrangère \(FLE\) Toutes compétences B1 à C1 \(S1\) - L-FLE-TTCB1B2C1-S1 - 2 ECTS](#)

[EC : Allemand \(4 S1\) / German - L-ALL-4-S1 - 2 ECTS](#)

[EC : Tandem et Intercompréhension en Langues Etrangères \(4 S1\) / Tandem and intercomprehension - L-TAN-4-S1 - 2 ECTS](#)

[EC : EPS4-S1 - Education Physique et Sportive / Physical Education - EPS4-S1 - 1 ECTS](#)

[EC : Options Sciences Humaines et Sociales, S1 Série 2 / Social and Human Sciences Options, S1 Series 2 - S-SERIE2-S1 - 2 ECTS](#)

UE : Amélioration continue et innovation / Continuous improvement and innovation - 5 ECTS

[EC : Ecologie industrielle et économie circulaire / Industrial ecology and circular economy - EIE - 2 ECTS](#)

[EC : Design thinking / Design thinking - DTH - 2 ECTS](#)

[EC : Qualité / Quality - QLT - 1 ECTS](#)

UE : Options 4GI / 4th year options - 2 ECTS

[EC : Optimisation exacte et approchée / Exact and approached optimization - OEA - 2 ECTS](#)

[EC : Data-driven decision making / Data-driven decision making - DDD - 2 ECTS](#)

[EC : Initiation à la recherche / Introduction to scientific research - INR - 2 ECTS](#)

[EC : Sûreté de fonctionnement / Dependability - SDF - 2 ECTS](#)

UE : Pilotage des opérations industrielles / Management of industrial operations - 5 ECTS

[EC : Ordonnancement / Scheduling - ORD - 2 ECTS](#)

[EC : Pilotage et conduite d'un système industriel / Piloting and management of an industrial system - PCS - 3 ECTS](#)

SEMESTRE : 2ème semestre / 2nd semester - 30 ECTS

PARCOURS : Parcours standard / Standard track - 30 ECTS

UE : Gestion et amélioration des performances de l'entreprise / Management and improvement of business performance - 5 ECTS

[EC : Lean / Lean - LEA - 3 ECTS](#)

[EC : Budget et contrôle de gestion / Budget and control management - BCG - 2 ECTS](#)

UE : Projets en systèmes d'information / Information systems projects - 6 ECTS

[EC : Projet ERP / ERP Project - ERP - 3 ECTS](#)

[EC : Projet SCADA-MES / SCADA-MES project - PSM - 3 ECTS](#)

UE : Education physique et sportive / Physical Education - 1 ECTS

[EC : EPS4-S2 - Education Physique et Sportive / Physical Education - EPS4-S2 - 1 ECTS](#)

UE : Stage industriel / Industrial internship - 15 ECTS

[EC : Stage industriel / Industrial internship - STI - 14 ECTS](#)

[EC : Analyse sociologique des organisations / Sociological analysis of organizations - ASO - 1 ECTS](#)

UE : Projets / Projects - 3 ECTS

[EC : Projet collectif S2 / Collective project S2 - PCO - 3 ECTS](#)

UE : Remise à niveau / Refresher - 3 ECTS

[EC : Systèmes d'informations d'entreprise / Business Information Systems - SIE - 1 ECTS](#)

[EC : Pilotage et conduite d'un système industriel / Piloting and management of an industrial system - PCS - 1 ECTS](#)

[EC : Introduction à la science des données / Introduction to data science - ISD - 1 ECTS](#)

PARCOURS : Parcours hors GI - ECTS

UE : Semestre hors GI sans ECTS / Semester outside GI with no ECTS - 0 ECTS

[EC : Semestre sans activité pédagogique / Semester with no pedagogical activity - - 0 ECTS](#)

[EC : Césure / Gap year - - 0 ECTS](#)

[EC : Projet labo pour étudiant d'échange / Research project for exchange student - LABECH15 - 15 ECTS](#)

[EC : Stage pour étudiant d'échange / Internship for exchange student - STAGECH30 - 30 ECTS](#)

[EC : Stage pour étudiant d'échange / Internship for exchange student - STAGECH15 - 15 ECTS](#)

[EC : Projet labo pour étudiant d'échange / Research project for exchange student - LABECH30 - 30 ECTS](#)

ANNEE : 5ème année / 5th year - ECTS

SEMESTRE : 1er semestre / 1st semester - 30 ECTS

PARCOURS : Parcours standard / Standard track - 30 ECTS

UE : Industrie 4.0 / Industry 4.0 - 12 ECTS

[EC : Cybersécurité / Cybersecurity - CYB - 2 ECTS](#)

[EC : Industrie du futur - Projet / Industry of the future - Project - IFU - 2 ECTS](#)

[EC : Knowledge management / Knowledge management - KNM - 2 ECTS](#)

[EC : Optimisation du transport et logistique / Transportation and logistics optimization - OTL - 3 ECTS](#)

[EC : Data science / Data science - DSC - 3 ECTS](#)

UE : Excellence opérationnelle / Operational excellence - 12 ECTS

[EC : Aide à la décision multicritères / Multicriteria decision support - ADM - 2 ECTS](#)

[EC : Achats et Pilotage Fournisseurs / Sourcing process and supplier survey - APF - 3 ECTS](#)

[EC : Maintenance / Maintenance - MTC - 2 ECTS](#)

[EC : Evaluation de performance / Performance evaluation - EVP - 2 ECTS](#)

[EC : Internal Supply chain and facility design / Internal Supply chain and facility design - IFD - 3 ECTS](#)

UE : Management de l'entreprise / Business management - 11 ECTS

[EC : Ethique de l'ingénieur / Engineering Ethics - ETH - 2 ECTS](#)

[EC : Responsabilité Sociétale et Environnementale / Social and Environmental Responsibility - RSE - 3 ECTS](#)

[EC : Strategic Supply chain / Strategic Supply chain - SSC - 3 ECTS](#)

[EC : Management pour l'ingénieur / Management for engineering - MGI - 3 ECTS](#)

UE : Projet de groupe / Group project - 3 ECTS

[EC : Projet collectif 5GI / Collective project S1 - PCO - 3 ECTS](#)

[EC : Projet Industriel / Industriel project - PRI - 3 ECTS](#)

[EC : Recherche en Génie Industriel / Research in Industrial Engineering - RGI - 3 ECTS](#)

UE : Humanités et activités physiques et sportives / Humanities and sports - 4 ECTS

[EC : EPS5-S1 - Education Physique et Sportive / Physical Education - EPS5-S1 - 1 ECTS](#)

[EC : Semaine de Rentrée / Welcome week - - 0 ECTS](#)

[EC : Projet Personnel en Humanités / Personal Project in Humanities - S-PPH-S1 - 1 ECTS](#)

[EC : Options Sciences Humaines et Sociales, S1 Série 5 / Social and Human Sciences Options, S1 Series 5 - S-SÉRIE5-S1 - 2 ECTS](#)

PARCOURS : Parcours DD Turin - 30 ECTS

UE : Master DD Polito / - 22 ECTS

[EC : Master Thesis DD Polito / Master DD Polito - MAS - 22 ECTS](#)

UE : UE à choix pour DD POLITO IN / Option for DD POLITO - 8 ECTS

[EC : Responsabilité Sociétale et Environnementale / Social and Environmental Responsibility - RSE - 3 ECTS](#)

[EC : Cybersécurité / Cybersecurity - CYB - 2 ECTS](#)

[EC : Internal Supply chain and facility design / Internal Supply chain and facility design - IFD - 3 ECTS](#)

[EC : Evaluation de performance / Performance evaluation - EVP - 2 ECTS](#)

[EC : Management pour l'ingénieur / Management for engineering - MGI - 3 ECTS](#)

[EC : Optimisation du transport et logistique / Transportation and logistics optimization - OTL - 3 ECTS](#)

[EC : Achats et Pilotage Fournisseurs / Sourcing process and supplier survey - APF - 3 ECTS](#)

[EC : Data science / Data science - DSC - 3 ECTS](#)

[EC : Projet Industriel / Industriel project - PRI - 3 ECTS](#)

[EC : Knowledge management / Knowledge management - KNM - 2 ECTS](#)

[EC : Industrie du futur - Projet / Industry of the future - Project - IFU - 2 ECTS](#)

[EC : Strategic Supply chain / Strategic Supply chain - SSC - 3 ECTS](#)

[EC : Aide à la décision multicritères / Multicriteria decision support - ADM - 2 ECTS](#)

[EC : Maintenance / Maintenance - MTC - 2 ECTS](#)

[EC : Ethique de l'ingénieur / Engineering Ethics - ETH - 2 ECTS](#)

PARCOURS : Parcours hors GI - ECTS

UE : Semestre hors GI sans ECTS / Semester outside GI with no ECTS - 0 ECTS

[EC : Césure / Gap year - - 0 ECTS](#)

[EC : Semestre sans activité pédagogique / Semester with no pedagogical activity - - 0 ECTS](#)

[EC : DD INSTN semestre 1/2 / DD INSTN semester 1/2 - - 0 ECTS](#)

[EC : DD TOHOKU semestre 1/5 cours japonais / DD TOHOKU semester 1/5 cours japonais - - 0 ECTS](#)

[EC : DD ETS semestre 1/3 / DD ETS semester 1/3 - - 0 ECTS](#)

[EC : DD POLITO semestre 1/3 / DD POLITO semester 1/3 - - 0 ECTS](#)

[EC : DD KAIST semestre 1/3 / DD KAIST semester 1/3 - - 0 ECTS](#)

[EC : DD PUC RIO semestre 1/4 / DD PUC RIO semester 1/4 - - 0 ECTS](#)

[EC : DD SHERBROOKE semestre 1/3 / DD SHERBROOKE semester 1/3 - - 0 ECTS](#)

[EC : DD UPC ESTIEB semestre 1/4 / DD UPC ESTIEB semester 1/4 - - 0 ECTS](#)

[EC : DD EPM semestre 1 / DD EPM semester 1 - - 0 ECTS](#)

[EC : DD IIT CHICAGO semestre 1/3 / DD IIT CHICAGO semester 1/3 - - 0 ECTS](#)

UE : Semestre hors GI avec 30 ECTS / Semester outside GI with 30 ECTS - 30 ECTS

[EC : DD ETS semestre 3/3 \(PFE\) / DD ETS semester 3/3 \(PFE\) - - 30 ECTS](#)

[EC : Projet de fin d'études 0ECTS / Final project assignment 0ECTS - - 0 ECTS](#)

[EC : Projet de fin d'études / Final project assignment - - 30 ECTS](#)

[EC : DD IIT CHICAGO semestre 3/3 \(PFE\) / DD IIT CHICAGO semester 3/3 \(PFE\) - - 0 ECTS](#)

[EC : DD POLITO semestre 3/3 \(PFE\) / DD POLITO semester 3/3 \(PFE\) - - 0 ECTS](#)

[EC : Projet de fin d'études / Final project assignment - - 30 ECTS](#)

[EC : DD EPM MAITRISE COURS semestre 3/3 \(PFE\) / DD EPM MAITRISE COURS semester 3/3 \(PFE\) - - 30 ECTS](#)

[EC : Projet de fin d'études 0ECTS / Final project assignment 0ECTS - - 0 ECTS](#)

[EC : Projet de fin d'études / Final project assignment - - 30 ECTS](#)

[EC : DD KAIST semestre 3/3 \(PFE\) / DD KAIST semester 3/3 \(PFE\) - - 0 ECTS](#)

[EC : Projet de fin d'études 0ECTS / Final project assignment 0ECTS - - 0 ECTS](#)

[EC : DD SHERBROOKE semestre 3/3 \(PFE\) / DD SHERBROOKE semester 3/3 \(PFE\) - - 30 ECTS](#)

[EC : DD UPC ESTIEB semestre 3/4 / DD UPC ESTIEB semester 3/4 - - 30 ECTS](#)

[EC : DD PUC RIO semestre 3/4 / DD PUC RIO semester 3/4 - - 30 ECTS](#)

UE : UE IFP School - 30 ECTS

[EC : Double Diplôme IFP - semestre 1 - 30 ECTS - - 30 ECTS](#)

UE : Semestre hors GI sans ECTS / Semester outside GI with no ECTS - 0 ECTS

[EC : Attente certification niveau français / Waiting for French level certification - - 0 ECTS](#)

[EC : Attente de validation de la dernière année - DD JAUME IN / Waiting for validation of the last year - DD JAUME IN - - 0 ECTS](#)

[EC : DD TOHOKU semestre 3/5 / DD TOHOKU semester 3/5 - - 0 ECTS](#)

[EC : Attente certification niveau anglais / Waiting for English level certification - - 0 ECTS](#)

[EC : DD EPM MAITRISE RECHERCHE semestre 3/4 \(PFE1/2\) / DD EPM MAITRISE RECHERCHE semester 3/4 \(PFE1/2\) - - 0 ECTS](#)

[EC : Attente de validation de la dernière année - DD UPC IN / Waiting for validation of the last year - DD UPC IN - - 0 ECTS](#)

UE : Semestre hors GI avec 30 ECTS / Semester outside GI with 30 ECTS - 30 ECTS

[EC : Projet de fin d'études 0ECTS / Final project assignment 0ECTS - - 0 ECTS](#)

[EC : DD TOHOKU semestre 5/5 \(PFE2/2\) / DD TOHOKU semester 5/5 \(PFE2/2\) - - 30 ECTS](#)

UE : Semestre hors GI avec 30 ECTS / Semester outside GI with 30 ECTS - 30 ECTS

[EC : Echange académique S1 / Academic exchange S1 - ECH - 30 ECTS](#)

PARCOURS : Parcours asynchrone - 30 ECTS

UE : Projet de fin d'études / Final project assignment - 30 ECTS

[EC : Projet de fin d'études / Final project assignment - - 30 ECTS](#)

SEMESTRE : 2ème semestre / 2nd semester - 30 ECTS

PARCOURS : Filière Etudiant Entreprendre (FEE) / Entrepreneur Student Track - 30 ECTS

PARCOURS : Parcours standard / Standard track - 30 ECTS

UE : Contrat LEAN / LEAN Contract - 30 ECTS

[EC : Contrat lean / projet en entreprise - - 30 ECTS](#)

[EC : Contrat lean / enseignements - - 0 ECTS](#)

UE : Projet de fin d'études / Final project assignment - 30 ECTS

[EC : Participation à la Vie du département / Contribution to the department activities - - 1 ECTS](#)

[EC : Projet de fin d'études / Final project assignment - PFE30 - 29 ECTS](#)

PARCOURS : Parcours asynchrone - 30 ECTS

UE : Semestre d'échange en asynchrone (S2) - 30 ECTS

[EC : Echange académique S2 / Academic exchange S2 - - 30 ECTS](#)

PARCOURS : Parcours hors GI - ECTS

UE : Semestre hors GI sans ECTS / Semester outside GI with no ECTS - 0 ECTS

[EC : Semestre sans activité pédagogique / Semester with no pedagogical activity - - 0 ECTS](#)

[EC : Attente de validation de la dernière année - DD JAUME IN / Waiting for validation of the last year - DD JAUME IN - - 0 ECTS](#)

[EC : Attente de validation de la dernière année - DD UPC IN / Waiting for validation of the last year - DD UPC IN - - 0 ECTS](#)

[EC : DD PUC RIO semestre 2/4 / DD PUC RIO semester 2/4 - - 0 ECTS](#)

[EC : DD UPC ESTIEB semestre 2/4 / DD UPC ESTIEB semester 2/4 - - 0 ECTS](#)

[EC : DD TOHOKU semestre 2/5 / DD TOHOKU semester 2/5 - - 0 ECTS](#)

[EC : Césure / Gap year - - 0 ECTS](#)

UE : Semestre hors GI avec 30 ECTS / Semester outside GI with 30 ECTS - 30 ECTS

[EC : DD UPC ESTIEB semestre 4/4 \(PFE\) / DD UPC ESTIEB semester 4/4 \(PFE\) - - 30 ECTS](#)

[EC : Projet de fin d'études 0ECTS S2 / Final project assignment 0ECTS S2 - - 0 ECTS](#)

[EC : DD EPM MAITRISE RECHERCHE semestre 3/4 \(PFE2/2\) / DD EPM MAITRISE RECHERCHE semester 3/4 \(PFE1/2\) - - 30 ECTS](#)

[EC : Projet de fin d'études 0ECTS S2 / Final project assignment 0ECTS S2 - - 0 ECTS](#)

[EC : Projet de fin d'études 0ECTS S2 / Final project assignment 0ECTS S2 - - 0 ECTS](#)

[EC : DD PUC RIO semestre 4/4 \(PFE\) / DD PUC RIO semester 4/4 \(PFE\) - - 30 ECTS](#)

[EC : DD TOHOKU semestre 4/5 \(PFE1/2\) / DD TOHOKU semester 4/5 \(PFE1/2\) - - 30 ECTS](#)

UE : Semestre hors GI avec 20 ECTS / Semester outside GI with 20 ECTS - 20 ECTS

[EC : Echange académique S2 / Academic exchange S2 - - 30 ECTS](#)

UE : Semestre hors GI avec 30 ECTS / Semester outside GI with 30 ECTS - 30 ECTS

[EC : DD POLITO semestre 2/3 / DD POLITO semester 2/3 - - 30 ECTS](#)

[EC : Projet de fin d'études 0ECTS S2 / Final project assignment 0ECTS S2 - - 0 ECTS](#)

[EC : Double Diplôme IFP - semestre 2 - 30 ECTS - - 30 ECTS](#)

[EC : DD INSTN semestre 2/2 \(PFE\) / DD INSTN semester 2/2 PFE - - 30 ECTS](#)

[EC : Projet de fin d'études 0ECTS S2 / Final project assignment 0ECTS S2 - - 0 ECTS](#)

[EC : DD EPM semestre 2 / DD EPM semester 2 - - 30 ECTS](#)

[EC : DD IIT CHICAGO semestre 2/3 / DD IIT CHICAGO semester 2/3 - - 30 ECTS](#)

[EC : DD KAIST semestre 2/3 / DD KAIST semester 2/3 - - 30 ECTS](#)

[EC : DD SHERBROOKE semestre 2/3 / DD SHERBROOKE semester 2/3 - - 30 ECTS](#)

[EC : DD ETS semestre 2/3 / DD ETS semester 2/3 - - 30 ECTS](#)

[EC : Echange académique S2 / Academic exchange S2 - - 30 ECTS](#)

UE : Semestre hors GI avec 24 ECTS / Semester outside GI with 24 ECTS - 24 ECTS

[EC : Echange académique S2 / Academic exchange S2 - - 30 ECTS](#)

UE : Semestre hors GI sans ECTS / Semester outside GI with no ECTS - 0 ECTS

[EC : Attente diplomation / Waiting for graduation - - 0 ECTS](#)

UE : Semestre hors GI sans ECTS / Semester outside GI with no ECTS - 0 ECTS

[EC : Attente diplomation / Waiting for graduation - - 0 ECTS](#)

[EC : Attente certification niveau anglais / Waiting for English level certification - - 0 ECTS](#)

[EC : Attente certification niveau français / Waiting for French level certification - - 0 ECTS](#)

IDENTIFICATION

CODE : GI-0-S1-EC-VAC
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

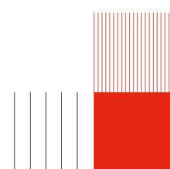
Campus LyonTech La Doua

20, avenue Albert Einstein - 69621 Villeurbanne cedex - France

Tél. + 33 (0)4 72 43 83 83 - Fax + 33 (0)4 72 43 85 00

www.insa-lyon.fr

membre de



IDENTIFICATION

CODE : GI-0-S1-EC-CESURE
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

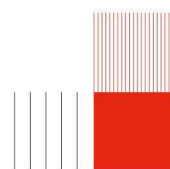
Campus LyonTech La Doua

20, avenue Albert Einstein - 69621 Villeurbanne cedex - France

Tél. + 33 (0)4 72 43 83 83 - Fax + 33 (0)4 72 43 85 00

www.insa-lyon.fr

membre de



IDENTIFICATION

CODE : GI-3-S1-EC-ECH
ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

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www.insa-lyon.fr

membre de



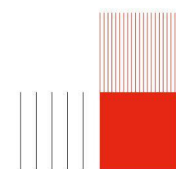
IDENTIFICATIONCODE : GI-3-S1-EC-APS
ECTS : 3**HOURS**Cours : 0h
TD : 42h
TP : 4h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 48h
Travail personnel : 12h
Total : 60h**ASSESSMENT METHOD**IE (interrogation écrite)
EE (évaluation écrite: poster, rapports.)
EO (évaluation orale) Oral**TEACHING AIDS****TEACHING LANGUAGE**

French

CONTACTMME SUBAI Corinne :
corinne.subai@insa-lyon.fr**AIMS**

This course belongs to teaching unit Production management GI-3-S1-UE-PIPR and contributes to the following skills :

- Security concepts
- Management of automated systems
- Choice of equipment and machines

CONTENTENVIRONMENT, COMMAND CHAINS AND REALIZATION -
Hydraulics, actuators, modulators
Measuring chains
Electric actuators
Real case study, choice of actuators and sensors (guiding principles)**BIBLIOGRAPHY**Catalogues et documents constructeur
Systèmes automatiques tome 1 et 2, J.-P. Hautier et J.-P. Caron Ellipses**PRE-REQUISITES**

IDENTIFICATION

CODE : GI-3-S1-EC-GIN
ECTS : 4

HOURS

Cours : 0h
TD : 14h
TP : 34h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 50h
Travail personnel : 16h
Total : 80h

ASSESSMENT METHOD

Exercices (TD) + final exam
TP : MCQ

TEACHING AIDS

Handout

TEACHING LANGUAGE

French

CONTACT

M. MONTEIRO Thibaud :
thibaud.monteiro@insa-lyon.fr

AIMS

This course belongs to teaching unit Pilotage de la production (GI-3-S1-UE-PIPR) and contributes to the following skills :

- C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
- C4 Sizing the hardware and / or software of a system (Level 3)
- C5 Managing a production system and react to malfunctions (Level 2)
- C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 3)
- C8 Managing supply in connection with the planning and inventory management policy (Level 3)

By allowing the engineering students to work and be evaluated on the following knowledge :

- (C8)

To be able to :

- (C2, C4, C6)
- (C8)
- (C5)
- Place the functions and objectives of production management for a company
- Manage supplies and inventory of a business
- Choose the method of management best suited to a given context
- Calculate the loads and manage production resources in the medium and long term
- TP Prelude : This lab is designed to introduce students to the various functions and the basics of production management."

CONTENT

- Introduction to production management
- Master data for production management
- Modelling of production systems and manufacturing processes for production management
- Different levels of decision making in production management: SOP, MPS, MRP, ...
- Main approaches for production planning: Inventory management, MRP, Kanban, ...

TP Prelude: 10 sessions in a scenario based on an industrial case cover the various functions by using a tutorial MIS (Prelude):

- Master data (material, BOM, routing, work centers...)
- Demand management : Master planning and sales orders
- Material requirement planning
- Production Order management
- Purchase order management
- Scheduling and production order follow-up

Education provided in French

BIBLIOGRAPHY

"Vincent GIARD : Gestion de la production et des flux. Economica 2003
Jacques Erschler et Bernard Grabot : Organisation et Gestion de la Production. Traité IC2 Hermes 2001
Lionel Dupont : La Gestion Industrielle. Hermès 1998
Management industriel et logistique, Gérard Baglin"

PRE-REQUISITES

Probabilities and statistics, computer sciences

IDENTIFICATION

CODE HU-0-S1-EC-L-FLE-
COMPB2

ECTS : 2

HOURS

Cours :	0h
TD :	26h
TP :	0h
Projet :	0h
Evaluation :	0h
Face à face pédagogique :	26h
Travail personnel :	0h
Total :	26h

ASSESSMENT METHOD

- Continuous assessment: 50%
- Final evaluation: 50%
- Attendance required

TEACHING AIDS

Various authentic documents
(paper, audio, video and
multimedia)

TEACHING LANGUAGE

French

CONTACT

Mme Aumeunier Lise :
elisabeth.aumeunier@insa-lyon.fr

Mme Fradois Delphine :
delphine.fradois@insa-lyon.fr

AIMS

Depending on the level:

- Review level B1 of CEFR and improve your knowledge of B2 level
- or
- Improve your knowledge of B2 level and begin learning level C1

CONTENT

Systematic practice of oral and written comprehension techniques, as well as narrative, summary, synthesis, and argumentation techniques. The program is consistent with the exercises required for the TCF.

Level B1 to review

Level B2

- Write a simple and coherent text on familiar topics
- Write personal letters to describe experiences and impressions.
- Write clear and detailed texts on a wide range of topics related to one's interests.
- Write an essay or report, conveying information or presenting reasons for or against a given opinion.
- Write a clear, well-structured text and develop a point of view.
- Write about complex topics in an essay or report.
- Synthesize ideas.
- Adopt a style appropriate to the audience.

BIBLIOGRAPHY

- La documentation française <https://www.vie-publique.fr/publications-de-la-documentation-francaise>
- Production écrite B1/B2 (Didier)
- Ecrire pour convaincre (Hachette)
- Vocabulaire expliqué du Français(CI)
- Francoscopie (Larousse)

Online resources :

<https://fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B1 Level

IDENTIFICATION

CODE : HU-3-S1-EC-L-JAP
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

M. Mihara Norio :
norio.mihara@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop
and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex
tasks that require the students to engage in the 5 linguistic activities, at a level and with
linguistic input that are appropriate for the group. In-class and/or guided independent
study of the forms and functions of the language is regular and adapted to the level of
the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar
training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATIONCODE HU-0-S1-EC-L-FLE-
CIVB2C1

ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Project based evaluation.

TEACHING AIDS

Various authentic materials

TEACHING LANGUAGE

French

CONTACTMme Aumeunier Lise :
elisabeth.aumeunier@insa-lyon.frMme Fradois Delphine :
delphine.fradois@insa-lyon.fr**AIMS**

Humanities Skills:

2. Work, learn, and develop independently

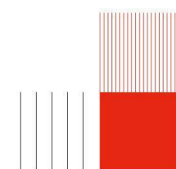
3. Interact with others and work in a team

4. Demonstrate creativity, innovation, and initiative

7. Work in an international and intercultural context

Language Activities: Written Expression, Written Expression, Oral Expression, Listening
Comprehension, Oral ExpressionAn intercultural approach for a better understanding of French civilization and culture
with the aim of fostering student autonomy and facilitating their integration
into their studies, student life, and social and professional life.**CONTENT**The work will be organized around various activities: debates, presentations, projects,
global simulations, forums, etc. on various past and present themes (society, culture,
history, etc.) and will integrate an intercultural dimension.**BIBLIOGRAPHY**You will find a selection of resources available online at the following address [https://
fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie](https://fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie)**PRE-REQUISITES**

B2 level



IDENTIFICATION

CODE : HU-3-S1-EC-L-CHI
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

Mme Jiang Chunyan :
chunyan.jiang-huang@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

-- understand and apply social codes from different countries

--have a conversation appropriate de the context

--confront communicational challenges of increasing difficulty

--produce diverse types of written expression

--understand authentic documents (written and audiovisual)

--talk about a society or a social phenomenon

--apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks

that require the students to engage in the 5 linguistic activities, at a level and with linguistic

input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

--language learning methodology (RC, OC, WP, vocabulary learning and grammar training)

--oral presentation skills

--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-3-S1-EC-L-ANG-A2
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation consists in continuous assessment of the student's preparation and oral work in class, as well as written work and oral presentations

TEACHING AIDS

Teaching documents are chosen by the instructor according to the target level:

-Various instructional and authentic English-language materials

-Audio-visual documents

TEACHING LANGUAGE

English

CONTACT

Mme Jouffroy Jeannie :
jeannie.jouffroy@insa-lyon.fr

AIMS

This EC contributes to the development of the following transversal skills :

CT3: INTERACT WITH OTHER PERSONS & WORK ON A TEAM
3.2 Put original discourse into context using explicit references.
3.3 Master non-verbal communication: posture and gesture.

CT7: WORK IN AN INTERNATIONAL, INTERCULTURAL CONTEXT
7.1 Communicate and interact in a foreign language.

The student-engineer will practice and be evaluated on the following learning outcomes:

- knowing how to communicate in person, by email and/or by telephone on factual topics
- knowing how to give instructions, explain a procedure, make a dynamic oral presentation
- mastery of the vocabulary and syntax necessary to the successful accomplishment of the above tasks

CONTENT

Students will learn and put into practice the professional communication skills, both oral and written, useful to engineers. In the process, they will develop those of the following skill sets determined to best further their linguistic abilities and respond to their particular needs. The activities, associated with socializing skills in the world of work, may include:

BIBLIOGRAPHY

The Common European Framework of Reference for Languages, Council of Europe

PRE-REQUISITES

A2/B1 level

IDENTIFICATION

CODE : HU-3-S1-EC-L-ITA
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

Mme Cognet Anne :
anne.cognet@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-3-S1-EC-L-POR
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

Mme Strelow Antunes Isabel :
isabel.strelow-antunes@insa-
lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES
It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop
and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex
tasks that require the students to engage in the 5 linguistic activities, at a level and with
linguistic input that are appropriate for the group. In-class and/or guided independent
study of the forms and functions of the language is regular and adapted to the level of
the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar
training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-3-S1-EC-L-ARA
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

M. Garay Oyarzo Edicto :
edicto.garay-oyarzo@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :
CT7: Work in an international, intercultural context
--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning
outcomes:

The student CAN:
-- understand and apply social codes from different countries
--have a conversation appropriate de the context
--confront communicational challenges of increasing difficulty
--produce diverse types of written expression
--understand authentic documents (written and audiovisual)
--talk about a society or a social phenomenon
--apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex
tasks that require the students to engage in the 5 linguistic activities, at a level and with
linguistic input that are appropriate for the group. In-class and/or guided independent
study of the forms and functions of the language is regular and adapted to the level of
the group.

In the first semester, themes covered include:

--language learning methodology (RC, OC, WP, vocabulary learning and grammar
training)
--oral presentation skills
--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-3-S1-EC-L-ANG
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation consists in continuous assessment of the student's preparation and oral work in class, as well as written work and oral presentations'

TEACHING AIDS

Teaching documents are chosen by the instructor according to the target level:
-Various instructional and authentic English-language materials
-Audio-visual documents

TEACHING LANGUAGE

English

CONTACT

Mme JOUFFROY Jeannie :
jeannie.jouffroy@insa-lyon.fr
M. HODGSON David :
david.hodgson@insa-lyon.fr

AIMS

This EC contributes to the development of the following transversal skills :

CT3: INTERACT WITH OTHER PERSONS & WORK ON A TEAM
3.2 Put original discourse into context using explicit references.
3.3 Master non-verbal communication: posture and gesture.

CT7: WORK IN AN INTERNATIONAL, INTERCULTURAL CONTEXT
7.1 Communicate and interact in a foreign language.

The student-engineer will practice and be evaluated on the following learning outcomes:

--knowing how to communicate in person, by email and/or by telephone on factual topics
--knowing how to give instructions, explain a procedure, make a dynamic oral presentation
--mastery of the vocabulary and syntax necessary to the successful accomplishment of the above tasks

CONTENT

Students will learn and put into practice the professional communication skills, both oral and written, useful to engineers. In the process, they will develop those of the following skill sets determined to best further their linguistic abilities and respond to their particular needs. The activities, associated with socializing skills in the world of work, may include:

--Holding a conversation on familiar topics
--Telling a story in a conversational context
--Initiating and respond to emails
--Hold a phone conversation on factual topics
--Giving instructions/ Explain a procedure
--Making a dynamic oral presentation using notes if needed on a topic of current events, travel, culture, other
--Preparing a personal or professional time abroad
--Organising after-work activities with colleagues
--Hosting clients and colleagues at lunch
--Learning and explaining professional rules of conduct as well as the company culture

BIBLIOGRAPHY

Le Cadre européen commun de référence pour les langues, le Conseil d'Europe

PRE-REQUISITES

A2/B1 level in English

IDENTIFICATION

CODE HU-0-S1-EC-L-FLE-
TTCB1B2

ECTS : 2

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

-Continuous assessment: 50%
-Final evaluation: 50%
-Attendance required

TEACHING AIDS

Various multimedia documents

TEACHING LANGUAGE

French

CONTACT

Mme Aumeunier Lise :
elisabeth.aumeunier@insa-lyon.fr

Mme Fradois Delphine :
delphine.fradois@insa-lyon.fr

AIMS

Depending on the level:

- Review level B1 of CEFR and improve your knowledge of B2 level
- or
- Improve your knowledge of B2 level and begin learning level C1

CONTENT

Systematic practice of oral and written comprehension techniques, as well as narrative, summary, synthesis, and argumentation techniques. The program is consistent with the exercises required for the TCF.

Level B1 to review

Level B2

- Write a simple and coherent text on familiar topics
- Write personal letters to describe experiences and impressions.
- Write clear and detailed texts on a wide range of topics related to one's interests.
- Write an essay or report, conveying information or presenting reasons for or against a given opinion.
- Write a clear, well-structured text and develop a point of view.
- Write about complex topics in an essay or report.
- Synthesize ideas.
- Adopt a style appropriate to the audience.

BIBLIOGRAPHY

Online resources :

<https://fle.insa-lyon.fr/fr/rubrique/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B1 level

IDENTIFICATION

CODE : HU-3-S1-EC-L-RUS
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

M. Iakovlev Maxime :
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AIMS

This EC belongs to the UE: HUMANITIES
It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-3-S1-EC-L-ALL
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

Mme Zuzik Mareike :
mareike.zyzik@insa-lyon.fr
Mme Vincensini Catherine :
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M. Bouet Christian :
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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context
--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (B1-B2+) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

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membre de



IDENTIFICATION

CODE : HU-3-S1-EC-L-TAN
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation in TANDEM consists in continuous assessment of their preparation and oral work in class, as well as written work and an oral presentation.

TEACHING AIDS

-Diverse authentic documents, both written and audiovisual

TEACHING LANGUAGE

French

CONTACT

Mme Davila Lozano Monica :
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Mme Strelow Isabel :
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Mme Raymond Camille :
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Mme Fradois Delphine :
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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills valued by INSA:

CT7: Work in an international, intercultural context

7.1 Communicate in a foreign language

7.2 Decode cultural references in speech, attitudes and behaviors.

7.3 Put one's own values, beliefs and behaviors into a cultural perspective.

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

THE STUDENT CAN:

--talk about themselves and their country

--talk about a social phenomenon OR

--talk about a cultural phenomenon OR

--talk about a scientific phenomenon OR

--have a conversation appropriate to the context; communicate in an appropriate manner

--give an oral presentation AND/OR write clear expository prose

--learn and apply usage and vocabulary discovered in class

--understand and apply diverse linguistic and social codes

By the end of the semester, the student will be able to...

- Communicate and interact in one or more foreign languages, both orally and in writing, adapting to the context and the interlocutors, from the transmission of simple information to mediation in more complex communication situations. (Communication skills)

- Identify, understand, and utilize the cultural codes associated with their own language and those of their interlocutors in order to adopt appropriate behavior and communication in diverse intercultural contexts. (Intercultural skills)

- Produce written or oral messages in their native language or in the target language. (Linguistic skills)

CONTENT

The TANDEM modules offer linguistic and cultural exchanges between French-speaking students and German-speaking, English-speaking, Spanish-speaking and Portuguese-speaking students in each of the languages.

The INTERCOMPREHENSION module is all about learning to understand native speakers of a Romance language different to one's own. Each student speaks their native language and learns to understand the others.

The work is carried out by organizing a debate around a controversial subject, by writing a story, by presenting an element belonging to the specific culture, by playing board games ... in short, by interacting permanently and in any form possible with other students fluent in the different working languages.

Of course, oral comprehension is also at the heart of this course. It is a living, authentic learning in the spirit of sharing. Everyone has something to learn and to pass on.

BIBLIOGRAPHY

PRE-REQUISITES

FOR THE TANDEMS

For francophone students:

.B2 in German, English, Spanish

.B1 in Portuguese

For anglophone students: B1 minimum in French

FOR INTERCOMPREHENSION

Must be native speaker of a romance language (French, Spanish, Catalan, Italian, Portuguese, Romanian)

IDENTIFICATION

CODE : HU-3-S1-EC-L-ESP
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

The average is made up of half (50%) continuous assessment (assessments of the 5 skills carried out in class) and a common exam (50% of the average) which assesses Written Comprehension, Oral Comprehension, Grammatical and Lexical Skills and Written Expression. This common exam does not have the value of level validation.

TEACHING AIDS

Authentic and/or didactic documents related to the selected topics.
- "Pink" grammar and exercise booklets
- "Yellow" booklet: conjugation guides
- The CRL

TEACHING LANGUAGE

French

CONTACT

M. Suarez Lopez Gonzalo :
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AIMS

The targeted and mobilized skills are those of both the INSA Humanities Skills Framework (specifically skills 3 and 7) and the CEFR.
Consistent description of the CEFR skills.
<http://www.sciencespo-lille.eu/sites/default/files/cecrl.pdf>

At the end of the semester, the student will be able to...

- Communicate and interact in Spanish, both orally and in writing, adapting to the context and the interlocutors, from the transmission of simple information to mediation in more complex communication situations. (COMMUNICATIONAL COMPETENCY)
- Identify, understand, and utilize the cultural codes associated with the Spanish language and its speakers in order to adopt appropriate behavior and communication in diverse intercultural contexts. (INTERCULTURAL COMPETENCY)
- Understand spoken and written messages in Spanish in communication situations corresponding to the targeted proficiency level, in accordance with the Common European Framework of Reference for Languages (CEFR).
- Produce spoken and written messages in Spanish, effectively utilizing the linguistic resources specific to the targeted proficiency level (CEFR).

CONTENT

The five skills recognized by the Common European Framework of Reference for Languages (CEFR) are practiced and assessed several times throughout the semester/year, ensuring regular practice of the various skills and knowledge acquired.

The themes, grammar, and vocabulary covered in class are adapted to the target level (indicated in the group code) and are at the discretion of the teacher.

BIBLIOGRAPHY

Web resources:

- for listening comprehension, grammar, and vocabulary training (all levels): <http://www.ver-taal.com/index.htm>
- grammar and vocabulary exercises (all levels): <https://www.espanolfacile.com/>
- <https://moodle.insa-lyon.fr/course/index.php?categoryid=353>

PRE-REQUISITES

None. Courses range from beginner to advanced level. Each student will be placed in a group corresponding to their level, either through a test at the beginning of the year (for new students) or based on their level from the previous year for students already attending INSA

IDENTIFICATIONCODE : GI-3-S1-EC-COM-HU
ECTS : 1**HOURS**

Cours :	0h
TD :	16h
TP :	0h
Projet :	0h
Evaluation :	0h
Face à face pédagogique :	16h
Travail personnel :	4h
Total :	20h

ASSESSMENT METHOD**TEACHING AIDS****TEACHING LANGUAGE**French
English**CONTACT**M. CHAUMARD Davyd :
davyd.chaumard@insa-lyon.fr**AIMS**

This course belongs to teaching unit Humanities and Sports Education (GI-3-S1-UE-HUEPS) and contributes to the following skills :

- to develop a practical awareness of the various channels of communication, through individual and group exercises focusing on drama
- to explore one's personal resources and limits (role games and improvisations)
- to analyse individual participation in a group situation
- to understand the methods of the social sciences, which provide a basis for the analysis of the activities
- cooperation and efficiency in a team job
- to know how preparing for 4GIs group project

CONTENT

Working with professional actors

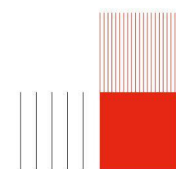
practicing communication : from experience to concepts

BIBLIOGRAPHY

provided by the professor, according to the subject of research chosen by each team of students

PRE-REQUISITES

Ability to use oral, written and audio-visual modes of expression in French



IDENTIFICATION

CODE : GI-3-S1-EC-REN
ECTS : 0

HOURS

Cours : 4h
TD : 2h
TP : 2h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 8h
Travail personnel : 0h
Total : 8h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

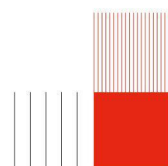
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IDENTIFICATION

CODE : CDS-3-S1-EC-EPS
ECTS : 1

HOURS

Cours : 0h
TD : 20h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 0h
Total : 20h

ASSESSMENT METHOD

Assessment in Physical Education concerns the teaching of Sports and Artistic Physical Activities (APSA), and will take the form of continuous assessment with half-yearly marking.

The mark depends on the degree of acquisition of the skills expected in each of the activities, and the progress made over all the sessions in the cycle. The mark takes into account :

Individual and/or team performance
mastery of execution
Progress in the sports project
Responsibility and autonomy

TEACHING AIDS

All physical, sporting, artistic and competitive activities

TEACHING LANGUAGE

French

CONTACT

Mme JAUSSAUD Marie :
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AIMS

This EC is part of the Teaching Unit: SHS and contributes to the development of the School's transversal competences

1* Auto-evaluating one's own performance

Knowledges :

- Fundamentals, principles of action and terminology of sports activities
- Criteria for observation, achievement and success.

Abilities :

- Assess your level of practice
- Build up a warm-up
- Set goals for progress
- Manage physical and mental potential

2* Work, learn and develop independently

Knowledge :

- PSAA rules
- Observation criteria
- Principles of warm-up and cool-down

Abilities :

- Mobilise resources
- Analyse, observe, question
- Take on different roles (referee, choreographer)

3* Interact with others, work as part of a team

Knowledges :

- Roles and functions in each sports activity

Abilities :

- Communicate appropriately: verbal, non-verbal and postural communication.
- Integrate into a group
- Take part in and develop a group project
- Take the initiative
- Be a good listener

4* Be creative, innovative and enterprising

Knowledge :

- Artistic disciplines

Abilities :

- Draw on knowledge and resources from different artistic fields to produce an original work.
- Mobilise the imagination and sensibility and make them visible through dance movement
- Access the symbolism of the body

5* Act responsibly in a complex world

Knowledge :

- Safety and operating rules

Abilities :

- Identify uncertainties and risks and act to reduce them
- Integrate a responsible dimension into their actions
- Show respect and fair play in a power struggle

6* Working in an international context

Knowledge :

- Socio-cultural differences

Abilities :

- Integrate cultural diversity into group work
- Act with respect for self and others

CONTENT

Physical Education and Sport lessons are organised around traditional Physical Education lessons, or advanced lessons, or appropriate practices (EPSA), or competitive practices within the framework of the Section Sportive Haut Niveau.

1. Physical Education lessons :

Students choose one or two physical and sporting activities per year from among the activities offered by the sports centre (individual, group, dual).

2. Appropriate Physical Education lessons: For all students who are exempt from physical activity for at least 2 months:

Swimming, Body-building, Nordic Walking, Somatic Exercise, Sophrology, Wheelchair Basketball, Pilates, Table Tennis, etc.

3. Advanced Physical Education courses :

Specialisation in a sporting activity, University training and competitions

4. SSHN (High-Level Athlete section)

University training and competitions

EPS 3GEN and GENEPI :

1st course in Hauteville in October: 2 days: outdoor activities Objective: Create team cohesion

1st term: PE lessons on Wednesdays from 8.00 to 9.30am: 9 team sports sessions

BIBLIOGRAPHY

PRE-REQUISITES

- EPS: none
- Appropriate Physical Education: subject to medical advice
- Advanced courses and competitive practice: previous practice required subject to specific selection according to each activity
- SHN: ministerial list

Levels 1 and 2: Physical Education, Appropriate physical education

Level 3: Advanced courses and competitive practice, SHN

INSA LYON

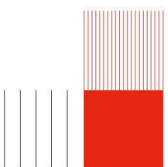
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IDENTIFICATION

CODE : GI-3-S1-EC-PSX
ECTS : 4

HOURS

Cours : 2h
TD : 50h
TP : 4h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 60h
Travail personnel : 20h
Total : 84h

ASSESSMENT METHOD

IE (interrogation écrite)
IE (interrogation écrite)
IE (interrogation écrite)
EE+EO (évaluation écrite+orale)

TEACHING AIDS

Lecture handout and exercises
(part Probability & statistics and
part Design of experiments)
Use of spreadsheet files.
Course materials available
electronically.

TEACHING LANGUAGE

French
English

CONTACT

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AIMS

This course belongs to teaching unit Informatique et mathématiques décisionnelles (GI-3-S1-UE-IMAD) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 3)
- A2 Operate a model of a real or virtual system (Level 3)
- A3 Implement an experimental approach (Level 3)
- A5 Process data (Level 3)
- A6 Report on an analysis or on a scientific approach (Level 3)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 2)
- C8 Managing supply in connection with the planning and inventory management policy (Level 2)
- C10 Developing and implementing an action plan as part of a quality and continuous improvement approach (Level 2)
- C15 Managing a project based on a master plan (planning, budgeting, definition of monitoring indicators) (Level 1)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Notion of probability, Principles of combinatorial analysis, Probability calculus (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Notions of conditional probabilities and independent events (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Notion of random variable, Main indicators for random variables: expected value, variance... (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Main probability distributions (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Central limit theorem (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Notions of sample, population, statistical inference (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Client and supplier risks (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Ponctual estimation, Confidence intervals (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- One-tailed and two-tailed statistical tests, Comparison tests, goodness-of-fit tests, independence tests (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Implementation of a design of experiments (C10)

To be able to :

- Computing the probability of a complex event, based on simple events or by combinatorial analysis (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Modelling a random phenomenon choosing the appropriate probability distribution (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Computing and interpreting the main indicators related to a random variable (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Determining the average behaviour of a phenomenon considering a large number of random trials (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Carrying out a statistical test to validate a hypothesis related to a population (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Estimating the parameters of a statistical population based on a sample of values (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Carrying out a statistical test to validate a hypothesis related to a population (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Planning the tests to be conducted (A2, A3, C15)
- Carrying out the measurements (A1, C1)
- Developing a computing tool to process a design of experiments (A3)

Knowledge :

- Notion of probability, Principles of combinatorial analysis, Probability calculus
- Notions of conditional probabilities and independent events
- Notion of random variable, Main indicators for random variables: expected value, variance...
- Main probability distributions
- Central limit theorem
- Notions of sample, population, statistical inference
- Client and supplier risks
- Ponctual estimation, Confidence intervals
- One-tailed and two-tailed statistical tests, Comparison tests, goodness-of-fit tests, independence tests
- Optimal organization of experimentation
- Regression model and mathematical predictor
- Statistical model associated with the analysis of variance
- Implementation of a design of experiments

Capacities :

- Computing the probability of a complex event, based on simple events or by combinatorial analysis

- Modelling a random phenomenon choosing the appropriate probability distribution
- Computing and interpreting the main indicators related to a random variable
- Determining the average behaviour of a phenomenon considering a large number of random trials
- Computing the client / supplier risks according to a quality control procedure
- Estimating the parameters of a statistical population based on a sample of values
- Carrying out a statistical test to validate a hypothesis related to a population
- Planning the tests to be conducted
- Carrying out the measurements
- Developing a computing tool to process a design of experiments
- Analyzing the results obtained
- Finding an optimal setup condition

CONTENT

The aim of this course is to introduce the engineering tools and techniques needed to model and analyse phenomena and systems involving uncertainties, in particular through experimental approaches such as design of experiments. It is based on a description of standard mathematical concepts, coming from, among others, probability theory, descriptive statistics and analysis of variance. Numerous examples and application exercises are provided for each part of the course.

The knowledge and skills acquired are also put into practice through a case study based on a quality control problem, enabling the application of estimation methods and hypothesis testing, and through the implementation of an experiment for setting up a machine (a catapult), with a Taguchi's fractional design.

These different activities also lead the students to use and develop data processing tools on spreadsheets.

BIBLIOGRAPHY

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Y. Dodge : Premiers pas en statistique. Springer. 2003.

S. Morgenthaler : Introduction à la statistique. Presses Polytechniques et Universitaires Romandes. 1997.

P. Bogaert : Probabilités pour scientifiques et ingénieurs: Introduction au calcul des probabilités, 2ème édition. De Boeck Supérieur. 2020.

G. Sado, M.C. Sado : Les plans d'expériences. De l'expérimentation à l'assurance qualité, Nlle édition. AFNOR. 2000.

M. Vigier : Pratique des plans d'expériences. Méthodologie Taguchi. Les Éditions d'Organisation. 1988.

J. Goupy : Introduction aux Plans d'expériences, 2ème édition. Dunod. 2001.

G. Lasnier : Plans d'expériences en gestion industrielle. Hermes science publ. - Lavoisier. 2003.

PRE-REQUISITES

None

INSA LYON

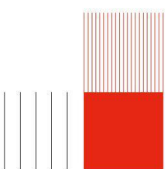
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IDENTIFICATION

CODE : GI-3-S1-EC-APM
ECTS : 4

HOURS

Cours : 0h
TD : 30h
TP : 32h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 66h
Travail personnel : 16h
Total : 82h

ASSESSMENT METHOD

1 x 1h50 = written exam

TEACHING AIDS

Course material

TEACHING LANGUAGE

French

CONTACT

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M. DUMITRESCU Emil :
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AIMS

This course belongs to teaching unit Informatique et mathématiques décisionnelles (GI-3-S1-UE-IMAD) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 2)
- A2 Operate a model of a real or virtual system (Level 2)
- A4 Design a system that meets a set of specifications (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
- C3 Evaluating, prototyping and simulating a system (Level 2)

- * Develop an experimental approach (Level 2)
- * Identify, formulate and solve a problem of complex engineering (Level 2)
- the following School skills : specific GI
- * evaluate, prototype and simulate a system (Level 2)
- * size the hardware and / or software of a system (Level 2)
- the following transversal School skills
- * Work, learn, evolve in an autonomous way (Level 2)
- * Interact with others, work in team (Level 3)

which decline the following professional skills :

- * Design and size systems required for a given activity of production, service, distribution ... (Level 2)
- * Identify and correct discrepancies in a continuous improvement process (material, human and information systems) by the implementation of analytical and simulation tools (Level 2)
- * Ensure quality reporting through the establishment of appropriate indicators, based on a systemic view of organizations (matrix, silo, project) for any scope and any type of activity (Level 2)"

CONTENT

- Adopt a structured programming approach to solving a complex problem
- Choose the solution methods and data structures appropriate for the problem
- Build and analyze algorithms in a pseudo-language
- Translate a UML class diagram into Python
- Transcribe an algorithm written in a pseudo-language into a Python program

BIBLIOGRAPHY

V. FELEA, Introduction à l'informatique, Apprendre à concevoir des algorithmes, Ed. Vuibert, 2013
L. DEBRAUWER, Algorithmique, Ed. ENI, 2008
T.CORMEN, C. LEISERSON, R.RIVEST, C. STEIN, Introduction à l'algorithmie, Ed. Dunod, 2002
DELANNOY C., Programmer en Java, Ed Eyrolles, 2001
OUSSALAH C., Ingénierie objet, Interéditions, 1997

PRE-REQUISITES

INSA LYON

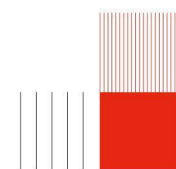
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membre de



IDENTIFICATION

CODE : GI-3-S1-EC-ROO
ECTS : 3

HOURS

Cours : 0h
TD : 18h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 32h
Travail personnel : 8h
Total : 40h

ASSESSMENT METHOD

IE (interrogation écrite)
EE (évaluation écrite: poster, rapports)

TEACHING AIDS

Available on caseine.org:
- lecture notes (in English and French)
- virtual programming labs to implement and test the programs as self-training (in English)
- quiz to check your understanding (in English)
- French / English glossary + crossword puzzles

TEACHING LANGUAGE

French
English

CONTACT

Mme LADIER Anne-Laure :
anne-laure.ladier@insa-lyon.fr

AIMS

This course belongs to teaching unit Informatique et mathématiques décisionnelles (GI-3-S1-UE-IMAD) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 2)
- A2 Operate a model of a real or virtual system (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
- C3 Evaluating, prototyping and simulating a system (Level 1)
- C4 Sizing the hardware and / or software of a system (Level 1)
- C5 Managing a production system and react to malfunctions (Level 1)
- C8 Managing supply in connection with the planning and inventory management policy (Level 1)
- C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 1)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Linear programming, simplex algorithm, duality (A1, A2, C2)
- Integer linear programming, branch&bound algorithm (A1, C2)
- Use of Excel Solver and OpenSolver for decision support (xA2, C2, C3)

To be able to :

- Specify a linear program in integer or real variables from the description of a problem expressed in natural language (C2, C3)
- Design a decision support tool for the management of a production or transport process
- Design a decision support tool for the management of a production or transport process (A1, A2, C2, C4, C5, C8, C9)
- Identify the classic problem class to which a given real problem relates (A1, C2)

CONTENT

Introduction to operations research and its industrial applications
Modeling in linear programming, graphical solving
Simplex algorithm
Duality and sensitivity analysis
Branch & Bound
Modeling in integer linear programming; modeling tips; concepts of complexity and formulation quality
Backpack and bin-packing problem
Modeling in PL/PLNE with Excel: application to a lot-sizing problem with capacity constraints
Coverage and partition issues
Traveling Salesman (TSP) and Vehicle Routing (VRP) Issues

BIBLIOGRAPHY

- Dinkel J. J., G. A. Kochenberger and D. R. Plane, Management Science Text and Applications, Irwin Editor, 1978, ISBN 0-256-02037-X
- Taha H. A., Operations Research an introduction, Sixth edition, Prentice Hall, 1997, ISBN 0-13-272915-b

PRE-REQUISITES

- Linear algebra (matrix calculation),
- Production management (MRP, concept of independent / dependent need).
- Use of simple Excel functions

IDENTIFICATIONCODE : GI-3-S1-EC-PFI
ECTS : 3**HOURS**Cours : 0h
TD : 40h
TP : 8h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 48h
Travail personnel : 0h
Total : 48h**ASSESSMENT METHOD**Written Assessment
Supervised Practical Work**TEACHING AIDS**Teaching Method
A project provides a common thread
- 1 lecture session followed by 2 tutorial sessions
- Work in groups of 2 people
The practical work is organized in groups of 3 students and takes place on a machine tool**TEACHING LANGUAGE**

French

CONTACTM. ARNAUD Frédéric :
frederic.arnaud@insa-lyon.fr**AIMS**

This course belongs to teaching unit Conception de produits et systèmes industriels (GI-3-S1-UE-CPSI) and contributes to the following skills :

A1 Analyze a system (real or virtual) or a problem (Level 1)

A3 Implement an experimental approach (Level 3)

A6 Report on an analysis or on a scientific approach (Level 3)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)

C3 Evaluating, prototyping and simulating a system (Level 1)

C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 3)

C13 Considering technological and methodological innovation (Level 2)

B2 work, learn, progress autonomously (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Mold design;

- Precedence constraints in machining;

- Isostatism;

- Cutting parameters.

To be able to :

- (A1, A3, A6, C2, C3, C6, C13, B2)

Design a mechanical manufacturing sequence by choosing a suitable material, designing a metal mold, respecting the geometric constraints, taking into account the economic constraints and machine capabilities

CONTENT

Mold design; precedence constraints in machining; Isostatism; cutting parameters

BIBLIOGRAPHY

Précis de construction mécanique. Projets-méthodes, Production, Normalisation ISBN : 2-09-194002-X

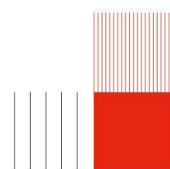
Fabrication par usinage. 2e édition ISBN : 978-2-10-051626-1

PRÉCIS DE FONDERIE MÉTHODOLOGIE, PRODUCTION ET NORMALISATION ISBN : 2-09-194019-4

Fonderie de précision à modèle perdu ISBN : 2-85330-100-1

PRE-REQUISITES

None



IDENTIFICATIONCODE : GI-3-S1-EC-RDM
ECTS : 3**HOURS**Cours : 0h
TD : 30h
TP : 2h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 34h
Travail personnel : 12h
Total : 46h**ASSESSMENT METHOD**EE (évaluation écrite: poster, rapports) : mini-project coefficient 1
IE (interrogation écrite) : written test 1h50 coefficient 2**TEACHING AIDS**

Duplicated course material

TEACHING LANGUAGE

French

CONTACTMme WALTER-LE BERRE Hélène
:
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This course belongs to teaching unit Conception de produits et systèmes industriels (GI-3-S1-UE-CPSI) and contributes to the following skills :

C2 Modeling and designing an information, decision and production system, of goods and services (Level 1)

C3 Evaluating, prototyping and simulating a system (Level 1)

C4 Sizing the hardware and / or software of a system (Level 1)

CONTENT

Classes and directed works : Strength Of Materials :

- * Stress modelling under simple load
- * Static balance and torque of mechanical actions
- * Stress tensor, strain tensor and strength criteria
- * Constitutive laws (Hooke's laws)
- * Bending deflection
- * Introduction to the Finite Element method
- * Application to a mini-project

- To Acquire the fundamental concepts of Strength Of Materials in order to analyze the behavior of mechanical components.

BIBLIOGRAPHYS. TIMOSHENKO, Résistance Des Matériaux, Librairie Polytechnique Béranger, 1963.
P. AGATI, F. LEROUGE et M. ROSSETTO, Résistance des Matériaux, Dunod, 1999.
D. GAY et J. GAMBELIN, Dimensionnement des Structures, Hermes, 1999.**PRE-REQUISITES**

- * Mechanics : static balance
- * Mathematics : integral and differential calculation bases and matrix algebra



IDENTIFICATION

CODE : GI-3-S2-EC-GFL
ECTS : 2

HOURS

Cours : 0h
TD : 4h
TP : 24h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 28h
Travail personnel : 8h
Total : 40h

ASSESSMENT METHOD

Report and simulation model

TEACHING AIDS

Moodle

TEACHING LANGUAGE

French
English

CONTACT

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M. MONTEIRO Thibaud :
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AIMS

This course belongs to teaching unit Gestion de la chaîne logistique (GI-3-S2-UE-GECL) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 3)
- A2 Operate a model of a real or virtual system (Level 3)
- A3 Implement an experimental approach (Level 3)
- A6 Report on an analysis or on a scientific approach (Level 1)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 3)
- C4 Sizing the hardware and / or software of a system (Level 2)
- C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Physical behavior of a chain of Actuators and Sensors
- To be able to :
 - (A1, A2, A3, A6, C1, C2, C3, C4, C9)
 - Apply a "simulation project" approach
 - Trace flows
 - Analyze average flows
 - Find the bottleneck
 - Make stocks functional
 - Sizing a production line
 - Propose modifications to optimize flows
 - Implement an experimental approach to validate the proposed modifications

CONTENT

Apply a "simulation project" approach

1. Define Performance Indicators
2. Find the data required for the simulation
3. Model the problem by simplifying it
4. Validate the model

Finding a Bottleneck

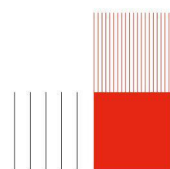
Propose changes to optimize flows

Implementing tools and fundamentals of industrial engineering in a simulation project: Stock management, Experimental design, Stochastic processes, Queueing systems...

BIBLIOGRAPHY

PRE-REQUISITES

Production management,
Experimental design,
Stochastic processes,
Queueing systems



IDENTIFICATION

CODE : GI-3-S2-EC-ISC
ECTS : 3

HOURS

Cours : 6h
TD : 28h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 48h
Travail personnel : 12h
Total : 60h

ASSESSMENT METHOD

IE (interrogation écrite) 80%
ES (évaluation en situation) 20%

TEACHING AIDS

Course and TD support
Exercices + Case studies

TEACHING LANGUAGE

French

CONTACT

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anne-laure.ladier@insa-lyon.fr

AIMS

This course belongs to teaching unit Supply chain management (GI-3-S2-UE-GECL) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 2)
- A2 Operate a model of a real or virtual system (Level 2)
- A5 Process data (Level 2)
- A6 Report on an analysis or on a scientific approach (Level 2)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
- C3 Evaluating, prototyping and simulating a system (Level 2)
- C8 Managing supply in connection with the planning and inventory management policy (Level 2)
- C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 2)
- C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 1)
- B5 act responsibly in a complex world (Level 1)
- B6 position oneself, work, and evolve within a company or a socio-productive organization (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Forecasting methods, N&OP and their link with Demand management (A1, A2, C2, C8)
- Supply chain models (C2, C8, C9, B6)
- VBA syntax and event driven programming paradigm (C2)
- Bullwhip effect in the supply chain (C3)

To be able to :

- Identify the main functions of an industrial company, their operational logic, and their management methods. (C11, B6)
- Characterize the relationships of the company with its environment and within its supply chain, and the associated processes. (C2, C8, C9, B5, B6)
- Import and filter data using VBA syntax (A1, A2, A5, C1, C2)
- Manipulate event driven programming with VBA (user interaction and time trigger) (A2, A5, C2)
- Understand the problem of forecast management (A1, A2, C2, C8)
- Understand the stakes of the sales and operations planning (S&OP) and its link with (A1, A2, C2, C8)
- Select and apply a forecasting method adapted to a given context (A1, C1)
- To calculate the intermediate management balances, indicators, and ratios derived from the income statement and balance sheet. (A1, A6, C1, C11)

CONTENT

- Corporate structure
- Introduction to supply chain, SCOR Model
- Forecasting methods, S&OP, demand management
- case study: forecasting and statistical models & VBA programming for data import and filtering
- business finance (income statement, balance sheet), sustainable finance
- Beer game

BIBLIOGRAPHY

- IDRISSI N, P. KNOCKAERT et M. CATTAN, Maîtriser les processus de l'entreprise. Organisation, 2001
- MARION A. /Collectif, Le diagnostic d'entreprise. Méthode et processus. Economica, 1999
- BAGLIN G., BRUEL O., GARREAU A., GREIF M., Van DELF C., Management Industriel et Logistique, Economica, 1996.
- GIARD V. Gestion de la production et des flux, 3e édition, Economica, 2003.
- JAVEL G. Organisation et gestion de la production, Masson, 1997.
- BOURBONNAIS R. et USUNIER J.C., Prévision des ventes, Economica 3rd éd. 2001.
- GOODRICH R., Applied statistical forecasting, Business Forecast System, 1992
- LEMANCQVIAUX P., La gestion prévisionnelle à court terme, Dunod, 1995

IDENTIFICATION

CODE : GI-3-S2-EC-GPC
ECTS : 4

HOURS

Cours : 0h
TD : 52h
TP : 2h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 56h
Travail personnel : 12h
Total : 68h

ASSESSMENT METHOD

EE+EO (évaluation écrite+orale)
IE (interrogation écrite)
IE (interrogation écrite)

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

M. ARNAUD Frédéric :
frederic.arnaud@insa-lyon.fr

AIMS

This course belongs to teaching unit Conception de produits et systèmes industriels (GI-3-S2-UE-CPSI) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 1)
- A3 Implement an experimental approach (Level 2)
- A4 Design a system that meets a set of specifications (Level 3)
- A6 Report on an analysis or on a scientific approach (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
- C4 Sizing the hardware and / or software of a system (Level 3)
- C10 Developing and implementing an action plan as part of a quality and continuous improvement approach (Level 3)
- C12 Changing organizations to meet new constraints or opportunities (Level 3)
- C14 Collectively managing a project: organization, communication, group coordination (Level 3)
- C15 Managing a project based on a master plan (planning, budgeting, definition of monitoring indicators) (Level 3)
- C16 Identifying, analyzing and controlling the risks inherent to a project (Level 2)
- C17 Identifying, formalizing and contractualising the requirements of a client, following their evolution and validating their compliance (requirements traceability) (Level 2)
- B3 interact with people and work in a team (Level 2)
- B4 demonstrate creativity, innovation, entrepreneurship (Level 1)

By allowing the engineering students to work and be evaluated on the following knowledge :

- General reliability of a mechanism
- Bearing calculations
- To be able to :
 - (C10, C12, C14, C15, C16, C17, B3, B4)
 - (A1, A3, A4, A6, C2, C4)
 - Manage a project
 - Design production equipment that meets productivity and safety expectations
 - Respect standards, costs and deadlines
 - Sizing a mechanically welded frame
 - Size bearings, belts
 - Sizing a tree statically and dynamically

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

IDENTIFICATION

CODE : GI-3-S2-EC-PSC

ECTS : 2

HOURS

Cours : 0h

TD : 26h

TP : 4h

Projet : 0h

Evaluation : 2h

Face à face pédagogique : 32h

Travail personnel : 0h

Total : 32h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

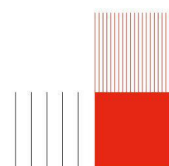
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membre de



IDENTIFICATION

CODE : GI-3-S2-EC-CSD
ECTS : 4

HOURS

Cours : 0h
TD : 22h
TP : 40h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 64h
Travail personnel : 16h
Total : 80h

ASSESSMENT METHOD

IE (written examination)
Report and defense

TEACHING AIDS

Paper version of the slideshow
projected in class.

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Pilotage des systèmes industriels (GI-3-S2-UE-PLSI) and contributes to the following skills :

A3 Implement an experimental approach (Level 2)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)

C3 Evaluating, prototyping and simulating a system (Level 2)

C13 Considering technological and methodological innovation (Level 1)

B2 work, learn, progress autonomously (Level 2)

B3 interact with people and work in a team (Level 3)

- The notion of stability of a system
- Performance concepts
- Single-input, single-output correctors

To be able to:

- Study the stability of a system
- Determine the system performance parameters and compare them to the specifications
- Synthesize a controller
- Simulate a dynamic system and its controller
- Controlling a physical system experimentally

CONTENT

BIBLIOGRAPHY

P. Borne et al., Analyse et régulation des processus industriels, Technip

PRE-REQUISITES



IDENTIFICATION

CODE : GI-3-S2-EC-CDA
ECTS : 3

HOURS

Cours : 0h
TD : 26h
TP : 20h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 48h
Travail personnel : 12h
Total : 60h

ASSESSMENT METHOD

ES (évaluation en situation)
ES (évaluation en situation)
ES (évaluation en situation)
ES (évaluation en situation)

TEACHING AIDS

The pedagogic form will insist on the transcription of analytical expression to plant reality.
Support are handout and annales

TEACHING LANGUAGE

French

CONTACT

MME SUBAI Corinne :
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AIMS

This course belongs to teaching unit Pilotage des systèmes industriels (GI-3-S2-UE-PI(SI)) and contributes to the following skills :

- C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 3)
- C4 Sizing the hardware and / or software of a system (Level 3)
- C8 Managing supply in connection with the planning and inventory management policy (Level 3)
- C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 3)
- C17 Identifying, formalizing and contractualising the requirements of a client, following their evolution and validating their compliance (requirements traceability) (Level 3)
- B2 Work, learn, progress autonomously (Level 3)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Modeling and designing an information, decision and production system, of goods and services (C2)
- Evaluating, prototyping and simulating a system (C3)
- Managing supply in connection with the planning and inventory management policy (C8)

To be able to :

- Identifying, formalizing and contractualising the requirements of a client, following their evolution and validating their compliance (requirements traceability) (C17)
- Sizing the hardware and / or software of a system (C4)
- Localizing and assigning the production, storage and transportation processes to different members of the supply chain (C9)
- Work, learn, progress autonomously (B2)

CONTENT

Part 1 : Petri Nets (PN)

- 1.1 Introduction to the resource/flow paradigm
- 1.2 Autonomous PN (definition, properties)
- 1.3 Non-autonomous PN (synchronised, temporal)
- 1.4 Analysis (dimensionning and performance assessment)

Partie 2 : Queueing Nets (QN)

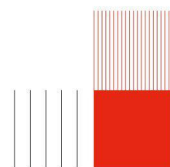
- 2.1 Introduction to the customer/server paradigm
- 2.1 Markov Chains and state equation
- 2.2 Performance indicators
- 2.3 Conventional Station
- 2.4 Markov Net limited to product form

BIBLIOGRAPHY

Du graficet aux réseaux de Petri, R. David, H. Alla HERMES 1989.
Modélisation des systèmes de production, J.M. Proth, X. Xie DUNOD 1992.
Introduction to DES, M. Cassandras, S. Lafortune, Wiley et sons 1999.
Modelling with generalized stochastic Petri Nets, A. Marsan, S. Donatelli Wiley et sons Ltd 1995.
Processus stochastiques, Alan Ruegg, méthodes mathématiques pour l'ingénieur, presses polytechniques romandes, Lausanne, 1989.
Network flow programming, by P. A. Jensen and J. W. Barnes, John Wiley and Sons, Inc., New York, 1980.

PRE-REQUISITES

Bachelor Eng.
Automatic Control
Discrete Mathematic
Graph Theory
Probability and Statistics



IDENTIFICATIONCODE : GI-3-S2-EC-COM-HU
ECTS : 1**HOURS**Cours : 0h
TD : 16h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 16h
Travail personnel : 4h
Total : 20h**ASSESSMENT METHOD****TEACHING AIDS****TEACHING LANGUAGE**French
English**CONTACT**M. CHAUMARD Davyd :
davyd.chaumard@insa-lyon.fr**AIMS**

This course belongs to teaching unit Humanities and Sports Education (GI-3-S2-UE-HUEPS) and contributes to the following skills :

- to develop a practical awareness of the various channels of communication, through individual and group exercises focusing on drama
- to explore one's personal resources and limits (role games and improvisations)
- to analyse individual participation in a group situation
- to understand the methods of the social sciences, which provide a basis for the analysis of the activities
- cooperation and efficiency in a team job
- to know how preparing for 4GIs group project

CONTENT

Working with professional actors

practicing communication : from experience to concepts

BIBLIOGRAPHY

provided by the professor, according to the subject of research chosen by each team of students

PRE-REQUISITES

Ability to use oral, written and audio-visual modes of expression in French

IDENTIFICATION

CODE HU-0-S2-EC-L-FLE-
TTCB1B2

ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-Continuous assessment: 50%
-Final evaluation: 50%
-Attendance required

TEACHING AIDS

Various authentic documents
(paper, audio, video and
multimedia)

TEACHING LANGUAGE

French

CONTACT

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Mme Fradois Delphine :
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AIMS

Depending on the level:

- Review level B1 of CEFR and improve your knowledge of B2 level
- or
- Improve your knowledge of B2 level and begin learning level C1

CONTENT

Systematic practice of oral and written comprehension techniques, as well as narrative, summary, synthesis, and argumentation techniques. The program is consistent with the exercises required for the TCF.

Level B1 to review

Level B2

- Write a simple and coherent text on familiar topics
- Write personal letters to describe experiences and impressions.
- Write clear and detailed texts on a wide range of topics related to one's interests.
- Write an essay or report, conveying information or presenting reasons for or against a given opinion.
- Write a clear, well-structured text and develop a point of view.
- Write about complex topics in an essay or report.
- Synthesize ideas.
- Adopt a style appropriate to the audience.

BIBLIOGRAPHY

- La documentation française <https://www.vie-publique.fr/publications-de-la-documentation-francaise>
- Production écrite B1/B2 (Didier)
- Ecrire pour convaincre (Hachette)
- Vocabulaire expliqué du Français(CI)
- Francoscopie (Larousse)

Online resources :

<https://fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B1 level

IDENTIFICATION

CODE HU-0-S2-EC-L-FLE-
CIVB2C1

ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESMENT METHOD

Project based evaluation.

TEACHING AIDS

Various authentic materials

TEACHING LANGUAGE

French

CONTACT

AIMS

Humanities Skills:

2. Work, learn, and develop independently

3. Interact with others and work in a team

4. Demonstrate creativity, innovation, and initiative

7. Work in an international and intercultural context

Language Activities: Written Expression, Written Expression, Oral Expression, Listening Comprehension, Oral Expression

An intercultural approach for a better understanding of French civilization and culture with the aim of fostering student autonomy and facilitating their integration into their studies, student life, and social and professional life.

CONTENT

The work will be organized around various activities: debates, presentations, projects, global simulations, forums, etc. on various past and present themes (society, culture, history, etc.) and will integrate an intercultural dimension.

BIBLIOGRAPHY

You will find a selection of resources available online at the following address <https://fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B2 level



IDENTIFICATION

CODE HU-0-S2-EC-L-FLE-
COMPB2

ECTS : 2

HOURS

Cours :	0h
TD :	26h
TP :	0h
Projet :	0h
Evaluation :	0h
Face à face pédagogique :	26h
Travail personnel :	0h
Total :	26h

ASSESSMENT METHOD

- Continuous assessment: 50%
- Final evaluation: 50%
- Attendance required

TEACHING AIDS

Various authentic documents
(paper, audio, video and
multimedia)

TEACHING LANGUAGE

French

CONTACT

AIMS

- Depending on the level:
- Review level B1 of CEFR and improve your knowledge of B2 level
- or
- Improve your knowledge of B2 level and begin learning level C1

CONTENT

Systematic practice of oral and written comprehension techniques, as well as narrative, summary, synthesis, and argumentation techniques. The program is consistent with the exercises required for the TCF.

Level B1 to review

Level B2

- Write a simple and coherent text on familiar topics
- Write personal letters to describe experiences and impressions.
- Write clear and detailed texts on a wide range of topics related to one's interests.
- Write an essay or report, conveying information or presenting reasons for or against a given opinion.
- Write a clear, well-structured text and develop a point of view.
- Write about complex topics in an essay or report.
- Synthesize ideas.
- Adopt a style appropriate to the audience.

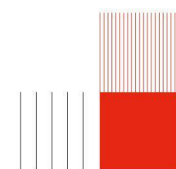
BIBLIOGRAPHY

- La documentation française <https://www.vie-publique.fr/publications-de-la-documentation-francaise>
- Production écrite B1/B2 (Didier)
- Ecrire pour convaincre (Hachette)
- Vocabulaire expliqué du Français(CI)
- Francoscopie (Larousse)

Online resources :
<https://fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B1 Level



IDENTIFICATION

CODE : HU-3-S2-EC-L-POR
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

Mme Strelow Antunes Isabel :
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lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES
It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop
and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex
tasks that require the students to engage in the 5 linguistic activities, at a level and with
linguistic input that are appropriate for the group. In-class and/or guided independent
study of the forms and functions of the language is regular and adapted to the level of
the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar
training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None



IDENTIFICATION

CODE : HU-3-S2-EC-L-TAN
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation in TANDEM consists in continuous assessment of their preparation and oral work in class, as well as written work and an oral presentation.

TEACHING AIDS

-Diverse authentic documents, both written and audiovisual

TEACHING LANGUAGE

French
English

CONTACT

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Mme Rivoire Camille :
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Mme Raymond Camille :
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Mme Fradois Delphine :
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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills valued by INSA:

CT7: Work in an international, intercultural context

7.1 Communicate in a foreign language

7.2 Decode cultural references in speech, attitudes and behaviors.

7.3 Put one's own values, beliefs and behaviors into a cultural perspective.

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

THE STUDENT CAN:

--talk about themselves and their country

--talk about a social phenomenon OR

--talk about a cultural phenomenon OR

--talk about a scientific phenomenon OR

--have a conversation appropriate to the context; communicate in an appropriate manner

--give an oral presentation AND/OR write clear expository prose

--learn and apply usage and vocabulary discovered in class

--understand and apply diverse linguistic and social codes

By the end of the semester, the student will be able to...

- Communicate and interact in one or more foreign languages, both orally and in writing, adapting to the context and the interlocutors, from the transmission of simple information to mediation in more complex communication situations. (Communication skills)

- Identify, understand, and utilize the cultural codes associated with their own language and those of their interlocutors in order to adopt appropriate behavior and communication in diverse intercultural contexts. (Intercultural skills)

- Produce written or oral messages in their native language or in the target language. (Linguistic skills)

CONTENT

The TANDEM modules offer linguistic and cultural exchanges between French-speaking students, English-speaking and Spanish-speaking students in each of the languages.

The INTERCOMPREHENSION module is all about learning to understand native speakers of a Romance language different to one's own. Each student speaks their native language and learns to understand the others.

The work is carried out by organizing a debate around a controversial subject, by writing a story, by presenting an element belonging to the specific culture, by playing board games ... in short, by interacting permanently and in any form possible with other students fluent in the different working languages.

Of course, oral comprehension is also at the heart of this course. It is a living, authentic learning in the spirit of sharing. Everyone has something to learn and to pass on.

BIBLIOGRAPHY

PRE-REQUISITES

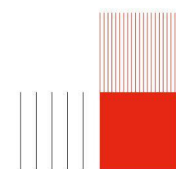
FOR THE TANDEMS

For francophone students: B2 in English or Spanish

For anglophone students: B1 minimum in French

FOR INTERCOMPREHENSION

Must be native speaker of a romance language (French, Spanish, Catalan, Italian, Portuguese, Romanian)



IDENTIFICATIONCODE : HU-3-S2-EC-L-ANG-A2
ECTS : 2**HOURS**Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h**ASSESSMENT METHOD**

Evaluation consists in continuous assessment of the student's preparation and oral work in class, as well as written work and oral presentations.

TEACHING AIDS

Teaching documents are chosen by the instructor according to the target level:

-Various instructional and authentic English-language materials

-Audio-visual documents

TEACHING LANGUAGE

French

CONTACT

Mme Jouffroy Jeannie :
jeannie.jouffroy@insa-lyon.fr

AIMS

To consolidate, acquire, and enrich linguistic knowledge and skills by developing the competences defined by the Common European Framework of Reference for Languages. Students are divided into groups based on level, and specific linguistic objectives are determined according to the student's level.

CONTENT

In order to develop the defined learning outcomes, the course will employ language activities that are varied (written comprehension and expression, oral comprehension and expression, oral interaction) and complex (projects, simulations, etc.).

The activities are designed to promote maximum language exposure and the use of the target language as a vehicle of culture and as a tool for work and communication.

The student will become more autonomous through group work and personal work.

BIBLIOGRAPHY

The Common European Framework of Reference for Languages, Council of Europe

PRE-REQUISITES

A2 level

IDENTIFICATION

CODE : HU-3-S2-EC-L-ARA
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

M. Garay Oyarzo Edicto :
edicto.garay-oyarzo@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES
It contributes to the development of the following transversal skills :
CT7: Work in an international, intercultural context
--CT7.1: Communicate in a foreign language
by allowing the student-engineer to develop and be evaluated on the following learning
outcomes:
The student CAN:
-- understand and apply social codes from different countries
--have a conversation appropriate de the context
--confront communicational challenges of increasing difficulty
--produce diverse types of written expression
--understand authentic documents (written and audiovisual)
--talk about a society or a social phenomenon
--apply grammar and vocabulary covered in class
The target level of acquisition (B1-B2+) is dependent on the level of group

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex
tasks
that require the students to engage in the 5 linguistic activities, at a level and with
linguistic
input that are appropriate for the group. In-class and/or guided independent study of the
forms and functions of the language is regular and adapted to the level of the group.
In the first semester, themes covered include:
--language learning methodology (RC, OC, WP, vocabulary learning and grammar
training)
--oral presentation skills
--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-3-S2-EC-L-ALL
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

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M. Bouet Christian :
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AIMS

This EC belongs to the UE: HUMANITIES
It contributes to the development of the following transversal skills :
CT7: Work in an international, intercultural context
--CT7.1: Communicate in a foreign language
by allowing the student-engineer to develop and be evaluated on the following learning outcomes:
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--talk about a society or a social phenomenon
--apply grammar and vocabulary covered in class
The target level of acquisition (B1-B2+) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.
In the first semester, themes covered include:
--language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
--oral presentation skills
--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

IDENTIFICATION

CODE : HU-3-S2-EC-L-RUS
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES
It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-3-S2-EC-L-JAP
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop
and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex
tasks that require the students to engage in the 5 linguistic activities, at a level and with
linguistic input that are appropriate for the group. In-class and/or guided independent
study of the forms and functions of the language is regular and adapted to the level of
the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar
training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-3-S2-EC-L-CHI
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

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chunyan.jiang-huang@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

-- understand and apply social codes from different countries

--have a conversation appropriate de the context

--confront communicational challenges of increasing difficulty

--produce diverse types of written expression

--understand authentic documents (written and audiovisual)

--talk about a society or a social phenomenon

--apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks

that require the students to engage in the 5 linguistic activities, at a level and with linguistic

input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

--language learning methodology (RC, OC, WP, vocabulary learning and grammar training)

--oral presentation skills

--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-3-S2-EC-L-ESP
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

The average is made up of half (50%) continuous assessment (assessments of the 5 skills carried out in class) and a common exam (50% of the average) which assesses Written Comprehension, Oral Comprehension, Grammatical and Lexical Skills and Written Expression. This common exam does not have the value of level validation

TEACHING AIDS

Authentic and/or didactic documents related to the selected topics.
- "Pink" grammar and exercise booklets
- "Yellow" booklet: conjugation guides
- The CRL

TEACHING LANGUAGE

French

CONTACT

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AIMS

The targeted and mobilized skills are those of both the INSA Humanities Skills Framework (specifically skills 3 and 7) and the CEFR.
Consistent description of the CEFR skills.
<http://www.sciencespo-lille.eu/sites/default/files/cecrl.pdf>

At the end of the semester, the student will be able to...

- Communicate and interact in Spanish, both orally and in writing, adapting to the context and the interlocutors, from the transmission of simple information to mediation in more complex communication situations. (COMMUNICATIONAL COMPETENCY)
- Identify, understand, and utilize the cultural codes associated with the Spanish language and its speakers in order to adopt appropriate behavior and communication in diverse intercultural contexts. (INTERCULTURAL COMPETENCY)
- Understand spoken and written messages in Spanish in communication situations corresponding to the targeted proficiency level, in accordance with the Common European Framework of Reference for Languages (CEFR).
- Produce spoken and written messages in Spanish, effectively utilizing the linguistic resources specific to the targeted proficiency level (CEFR).

CONTENT

The five skills recognized by the Common European Framework of Reference for Languages (CEFR) are practiced and assessed several times throughout the semester/year, ensuring regular practice of the various skills and knowledge acquired.

The themes, grammar, and vocabulary covered in class are adapted to the target level (indicated in the group code) and are at the discretion of the teacher.

BIBLIOGRAPHY

Web resources:

- for listening comprehension, grammar, and vocabulary training (all levels): <http://www.ver-taal.com/index.htm>
- grammar and vocabulary exercises (all levels): <https://www.espanolfacile.com/>
- <https://moodle.insa-lyon.fr/course/index.php?categoryid=353>

PRE-REQUISITES

None. Courses range from beginner to advanced level. Each student will be placed in a group corresponding to their level, either through a test at the beginning of the year (for new students) or based on their level from the previous year for students already attending INSA.

IDENTIFICATION

CODE : HU-3-S2-EC-L-ANG
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation consists in continuous assessment of the student's preparation and oral work in class, as well as written work and oral presentations'

TEACHING AIDS

Teaching documents are chosen by the instructor according to the target level:
-Various instructional and authentic English-language materials
-Audio-visual documents

TEACHING LANGUAGE

English

CONTACT

Mme JOUFFROY Jeannie :
jeannie.jouffroy@insa-lyon.fr
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AIMS

This EC contributes to the development of the following transversal skills :

CT3: INTERACT WITH OTHER PERSONS & WORK ON A TEAM
3.2 Put original discourse into context using explicit references.
3.3 Master non-verbal communication: posture and gesture.

CT7: WORK IN AN INTERNATIONAL, INTERCULTURAL CONTEXT
7.1 Communicate and interact in a foreign language.

The student-engineer will practice and be evaluated on the following learning outcomes:

--knowing how to communicate in person, by email and/or by telephone on factual topics
--knowing how to give instructions, explain a procedure, make a dynamic oral presentation
--mastery of the vocabulary and syntax necessary to the successful accomplishment of the above tasks

CONTENT

Students will learn and put into practice the professional communication skills, both oral and written, useful to engineers. In the process, they will develop those of the following skill sets determined to best further their linguistic abilities and respond to their particular needs. The activities, associated with socializing skills in the world of work, may include:

--Holding a conversation on familiar topics
--Telling a story in a conversational context
--Initiating and respond to emails
--Hold a phone conversation on factual topics
--Giving instructions/ Explain a procedure
--Making a dynamic oral presentation using notes if needed on a topic of current events, travel, culture, other
--Preparing a personal or professional time abroad
--Organising after-work activities with colleagues
--Hosting clients and colleagues at lunch
--Learning and explaining professional rules of conduct as well as the company culture

BIBLIOGRAPHY

The Common European Framework of Reference for Languages, Council of Europe

PRE-REQUISITES

A2/B1 level in English

IDENTIFICATION

CODE : HU-3-S2-EC-L-ITA
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

Mme Cognet Anne :
anne.cognet@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop
and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex
tasks that require the students to engage in the 5 linguistic activities, at a level and with
linguistic input that are appropriate for the group. In-class and/or guided independent
study of the forms and functions of the language is regular and adapted to the level of
the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar
training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : CDS-3-S2-EC-EPS
ECTS : 1

HOURS

Cours : 0h
TD : 20h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 0h
Total : 20h

ASSESSMENT METHOD

Assessment in Physical Education concerns the teaching of Sports and Artistic Physical Activities (APSA), and will take the form of continuous assessment with half-yearly marking.

The mark depends on the degree of acquisition of the skills expected in each of the activities, and the progress made over all the sessions in the cycle. The mark takes into account :

Individual and/or team performance
mastery of execution
Progress in the sports project
Responsibility and autonomy

TEACHING AIDS

All physical, sporting, artistic and competitive activities

TEACHING LANGUAGE

French

CONTACT

Mme JAUSSAUD Marie :
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AIMS

This EC is part of the Teaching Unit: SHS and contributes to the development of the School's transversal competences

1* Auto-evaluating one's own performance

Knowledges :

- Fundamentals, principles of action and terminology of sports activities
- Criteria for observation, achievement and success.

Abilities :

- Assess your level of practice
- Build up a warm-up
- Set goals for progress
- Manage physical and mental potential

2* Work, learn and develop independently

Knowledge :

- PSAA rules
- Observation criteria
- Principles of warm-up and cool-down

Abilities :

- Mobilise resources
- Analyse, observe, question
- Take on different roles (referee, choreographer)

3* Interact with others, work as part of a team

Knowledges :

- Roles and functions in each sports activity

Abilities :

- Communicate appropriately: verbal, non-verbal and postural communication.
- Integrate into a group
- Take part in and develop a group project
- Take the initiative
- Be a good listener

4* Be creative, innovative and enterprising

Knowledge :

- Artistic disciplines

Abilities :

- Draw on knowledge and resources from different artistic fields to produce an original work.
- Mobilise the imagination and sensibility and make them visible through dance movement
- Access the symbolism of the body

5* Act responsibly in a complex world

Knowledge :

- Safety and operating rules

Abilities :

- Identify uncertainties and risks and act to reduce them
- Integrate a responsible dimension into their actions
- Show respect and fair play in a power struggle

6* Working in an international context

Knowledge :

- Socio-cultural differences

Abilities :

- Integrate cultural diversity into group work
- Act with respect for self and others

CONTENT

Physical Education and Sport lessons are organised around traditional Physical Education lessons, or advanced lessons, or appropriate practices (EPSA), or competitive practices within the framework of the Section Sportive Haut Niveau.

1. Physical Education lessons :

Students choose one or two physical and sporting activities per year from among the activities offered by the sports centre (individual, group, dual).

2. Appropriate Physical Education lessons: For all students who are exempt from physical activity for at least 2 months:

Swimming, Body-building, Nordic Walking, Somatic Exercise, Sophrology, Wheelchair Basketball, Pilates, Table Tennis, etc.

3. Advanced Physical Education courses :

Specialisation in a sporting activity, University training and competitions

4. SSHN (High-Level Athlete section)

University training and competitions

EPS 3 GEN - GENEPI :

PE lessons on Wednesday from 8.00 to 9.30: 2 x 5 sessions: outdoor activities and 2nd 2-day course in Hauteville

BIBLIOGRAPHY

PRE-REQUISITES

- EPS: none
- Appropriate Physical Education: subject to medical advice
- Advanced courses and competitive practice: previous practice required subject to specific selection according to each activity
- SHN: ministerial list

Levels 1 and 2: Physical Education, Appropriate physical education

Level 3: Advanced courses and competitive practice, SHN

INSA LYON

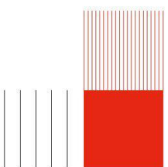
Campus LyonTech La Doua

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membre de



IDENTIFICATION

CODE : GI-3-S2-EC-REC
ECTS : 1

HOURS

Cours : 2h
TD : 6h
TP : 8h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 16h
Travail personnel : 8h
Total : 24h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

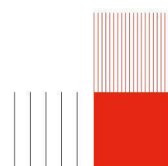
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IDENTIFICATION

CODE : GI-3-S2-EC-CBD
ECTS : 3

HOURS

Cours : 0h
TD : 30h
TP : 16h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 48h
Travail personnel : 8h
Total : 56h

ASSESSMENT METHOD

IE (interrogation écrite)
ES (évaluation en situation)

TEACHING AIDS

Slides

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Conception et gestion des systèmes d'informations (GI-3-S2-UE-CGSI) and contributes to the following skills :

A1 Analyze a system (real or virtual) or a problem (Level 2)

A2 Operate a model of a real or virtual system (Level 2)

A5 Process data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- (A1, C2)

To be able to :

- (A1, A5, C2, A2)

* Design and size systems required for a given activity of production, service, distribution

...

* Identify and correct discrepancies in a continuous improvement process (material, human and information systems) by the implementation of analytical and simulation tools

* Ensure quality reporting through the establishment of appropriate indicators, based on a systemic view of organizations (matrix, silo, project) for any scope and any type of activity"

CONTENT

Know :

- Architecture of a Data base system management

- Model entity - association

- Fonctionnal and multivaluate dependances

- Transitive closure notion, Minimal cover

- Normal forms

- Relationnal algebra, calculus

- SQL language

- Structure of a XML document, Wellformed XML document

- Document type definition, XSLT document

To be able to :

- Design an entity/association model

- Translate an entity/association model to relationnal model

- Write queries in relationnel algebra, in SQL

- Write a well formed XML document

- Validate a XML document from a DTD

- Transform a XML document with XSLT language

BIBLIOGRAPHY

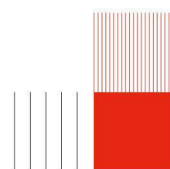
C. Chrismet, "Mise en oeuvre des bases de données" ; Ed. Eyrolles 1990

H.F. Korth, A. Silbershats, "Systèmes de gestion de base de données", Mc Graw-Hill, 1988

C. Marée, G. Ledant, "SQL : Initiation, programmation et maîtrise" - Armand Colin, 1994

K. Williams, M. Brundage : XML et les bases de données - Ed. Eyrolles 2001

PRE-REQUISITES



IDENTIFICATIONCODE : GI-3-S2-EC-RIP
ECTS : 2**HOURS**Cours : 0h
TD : 8h
TP : 16h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 24h
Travail personnel : 8h
Total : 40h**ASSESSMENT METHOD**

ES (évaluation en situation)

TEACHING AIDS**TEACHING LANGUAGE**

French

CONTACTMme ROBARDET Céline :
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M. MONCLA Ludovic :
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"This course belongs to teaching unit Conception et gestion des systèmes d'informations (GI-3-S2-UE-CGSI) and contributes to the following skills :

By allowing the engineering students to work and be evaluated on the following knowledge :

To be able to :

- * increase skills in object modeling (up to level 4)
- * be able to read/write in a file (parser, writing according to a specified formation)
- * be able to correct a program and validate it through unit tests.

From the point of view of learning around the management of the supply chain, this project will allow students to implement on a concrete instance an optimization strategy.

CONTENT

The student is placed in a situation of carrying out an IT project based on precise specifications. The project focuses on solving a constrained optimization problem. A validation program is provided to students in order to concretely assess the feasibility of their solution. Optimality is not guaranteed but a comparison of the results obtained by the different groups of students creates emulation. The objective is to allow students to acquire and deepen their programming skills. The language used is Python.

BIBLIOGRAPHY**PRE-REQUISITES**

IDENTIFICATION

CODE : GI-0-S2-EC-VAC
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

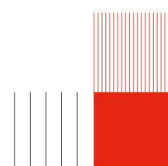
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IDENTIFICATION

CODE : GI-0-S2-EC-CESURE

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

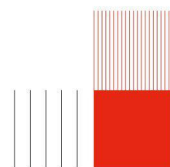
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IDENTIFICATION

CODE : GI-4-S1-EC-BIN
ECTS : 2

HOURS

Cours : 0h
TD : 4h
TP : 14h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 12h
Total : 32h

ASSESSMENT METHOD

Written exam at the end of the tutorial part ("TD")
Project evaluation : collective presentation and written report

TEACHING AIDS

Slides

TEACHING LANGUAGE

English

CONTACT

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AIMS

GI-4-S1-UE-GEDE
GI-4-S1-EC-BIN

General engineering skills:

- Develop an experimental approach;
- Identify, formulate and solve a complex engineering problem.

Specific Industrial Engineering skills:

- Observe, measure, analyze and interpret an activity or system from data
- Modeling, designing a system of information, decision-making and production of goods and services;
- Evaluate, prototype or simulate a system.

Abilities :

Collect customer needs for analysis; Identify sources of production data useful for an analytical approach; Process voluminous information for analysis; Exploit the data in the perspective of the analyzed activity: Carry out a project, working in a group.

Knowledge: operational and analytical database schemas; Elementary actions and sequencing for data transformation; Methods and tools for visualizing information from data.

CONTENT

DATA WAREHOUSE

1. Dimensional modeling for decision support systems
2. The Extract-Transform-Load (ETL) task : data cleaning, fusion and restructuration methods, ETL tools (Tableau)
3. Query tools and methods : reporting, scorecards, etc.

BIBLIOGRAPHY

Systèmes d'information décisionnels, E. Ferragu, 2013 (available at Doc'INSA)
- Piloter l'entreprise grâce au data warehouse, Jean-Michel Franco, Sandrine de Lignerolles, 2000
(available at the Industrial Eng. dept. local library)

PRE-REQUISITES

Relational Databases, SQL

IDENTIFICATION

CODE : GI-4-S1-EC-ISD
ECTS : 2

HOURS

Cours : 0h
TD : 18h
TP : 16h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 36h
Travail personnel : 12h
Total : 48h

ASSESSMENT METHOD

TEACHING AIDS

Course handouts

TEACHING LANGUAGE

English

CONTACT

MME ROBARDET Celine :
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AIMS

This course belongs to teaching unit Gestion et exploitation des données de l'entreprise (GI-4-S1-UE-GEDE) and contributes to the following skills :

A3 Implement an experimental approach (Level 2)
A5 Process data (Level 3)
A6 Report on an analysis or on a scientific approach (Level 3)
C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)
C4 Sizing the hardware and / or software of a system (Level 2)
C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 1)
B2 work, learn, progress autonomously (Level 2)
B3 interact with people and work in a team (Level 3)

By allowing the engineering students to work and be evaluated on the following skills:

- Mastering structured data processing tools: Excel and Stats (C4)
- Know how to make a decision based on numerical results (C1)
- Choose a processing method based on the data and objectives (A3, A5, A6, C1, C11, B2, B3)

CONTENT

DATA ANALYSIS

1.Data

Characteristics: origin, quality, coding

Preparation: restoration, filtering, synchronization, construction of classes

Dependence: distances, ultra metric, correlation

2.Tools :

Descriptive methods: principal components analysis, correspondences, preferences

Explanatory methods: regression, segmentation, canonical analysis, discriminated analysis

Structural methods: ultra metric and hierarchical classification, dynamic swarms.

3.Analysis :

Analysis methodologies

Interpretation rules

Case studies

BIBLIOGRAPHY

**** Data analysis ****

1. E.Diday, J.Lemaire, J.Pouget, F.Testu, Eléments d'analyse de données, Dunod 1982
2. G.Saporta, Probabilités, Analyse de données et Statistiques, Technip 1990
3. Jean-Pierre , Analyse interactive des données (ACP, AFC) avec Excel 2000 : théorie et pratique / Georgin. Presses universitaires de Rennes , 2002
4. Alain Morineau et Yves-Marie Chatelin , L'analyse statistique des données [Texte imprimé] : apprendre, comprendre et réaliser avec Excel, Ellipses , 2005 , 407 p

**** Data warehouse ****

1. Entrepôts de données. Guide pratique de modélisation dimensionnelle, 2ème édition, R. Kimball, M. Ross
2. Piloter l'entreprise grâce au data Warehouse, J.-M. Franco, S de Lignerolles

PRE-REQUISITES

Statistics, data base

IDENTIFICATIONCODE : GI-0-S1-EC-LABECH15
ECTS : 15**HOURS**Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 300h
Total : 300h**ASSESSMENT METHOD****TEACHING AIDS****TEACHING LANGUAGE**French
English**CONTACT**M. Dumitrescu Emil :
emil.dumitrescu@insa-lyon.fr**AIMS****CONTENT**

The reception and supervision of students takes place in a scientific laboratory. The student works on a well defined scientific problem, following a project approach, which requires managing objectives, planning, monitoring and deliverables. The host laboratories are DISP and AMPERE. Available on application and selection, and depending on subject availability.

BIBLIOGRAPHY**PRE-REQUISITES**

IDENTIFICATION

CODE : GI-4-S1-EC-ORD
ECTS : 2

HOURS

Cours : 0h
TD : 8h
TP : 14h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 24h
Travail personnel : 12h
Total : 40h

ASSESSMENT METHOD

IE (written test)
ES (situational assessment)

TEACHING AIDS

Handout for the lecture-tutorials part with exercises.
Project description.
Access to Opcenter APS software package.

TEACHING LANGUAGE

French
English

CONTACT

M. FONDREVELLE Julien :
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AIMS

This course belongs to teaching unit Pilotage des opérations industrielles (GI-4-S1-UE-POPI) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 3)
- A2 Operate a model of a real or virtual system (Level 3)
- A3 Implement an experimental approach (Level 2)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 3)
- C5 Managing a production system and react to malfunctions (Level 3)
- C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Production scheduling techniques and methods: single machine, parallel machines, flowshop, jobshop, hybrid configurations. (A1, A2, A3, C1, C2, C3, C5, C6)

To be able to :

- Characterizing production master data; (A1, A2, C1, C2, C6)
- Specifying a scheduling problem; (A1, A2, C1, C2, C3)
- Choosing or proposing a method to solve a production scheduling problem that is best suited to a given context, and applying it; (A2, A3, C2, C3, C5, C6)
- Analyzing the impacts of scheduling methods. (A1, A2, A3, C1, C3, C5, C6)

CONTENT

1- Lecture-tutorials part :

Introduction: Positioning scheduling within the overall production management process, characteristics of scheduling problems, problem classification.

Single-machine problems.

Parallel machine problems.

Flowshop problems.

Jobshop problems.

Hybrid problems.

2- Project: using a scheduling software package (Opcenter APS) to solve a case study

Study of production master data.

Analysis of software functionalities.

Consideration of additional constraints.

Software evaluation and critical analysis.

BIBLIOGRAPHY

BAGLIN G., BRUEL O., GARREAU A., GREIF M., Van DELF C., Management Industriel et Logistique, Economica, 1996.

GIARD V. Gestion de la production et des flux, 3e édition, Economica, 2003.

JAVEL G. Organisation et gestion de la production, Masson, 1997.

PINEDO M., Scheduling: Theory, Algorithms, and Systems, Prentice Hall, 2001.

RINNOOY KAN A.H.G., Machine scheduling problems: Classification, complexity and computations, Nijhoff, The Hague (1976).Hall, 2001

PRE-REQUISITES

Basics of Production Management (production organization, master data, production planning)

Basic knowledge about algorithms.

IDENTIFICATION

CODE : GI-4-S1-EC-QLT
ECTS : 1

HOURS

Cours : 0h
TD : 12h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 14h
Travail personnel : 6h
Total : 20h

ASSESSMENT METHOD

1 two-hour exam

TEACHING AIDS

Handout in English, exercises in English, Minitab software and an online tutorial in French

TEACHING LANGUAGE

French
English

CONTACT

M. MOYAUX Thierry :
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AIMS

GI-4-S1-UE-AMCI
GI-4-S1-EC-QLT
Quality - Maintenance

Understand the issues and tools of quality (quality process vs. quality tools) and maintenance (impact of the organisation of quality on the other functions of companies).

Understand the usefulness and the operation of some tools (control charts, Gage R&R, reliability, etc.).

This EC contributes to:

- * School skills in Engineering Sciences / Level (1 for sensitization, 2 for appropriation, 3 for mastery) :
 - . Solve questions asked by a scientific problem by being aware of limits / 3
 - . Develop an experimental approach / 3
 - . Implement a scientific approach / 2
- * School skills specific to GI:
 - . Observing, measuring, analyzing and interpreting an activity or a system from data / 3
 - . Managing a production system and react to malfunctions / 3
 - . Developing and implementing an action plan as part of a quality and continuous improvement approach / 3
 - . Collectively managing a project: organization, communication, group coordination / 1

CONTENT

FR or EN - possibility to integrate a group with lecture/practical work in english (but online tutorial in French only)

ONLINE TUTORIAL about Six Sigma to do without the assistance of the teacher

QUALITY

- * General introduction: relations of quality and maintenance
- * Introduction to quality: definitions, goals, voice of the client, overview of tools
- * Six sigma : Introduction, Etapes du DMAIC, Organisation.
- * Overview of ISO9001:2008
- * Control charts
- * Gage R&R

MAINTENANCE

- * Introduction: Definitions, Types of maintenance
- * TPM (Total Productive Maintenance), Overview, The 8 pillars with details on Autonomous maintenance in Pillar 2 and overview of RCM (Reliability-Centered Maintenance) in Pillar 3 (Planned maintenance), Lean Six Sigma
- * Types of maintenance, Impact of OEE (Overall Equipment Effectiveness), Frequency of change calculated from reliability function

BIBLIOGRAPHY

Bufferne J. (www.jean-bufferne.com), Le guide de la TPM, Éditions d'Organisation, Eyrolles, 2006.

Duret D., Pillet M., Qualité en production; De l'ISO 9000 à Six Sigma, 3e édition, Editions d'Organisation, 2005.

Pillet M., Six Sigma; Comment l'appliquer, Editions d'Organisation, 2004.

PRE-REQUISITES

Probability - Statistics

IDENTIFICATION

CODE : GI-4-S1-EC-EIE
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h

ASSESSMENT METHOD

Written examination
Report

TEACHING AIDS

Powerpoint slides
Sustainable development and
societal responsibility serious
game (CIPE)

TEACHING LANGUAGE

French
English

CONTACT

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AIMS

This course belongs to teaching unit Continuous improvement and innovation(GI-4-S1-UE-AMCI) and contributes to the following skills :

A1 Analyze a system (real or virtual) or a problem (Level 2)

A2 Operate a model of a real or virtual system (Level 2)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)

C3 Evaluating, prototyping and simulating a system (Level 2)

C10 Developing and implementing an action plan as part of a quality and continuous improvement approach (Level 1)

C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 2)

C12 Changing organizations to meet new constraints or opportunities (Level 2)

C20 Implementing Corporate Social Responsibility (CSR) (Level 1)

B5 act responsibly in a complex world (Level 2)

B6 position oneself, work, and evolve within a company or a socio-productive organization (Level 1)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Definitions of industrial ecology, circular economy and associated terminology (B5, B6)

- Material Flow Analysis (A1, A2, C1, C2, C3)

- Integrated Assessment Model (A1, A2, C1, C2, C3)

- Waste definition, associated waste management rules, legal aspects (A1, C10)

- Challenges of circular economy and reverse logistics from the industrial engineering point of view (C20, B5)

- Eco-parks, industrial symbiosis, synergy and mutualization (C11, C12)

To be able to :

- Propose solutions limiting the consumption and waste of resources and the production of waste (C10, C20)

- Diagnose sustainable development issues within an industrial company (C10, C11)

CONTENT

- Definitions

- Sustainable development and societal responsibility serious game

- Material Flow Analysis

- Best Available Techniques

- Waste

- Reverse logistics

- Ecoparks

- Disassembly problems

BIBLIOGRAPHY

PRE-REQUISITES

Notions of life cycle analysis

IDENTIFICATION

CODE : CDS-4-S1-EC-EPS
ECTS : 1

HOURS

Cours : 0h
TD : 20h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 0h
Total : 20h

ASSESSMENT METHOD

Assessment in Physical Education concerns the teaching of Sports and Artistic Physical Activities (APSA), and will take the form of continuous assessment with half-yearly marking.

The mark depends on the degree of acquisition of the skills expected in each of the activities, and the progress made over all the sessions in the cycle. The mark takes into account :

Individual and/or team performance
mastery of execution
Progress in the sports project
Responsibility and autonomy

TEACHING AIDS

All physical, sporting, artistic and competitive activities

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC is part of the Teaching Unit: SHS and contributes to the development of the School's transversal competences

1* Auto-evaluating one's own performance

Knowledges :

- Fundamentals, principles of action and terminology of sports activities
- Criteria for observation, achievement and success.

Abilities :

- Assess your level of practice
- Build up a warm-up
- Set goals for progress
- Manage physical and mental potential

2* Work, learn and develop independently

Knowledge :

- PSAA rules
- Observation criteria
- Principles of warm-up and cool-down

Abilities :

- Mobilise resources
- Analyse, observe, question
- Take on different roles (referee, choreographer)

3* Interact with others, work as part of a team

Knowledges :

- Roles and functions in each sports activity

Abilities :

- Communicate appropriately: verbal, non-verbal and postural communication.
- Integrate into a group
- Take part in and develop a group project
- Take the initiative
- Be a good listener

4* Be creative, innovative and enterprising

Knowledge :

- Artistic disciplines

Abilities :

- Draw on knowledge and resources from different artistic fields to produce an original work.
- Mobilise the imagination and sensibility and make them visible through dance movement
- Access the symbolism of the body

5* Act responsibly in a complex world

Knowledge :

- Safety and operating rules

Abilities :

- Identify uncertainties and risks and act to reduce them
- Integrate a responsible dimension into their actions
- Show respect and fair play in a power struggle

6* Working in an international context

Knowledge :

- Socio-cultural differences

Abilities :

- Integrate cultural diversity into group work
- Act with respect for self and others

CONTENT

Physical Education and Sport lessons are organised around traditional Physical Education lessons, or advanced lessons, or appropriate practices (EPSA), or competitive practices within the framework of the Section Sportive Haut Niveau.

1. Physical Education lessons :

Students choose one or two physical and sporting activities per year from among the activities offered by the sports centre (individual, group, dual).

2. Appropriate Physical Education lessons: For all students who are exempt from physical activity for at least 2 months:

Swimming, Body-building, Nordic Walking, Somatic Exercise, Sophrology, Wheelchair Basketball, Pilates, Table Tennis, etc.

3. Advanced Physical Education courses :

Specialisation in a sporting activity, University training and competitions

4. SSHN (High-Level Athlete section)

BIBLIOGRAPHY

PRE-REQUISITES

- EPS: none
 - Appropriate Physical Education: subject to medical advice
 - Advanced courses and competitive practice: previous practice required subject to specific selection according to each activity
 - SHN: ministerial list
- Levels 1 and 2: Physical Education, Appropriate physical education
Level 3: Advanced courses and competitive practice, SHN

INSA LYON

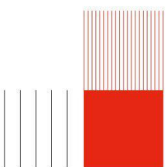
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IDENTIFICATION

CODE : HU-4-S1-EC-L-CHI
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

-- understand and apply social codes from different countries

--have a conversation appropriate de the context

--confront communicational challenges of increasing difficulty

--produce diverse types of written expression

--understand authentic documents (written and audiovisual)

--talk about a society or a social phenomenon

--apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks

that require the students to engage in the 5 linguistic activities, at a level and with linguistic

input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

--language learning methodology (RC, OC, WP, vocabulary learning and grammar training)

--oral presentation skills

--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-4-S1-EC-L-ESP
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

The average is made up of half (50%) continuous assessment (assessments of the 5 skills carried out in class) and a common exam (50% of the average) which assesses Written Comprehension, Oral Comprehension, Grammatical and Lexical Skills and Written Expression. This common exam does not have the value of level validation.

TEACHING AIDS

Authentic and/or didactic documents related to the selected topics.
- "Pink" grammar and exercise booklets
- "Yellow" booklet: conjugation guides
- The CRL

TEACHING LANGUAGE

French

CONTACT

M. Suarez Lopez Gonzalo :
gonzalo.suarez-lopez@insa-lyon.fr

AIMS

The targeted and mobilized skills are those of both the INSA Humanities Skills Framework (specifically skills 3 and 7) and the CEFR.
Consistent description of the CEFR skills.
<http://www.sciencespo-lille.eu/sites/default/files/cecrl.pdf>

At the end of the semester, the student will be able to...

- Communicate and interact in Spanish, both orally and in writing, adapting to the context and the interlocutors, from the transmission of simple information to mediation in more complex communication situations. (COMMUNICATIONAL COMPETENCY)
- Identify, understand, and utilize the cultural codes associated with the Spanish language and its speakers in order to adopt appropriate behavior and communication in diverse intercultural contexts. (INTERCULTURAL COMPETENCY)
- Understand spoken and written messages in Spanish in communication situations corresponding to the targeted proficiency level, in accordance with the Common European Framework of Reference for Languages (CEFR).
- Produce spoken and written messages in Spanish, effectively utilizing the linguistic resources specific to the targeted proficiency level (CEFR).

CONTENT

The five skills recognized by the Common European Framework of Reference for Languages (CEFR) are practiced and assessed several times throughout the semester/year, ensuring regular practice of the various skills and knowledge acquired.

The themes, grammar, and vocabulary covered in class are adapted to the target level (indicated in the group code) and are at the discretion of the teacher.

BIBLIOGRAPHY

Web resources:

- for listening comprehension, grammar, and vocabulary training (all levels): <http://www.ver-taal.com/index.htm>
- grammar and vocabulary exercises (all levels): <https://www.espanolfacile.com/>
- <https://moodle.insa-lyon.fr/course/index.php?categoryid=353>

PRE-REQUISITES

None. Courses range from beginner to advanced level. Each student will be placed in a group corresponding to their level, either through a test at the beginning of the year (for new students) or based on their level from the previous year for students already attending INSA.

IDENTIFICATION

CODE : HU-4-S1-EC-L-ANG
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation consists in continuous assessment of the student's preparation and oral work in class, as well as written work and oral presentations'

TEACHING AIDS

Teaching documents are chosen by the instructor according to the target level:
-Various instructional and authentic English-language materials
-Audio-visual documents

TEACHING LANGUAGE

English

CONTACT

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AIMS

This EC contributes to the development of the following transversal skills :

CT3: INTERACT WITH OTHER PERSONS & WORK ON A TEAM
3.2 Put original discourse into context using explicit references.
3.3 Master non-verbal communication: posture and gesture.

CT7: WORK IN AN INTERNATIONAL, INTERCULTURAL CONTEXT
7.1 Communicate and interact in a foreign language.

The student-engineer will practice and be evaluated on the following learning outcomes:

--knowing how to communicate in person, by email and/or by telephone on factual topics
--knowing how to give instructions, explain a procedure, make a dynamic oral presentation
--mastery of the vocabulary and syntax necessary to the successful accomplishment of the above tasks

The course is taught in level groups; themes and linguistic skills vary between levels.

CONTENT

Students will learn and put into practice the professional communication skills, both oral and written, useful to engineers. In the process, they will develop those of the following skill sets determined to best further their linguistic abilities and respond to their particular needs. The activities, associated with socializing skills in the world of work, may include:

--Holding a conversation on familiar topics
--Telling a story in a conversational context
--Initiating and respond to emails
--Hold a phone conversation on factual topics
--Giving instructions/ Explain a procedure
--Making a dynamic oral presentation using notes if needed on a topic of current events, travel, culture, other
--Preparing a personal or professional time abroad
--Organising after-work activities with colleagues
--Hosting clients and colleagues at lunch
--Learning and explaining professional rules of conduct as well as the company culture

BIBLIOGRAPHY

The Common European Framework of Reference for Languages, Council of Europe

PRE-REQUISITES

A2/B1 level in English

IDENTIFICATION

CODE : HU-4-S1-EC-L-ARA
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

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edicto.garay-oyarzo@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

-- understand and apply social codes from different countries

--have a conversation appropriate de the context

--confront communicational challenges of increasing difficulty

--produce diverse types of written expression

--understand authentic documents (written and audiovisual)

--talk about a society or a social phenomenon

--apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

--language learning methodology (RC, OC, WP, vocabulary learning and grammar training)

--oral presentation skills

--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-4-S1-EC-L-ALL
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

Mme Zuzik Mareike :
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M. Bouet Christian :
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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

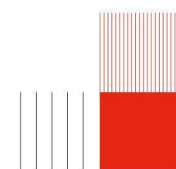
Campus LyonTech La Doua

20, avenue Albert Einstein - 69621 Villeurbanne cedex - France

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www.insa-lyon.fr

membre de



IDENTIFICATION

CODE : HU-4-S1-EC-L-JAP
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

M. Mihara Norio :
norio.mihara@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE HU-0-S1-EC-L-FLE-
CIVB2C1

ECTS : 2

HOURS

Cours : 0h

TD : 26h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 26h

Travail personnel : 0h

Total : 26h

ASSESSMENT METHOD

Project based evaluation.

TEACHING AIDS

Various authentic materials

TEACHING LANGUAGE

French

CONTACT

Mme Aumeunier Lise :
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Mme Fradois Delphine :
delphine.fradois@insa-lyon.fr

AIMS

Humanities Skills:

2. Work, learn, and develop independently

3. Interact with others and work in a team

4. Demonstrate creativity, innovation, and initiative

7. Work in an international and intercultural context

Language Activities: Written Expression, Written Expression, Oral Expression, Listening Comprehension, Oral Expression

An intercultural approach for a better understanding of French civilization and culture with the aim of fostering student autonomy and facilitating their integration into their studies, student life, and social and professional life.

CONTENT

The work will be organized around various activities: debates, presentations, projects, global simulations, forums, etc. on various past and present themes (society, culture, history, etc.) and will integrate an intercultural dimension.

BIBLIOGRAPHY

You will find a selection of resources available online at the following address <https://fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B2 level

IDENTIFICATION

CODE HU-0-S1-EC-L-FLE-
COMPB2

ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-Continuous assessment: 50%
-Final evaluation: 50%
-Attendance required

TEACHING AIDS

Various authentic documents
(paper, audio, video and
multimedia)

TEACHING LANGUAGE

French

CONTACT

Mme Aumeunier Lise :
elisabeth.aumeunier@insa-lyon.fr

Mme Fradois Delphine :
delphine.fradois@insa-lyon.fr

AIMS

Depending on the level:

- Review level B1 of CEFR and improve your knowledge of B2 level
- or
- Improve your knowledge of B2 level and begin learning level C1

CONTENT

Systematic practice of oral and written comprehension techniques, as well as narrative, summary, synthesis, and argumentation techniques. The program is consistent with the exercises required for the TCF.

Level B1 to review

Level B2

- Write a simple and coherent text on familiar topics
- Write personal letters to describe experiences and impressions.
- Write clear and detailed texts on a wide range of topics related to one's interests.
- Write an essay or report, conveying information or presenting reasons for or against a given opinion.
- Write a clear, well-structured text and develop a point of view.
- Write about complex topics in an essay or report.
- Synthesize ideas.
- Adopt a style appropriate to the audience.

BIBLIOGRAPHY

- La documentation française <https://www.vie-publique.fr/publications-de-la-documentation-francaise>
- Production écrite B1/B2 (Didier)
- Ecrire pour convaincre (Hachette)
- Vocabulaire expliqué du Français(CI)
- Francoscopie (Larousse)

Online resources :

<https://fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B1 Level

IDENTIFICATION

CODE HU-0-S1-EC-L-FLE-
TTCB1B2

ECTS : 2

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

-Continuous assessment: 50%
-Final evaluation: 50%
-Attendance required

TEACHING AIDS

Various multimedia documents

TEACHING LANGUAGE

French

CONTACT

Mme Aumeunier Lise :
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Mme Fradois Delphine :
delphine.fradois@insa-lyon.fr

AIMS

Depending on the level:

- Review level B1 of CEFR and improve your knowledge of B2 level
- or
- Improve your knowledge of B2 level and begin learning level C1

CONTENT

Systematic practice of oral and written comprehension techniques, as well as narrative, summary, synthesis, and argumentation techniques. The program is consistent with the exercises required for the TCF.

Level B1 to review

Level B2

- Write a simple and coherent text on familiar topics
- Write personal letters to describe experiences and impressions.
- Write clear and detailed texts on a wide range of topics related to one's interests.
- Write an essay or report, conveying information or presenting reasons for or against a given opinion.
- Write a clear, well-structured text and develop a point of view.
- Write about complex topics in an essay or report.
- Synthesize ideas.
- Adopt a style appropriate to the audience.

BIBLIOGRAPHY

Online resources :

<https://fle.insa-lyon.fr/fr/rubrique/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B1 level

IDENTIFICATION

CODE : HU-4-S1-EC-L-RUS
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (B1-B2+) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-4-S1-EC-L-ANG-A2
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation consists in continuous assessment of the student's preparation and oral work in class, as well as written work and oral presentations

TEACHING AIDS

Teaching documents are chosen by the instructor according to the target level:

-Various instructional and authentic English-language materials

-Audio-visual documents

TEACHING LANGUAGE

French

CONTACT

Mme Jouffroy Jeannie :
jeannie.jouffroy@insa-lyon.fr

AIMS

This EC contributes to the development of the following transversal skills :

CT3: INTERACT WITH OTHER PERSONS & WORK ON A TEAM
3.2 Put original discourse into context using explicit references.
3.3 Master non-verbal communication: posture and gesture.

CT7: WORK IN AN INTERNATIONAL, INTERCULTURAL CONTEXT
7.1 Communicate and interact in a foreign language.

The student-engineer will practice and be evaluated on the following learning outcomes:

- knowing how to communicate in person, by email and/or by telephone on factual topics
- knowing how to give instructions, explain a procedure, make a dynamic oral presentation
- mastery of the vocabulary and syntax necessary to the successful accomplishment of the above tasks

CONTENT

Students will learn and put into practice the professional communication skills, both oral and written, useful to engineers. In the process, they will develop those of the following skill sets determined to best further their linguistic abilities and respond to their particular needs. The activities, associated with socializing skills in the world of work, may include:

- Holding a conversation on familiar topics
- Telling a story in a conversational context
- Initiating and respond to emails
- Hold a phone conversation on factual topics
- Giving instructions/ Explain a procedure
- Making a dynamic oral presentation using notes if needed on a topic of current events, travel, culture, other
- Preparing a personal or professional time abroad
- Organising after-work activities with colleagues
- Hosting clients and colleagues at lunch
- Learning and explaining professional rules of conduct as well as the company culture

BIBLIOGRAPHY

The Common European Framework of Reference for Languages, Council of Europe

PRE-REQUISITES

A2/B1 level

IDENTIFICATION

CODE : HU-4-S1-EC-L-ITA
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-4-S1-EC-L-TAN
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation in TANDEM consists in continuous assessment of their preparation and oral work in class, as well as written work and an oral presentation.

TEACHING AIDS

-Diverse authentic documents, both written and audiovisual

TEACHING LANGUAGE

French

CONTACT

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Mme Strelow Isabel :
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Mme Fradois Delphine :
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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills valued by INSA:

CT7: Work in an international, intercultural context

7.1 Communicate in a foreign language

7.2 Decode cultural references in speech, attitudes and behaviors.

7.3 Put one's own values, beliefs and behaviors into a cultural perspective.

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

THE STUDENT CAN:

--talk about themselves and their country

--talk about a social phenomenon OR

--talk about a cultural phenomenon OR

--talk about a scientific phenomenon OR

--have a conversation appropriate to the context; communicate in an appropriate manner

--give an oral presentation AND/OR write clear expository prose

--learn and apply usage and vocabulary discovered in class

--understand and apply diverse linguistic and social codes

By the end of the semester, the student will be able to...

- Communicate and interact in one or more foreign languages, both orally and in writing, adapting to the context and the interlocutors, from the transmission of simple information to mediation in more complex communication situations. (Communication skills)

- Identify, understand, and utilize the cultural codes associated with their own language and those of their interlocutors in order to adopt appropriate behavior and communication in diverse intercultural contexts. (Intercultural skills)

- Produce written or oral messages in their native language or in the target language. (Linguistic skills)

CONTENT

The TANDEM modules offer linguistic and cultural exchanges between French-speaking students and German-speaking, English-speaking, Spanish-speaking and Portuguese-speaking students in each of the languages.

The INTERCOMPREHENSION module is all about learning to understand native speakers of a Romance language different to one's own. Each student speaks their native language and learns to understand the others.

The work is carried out by organizing a debate around a controversial subject, by writing a story, by presenting an element belonging to the specific culture, by playing board games ... in short, by interacting permanently and in any form possible with other students fluent in the different working languages.

Of course, oral comprehension is also at the heart of this course. It is a living, authentic learning in the spirit of sharing. Everyone has something to learn and to pass on.

BIBLIOGRAPHY

PRE-REQUISITES

FOR THE TANDEMS

For francophone students:

.B2 in German, English, Spanish

.B1 in Portuguese

For anglophone students: B1 minimum in French

FOR INTERCOMPREHENSION

Must be native speaker of a romance language (French, Spanish, Catalan, Italian, Portuguese, Romanian)

IDENTIFICATION

CODE : HU-4-S1-EC-L-POR
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

Mme Strelow Antunes Isabel :
isabel.strelow-antunes@insa-
lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES
It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop
and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex
tasks that require the students to engage in the 5 linguistic activities, at a level and with
linguistic input that are appropriate for the group. In-class and/or guided independent
study of the forms and functions of the language is regular and adapted to the level of
the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar
training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None



IDENTIFICATION

CODE : GI-4-S1-EC-PCO
ECTS : 4

HOURS

Cours : 0h
TD : 8h
TP : 0h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 12h
Travail personnel : 52h
Total : 80h

ASSESSMENT METHOD

Continuous assessment in an activity situation: evaluation of :
- Requirement analysis,
- Preliminary design,
- Preliminary realization,
- Planning and monitoring, and associated deliverables.
Collective project defense.
Individual assessments of project team members.

TEACHING AIDS

Numerous resources available on the course's Moodle page (presentation materials, process notes, etc.).

TEACHING LANGUAGE

French
English

CONTACT

M. FONDREVELLE Julien :
julien.fondrevelle@insa-lyon.fr

AIMS

This course belongs to teaching unit Collectiv project (GI-4-S1-UE-PROJ) and contributes to the following skills :Project management methodes and tools (following a project, steeps of a project, risk management, AMDEC, formalization of the customers objectives , redaction of specifications, redaction of plan realisation.

Place the project and its stakes in the organization ; organize and manage a project and work in team; meetings management ; manage relevant information; communicate, negotiate and defend a project

CONTENT

Project steps: Identification and expression of customer needs and constraints, Study of existing situation and benchmarking, Identification of risks (FMEA), Formulation of "quality" objectives and rules for verifying their achievement, Specifications (functional, metrological, pedagogical, technical, product, etc.), Delivery schedule and contractual commitment. Drawing up an Analysis of Requirements, a Project Management Plan and a Technical Proposal. Implementation of management dashboards. Task planning and identification of mobilized resources. Achievements and actions in the field (specific to the project). Prototype development. Obtaining and formalizing results, checking that initial objectives have been met.

BIBLIOGRAPHY

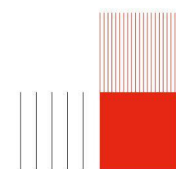
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- BOURGEOIS J-P. « Gestion de Projet » Technique de l'Ingénieur T 7700
- CHVIDCHENKO I., CHEVALLIER J. « Conduite et gestion de projet. Principes et pratiques pour petits et grands projets » Cepaduès Edition, 1997
- AFITEP « Le Management de Projet, Principes et Pratique » Afnor Gestion, 1991
- GIARD V. « La Gestion de Projet » Economica, 1991
- PMI Project management Body of Knowledge; Project Management Institute; www.pmi.org
- R. MARCINIAK , M. PAGERIE Gestion de projet . Guide pratique de tous vos projets et produits industriels, Editions WEKA
- J. MAISONNEUVE La dynamique des groupes, Paris Que-sais je ?, PUF, 1964
- B. KAYE Pédagogie de groupe sciences de l'éducation, Dunod, 1975
- R. MUCHIELLI Les méthodes actives dans la pédagogie des adultes, 1991 Gestion de Projets , Tome 1 et 2 , Edition WEKAP7

PRE-REQUISITES

WARNING : This course extends over two semesters. You will only be allowed to take it if you also take its counterpart in semester 2 (GI-4-S2-EC-PCO).

Prerequisites:

- Problem-solving methodology
- Quality tools
- Basic project management tools and methods



IDENTIFICATIONCODE : GI-4-S1-EC-OEA
ECTS : 2**HOURS**Cours : 0h
TD : 0h
TP : 28h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h**ASSESSMENT METHOD**

Lab: 100 %

TEACHING AIDS

Slides et Polycopié

TEACHING LANGUAGE

French

CONTACTM. VERCRAENE Samuel :
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M. MONTEIRO Thibaud :
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This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

Provide the basis of operational research (concepts, methods, models) for industrial engineers.

Enable the students to master linear programming, integer linear programming, graph theory, to model and solve industrial and logistic problems, using optimization softwares.

CONTENT

The first half of the course alternates between 2-hour lectures and practical sessions, introducing the main concepts of exact optimization and key metaheuristics. It covers a review of linear programming and the simplex method, as well as complexity, integer linear programming, path-based heuristics, and population-based heuristics.

The second half of the course is dedicated to a project. In 2024-25, it focused on a two-tier urban logistics problem, where a truck tour supplies depots, and bicycles then collect goods from the depots to deliver them to customers' homes. A code implementing an algorithmic solution to the problem is provided. Students are then required to replace the solution components with more advanced exact or heuristic methods.

BIBLIOGRAPHY**PRE-REQUISITES**Probability and statistics
Algorithmics
Programming
Integer Linear programming

IDENTIFICATION

CODE : GI-4-S1-EC-INR
ECTS : 2

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 2h
Travail personnel : 26h
Total : 50h

ASSESSMENT METHOD

written report and defense

TEACHING AIDS

Depending on the subject

TEACHING LANGUAGE

French
English

CONTACT

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AIMS

This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

A6 Communicate a scientific analysis or approach with scenarios adapted to their speciality (level 1)

B4 Be creative, innovative, entrepreneurial (Level 1)

C13 Take into account technological and methodological innovation (level 1)

In addition, it requires the mobilization of the following skills:

B2 Working, learning, evolving autonomously

B3 Interact with others, work in teams

B7 Working in an International and Intercultural Context

C1 Observe, measure, analyze and interpret an activity or system from data

C14 Leading a project collectively: organization, communication, animation, group coordination

C16 Identify, analyze and control the risks inherent in a project

By allowing the student to work and be evaluated on the following knowledge:

- Devices and actors of research in France and the place of engineers PhD (A6, B4, C13)

By allowing the student to work and be evaluated on the following abilities :

- Developing a scientific approach of questioning on a research problem (A6, B4, C13)

CONTENT

Research project, in pairs, allowing the student to learn and experiment a research/innovation approach, on a subject proposed and supervised by a teacher-researcher, according to the following steps:

- appropriation of the subject
- formalization of the research problem and proposal of possible solutions
- development of solutions and experiments
- analysis of results, proposal of perspectives

BIBLIOGRAPHY

Depending on the subject

PRE-REQUISITES

Depending on the subject

IDENTIFICATIONCODE : GI-4-S1-EC-DDD
ECTS : 2**HOURS**Cours : 0h
TD : 0h
TP : 28h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h**ASSESSMENT METHOD**Continuous control
Lab report and oral presentation.**TEACHING AIDS**

PDF ressources on moodle

TEACHING LANGUAGE

English

CONTACTMME ROBARDET Celine :
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This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

CONTENT

- Building a Data Warehouse from raw data
- Introduction to data modeling for machine learning
- Methodology for the application of machine learning
- Application: predictive maintenance, building a model for failure prediction
- Application: Construction and operation of a model for failure prediction.

BIBLIOGRAPHY**PRE-REQUISITES**Notions of relational databases
PYTHON programming
Concepts of Statistics

IDENTIFICATION

CODE : GI-4-S1-EC-SDF
ECTS : 2

HOURS

Cours : 0h
TD : 16h
TP : 12h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h

ASSESSMENT METHOD**TEACHING AIDS****TEACHING LANGUAGE**

French

CONTACT

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M. ZAMAI Eric :
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AIMS

This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

Knowledge :

- Defining a rationale for dimensioning
- MTBF MTTF MTTR
- State graph, Markov chain
- Fault Tree, Reliability Diagram
- Decision support

Capacities :

- Apprehend the phenomena of malfunctioning
- Preliminary Risk Analysis
- Lifetime, Reliability, Availability, Maintainability of material and organisational structures
- Contract, commitment and penalties
- Negotiate solutions to increase fault tolerance
- Performance indicators
- Dimensioning

CONTENT

Dependability assesment methods

BIBLIOGRAPHY

Sûreté de Fonctionnement des systèmes matériels, A. Villemeur, Edt Eyrolles
Reliability Eng., Elsevier

PRE-REQUISITES

Stochastic process
Markov Chains
Probability



IDENTIFICATION

CODE : GI-4-S1-EC-ECH
ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

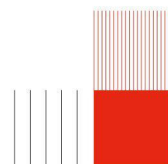
TEACHING LANGUAGE

French

CONTACT

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MME PELLET Lorraine :
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IDENTIFICATION

CODE : GI-0-S1-EC-CESURE
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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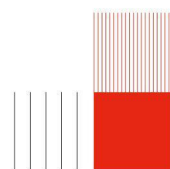
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membre de



IDENTIFICATION

CODE : GI-0-S1-EC-VAC
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

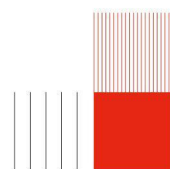
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IDENTIFICATION

CODE : GI-4-S1-EC-ISD
ECTS : 2

HOURS

Cours : 0h
TD : 18h
TP : 16h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 36h
Travail personnel : 12h
Total : 48h

ASSESSMENT METHOD

TEACHING AIDS

Course handouts

TEACHING LANGUAGE

English

CONTACT

MME ROBARDET Celine :
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AIMS

This course belongs to teaching unit Gestion et exploitation des données de l'entreprise (GI-4-S1-UE-GEDE) and contributes to the following skills :

A3 Implement an experimental approach (Level 2)
A5 Process data (Level 3)
A6 Report on an analysis or on a scientific approach (Level 3)
C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)
C4 Sizing the hardware and / or software of a system (Level 2)
C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 1)
B2 work, learn, progress autonomously (Level 2)
B3 interact with people and work in a team (Level 3)

By allowing the engineering students to work and be evaluated on the following skills:

- Mastering structured data processing tools: Excel and Stats (C4)
- Know how to make a decision based on numerical results (C1)
- Choose a processing method based on the data and objectives (A3, A5, A6, C1, C11, B2, B3)

CONTENT

DATA ANALYSIS

1.Data

Characteristics: origin, quality, coding

Preparation: restoration, filtering, synchronization, construction of classes

Dependence: distances, ultra metric, correlation

2.Tools :

Descriptive methods: principal components analysis, correspondences, preferences

Explanatory methods: regression, segmentation, canonical analysis, discriminated analysis

Structural methods: ultra metric and hierarchical classification, dynamic swarms.

3.Analysis :

Analysis methodologies

Interpretation rules

Case studies

BIBLIOGRAPHY

**** Data analysis ****

1. E.Diday, J.Lemaire, J.Pouget, F.Testu, Eléments d'analyse de données, Dunod 1982
2. G.Saporta, Probabilités, Analyse de données et Statistiques, Technip 1990
3. Jean-Pierre , Analyse interactive des données (ACP, AFC) avec Excel 2000 : théorie et pratique / Georgin. Presses universitaires de Rennes , 2002
4. Alain Morineau et Yves-Marie Chatelin , L'analyse statistique des données [Texte imprimé] : apprendre, comprendre et réaliser avec Excel, Ellipses , 2005 , 407 p

**** Data warehouse ****

1. Entrepôts de données. Guide pratique de modélisation dimensionnelle, 2ème édition, R. Kimball, M. Ross
2. Piloter l'entreprise grâce au data Warehouse, J.-M. Franco, S de Lignerolles

PRE-REQUISITES

Statistics, data base

IDENTIFICATION

CODE : GI-4-S1-EC-BIN
ECTS : 2

HOURS

Cours : 0h
TD : 4h
TP : 14h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 12h
Total : 32h

ASSESSMENT METHOD

Written exam at the end of the tutorial part ("TD")
Project evaluation : collective presentation and written report

TEACHING AIDS

Slides

TEACHING LANGUAGE

English

CONTACT

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AIMS

GI-4-S1-UE-GEDE
GI-4-S1-EC-BIN

General engineering skills:

- Develop an experimental approach;
- Identify, formulate and solve a complex engineering problem.

Specific Industrial Engineering skills:

- Observe, measure, analyze and interpret an activity or system from data
- Modeling, designing a system of information, decision-making and production of goods and services;
- Evaluate, prototype or simulate a system.

Abilities :

Collect customer needs for analysis; Identify sources of production data useful for an analytical approach; Process voluminous information for analysis; Exploit the data in the perspective of the analyzed activity: Carry out a project, working in a group.

Knowledge: operational and analytical database schemas; Elementary actions and sequencing for data transformation; Methods and tools for visualizing information from data.

CONTENT

DATA WAREHOUSE

1. Dimensional modeling for decision support systems
2. The Extract-Transform-Load (ETL) task : data cleaning, fusion and restructuration methods, ETL tools (Tableau)
3. Query tools and methods : reporting, scorecards, etc.

BIBLIOGRAPHY

Systèmes d'information décisionnels, E. Ferragu, 2013 (available at Doc'INSA)
- Piloter l'entreprise grâce au data warehouse, Jean-Michel Franco, Sandrine de Lignerolles, 2000
(available at the Industrial Eng. dept. local library)

PRE-REQUISITES

Relational Databases, SQL

IDENTIFICATION

CODE : GI-4-S1-EC-SIE
ECTS : 3

HOURS

Cours : 6h
TD : 16h
TP : 18h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 42h
Travail personnel : 18h
Total : 62h

ASSESSMENT METHOD

Situational assessment Group score

TEACHING AIDS

Project Subject, Company Data
Readings to be done before some sessions (theoretical complements)
Methodological guides presented during the sessions

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to the Management and Exploitation of Company Data (GI-4-S1-UE-GEDE) teaching unit and contributes to the following skills:

By allowing the student to work and be evaluated on the following knowledge:

- BPMN modeling
- Process Value Analysis
- Evaluation of the performance of a process
- Process mining
- The various industrial information systems

By allowing the student to work and be evaluated on the following abilities:

- Model a complex business process
- Diagnose a business process in terms of inefficiencies and non-value-added tasks/flows
- Identify and evaluate business process improvement solutions
- Specify the information system supporting a business process

CONTENT

This course is conducted as a team project and is based on an industrial case. The objective of the project is to carry out a preliminary study for the modernization of a manufacturing company's information system. This preliminary study includes several phases: (1) Analysis of the existing situation (AS-IS) through the modeling of current business processes, the diagnosis of these processes and the identification of improvement solutions; (2) Design of an improvement solution (TO-BE) through the modeling of new processes and the specification of a software/hardware solution to support the new organization (without however developing it).

BIBLIOGRAPHY

1. Marlon Dumas, Marcello La Rosa, Jan Mendling, Hajo A. Reijers. "Fundamentals of Business Process Management". Editions Springer, ISBN :9783662565094 / 3662565099, 2018, 527 pages
2. Heru Susanto, Fang-Yie Leu, Chin Kang Chen. "Business Process Reengineering: An ICT Approach". Editions Apple Academic Press, ISBN :9780429949302 / 0429949308, 2019, 248 pages
3. W.M.P. van der Aalst. "Process Mining: Data Science in Action". Editions Springer-Verlag, 2016. <http://www.springer.com/978-3-662-49850-7>

PRE-REQUISITES

- Process Modeling (BPMN)
- Object-oriented modeling (UML)



IDENTIFICATION

CODE : GI-4-S1-EC-PCO
ECTS : 4

HOURS

Cours : 0h
TD : 8h
TP : 0h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 12h
Travail personnel : 52h
Total : 80h

ASSESSMENT METHOD

Continuous assessment in an activity situation: evaluation of :
- Requirement analysis,
- Preliminary design,
- Preliminary realization,
- Planning and monitoring, and associated deliverables.
Collective project defense.
Individual assessments of project team members.

TEACHING AIDS

Numerous resources available on the course's Moodle page (presentation materials, process notes, etc.).

TEACHING LANGUAGE

French
English

CONTACT

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AIMS

This course belongs to teaching unit Collectiv project (GI-4-S1-UE-PROJ) and contributes to the following skills :Project management methodes and tools (following a project, steeps of a project, risk management, AMDEC, formalization of the customers objectives , redaction of specifications, redaction of plan realisation.

Place the project and its stakes in the organization ; organize and manage a project and work in team; meetings management ; manage relevant information; communicate, negotiate and defend a project

CONTENT

Project steps: Identification and expression of customer needs and constraints, Study of existing situation and benchmarking, Identification of risks (FMEA), Formulation of "quality" objectives and rules for verifying their achievement, Specifications (functional, metrological, pedagogical, technical, product, etc.), Delivery schedule and contractual commitment. Drawing up an Analysis of Requirements, a Project Management Plan and a Technical Proposal. Implementation of management dashboards. Task planning and identification of mobilized resources. Achievements and actions in the field (specific to the project). Prototype development. Obtaining and formalizing results, checking that initial objectives have been met.

BIBLIOGRAPHY

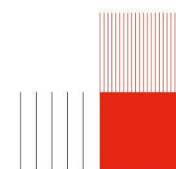
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- AFITEP « Le Management de Projet, Principes et Pratique » Afnor Gestion, 1991
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- PMI Project management Body of Knowledge; Project Management Institute; www.pmi.org
- R. MARCINIAK , M. PAGERIE Gestion de projet . Guide pratique de tous vos projets et produits industriels, Editions WEKA
- J. MAISONNEUVE La dynamique des groupes, Paris Que-sais je ?, PUF, 1964
- B. KAYE Pédagogie de groupe sciences de l'éducation, Dunod, 1975
- R. MUCHIELLI Les méthodes actives dans la pédagogie des adultes, 1991 Gestion de Projets , Tome 1 et 2 , Edition WEKAP7

PRE-REQUISITES

WARNING : This course extends over two semesters. You will only be allowed to take it if you also take its counterpart in semester 2 (GI-4-S2-EC-PCO).

Prerequisites:

- Problem-solving methodology
- Quality tools
- Basic project management tools and methods



IDENTIFICATION

CODE : GI-4-S1-EC-REN
ECTS : 0

HOURS

Cours : 2h
TD : 4h
TP : 2h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 8h
Travail personnel : 0h
Total : 8h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

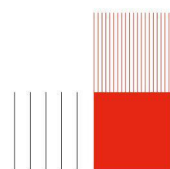
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membre de



IDENTIFICATION

CODE : HU-4-S1-EC-L-CHI
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

-- understand and apply social codes from different countries

--have a conversation appropriate de the context

--confront communicational challenges of increasing difficulty

--produce diverse types of written expression

--understand authentic documents (written and audiovisual)

--talk about a society or a social phenomenon

--apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks

that require the students to engage in the 5 linguistic activities, at a level and with linguistic

input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

--language learning methodology (RC, OC, WP, vocabulary learning and grammar training)

--oral presentation skills

--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-4-S1-EC-L-ITA
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-4-S1-EC-L-POR
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES
It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop
and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex
tasks that require the students to engage in the 5 linguistic activities, at a level and with
linguistic input that are appropriate for the group. In-class and/or guided independent
study of the forms and functions of the language is regular and adapted to the level of
the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar
training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-4-S1-EC-L-ANG-A2
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation consists in continuous assessment of the student's preparation and oral work in class, as well as written work and oral presentations

TEACHING AIDS

Teaching documents are chosen by the instructor according to the target level:

-Various instructional and authentic English-language materials

-Audio-visual documents

TEACHING LANGUAGE

French

CONTACT

Mme Jouffroy Jeannie :
jeannie.jouffroy@insa-lyon.fr

AIMS

This EC contributes to the development of the following transversal skills :

CT3: INTERACT WITH OTHER PERSONS & WORK ON A TEAM
3.2 Put original discourse into context using explicit references.
3.3 Master non-verbal communication: posture and gesture.

CT7: WORK IN AN INTERNATIONAL, INTERCULTURAL CONTEXT
7.1 Communicate and interact in a foreign language.

The student-engineer will practice and be evaluated on the following learning outcomes:

- knowing how to communicate in person, by email and/or by telephone on factual topics
- knowing how to give instructions, explain a procedure, make a dynamic oral presentation
- mastery of the vocabulary and syntax necessary to the successful accomplishment of the above tasks

CONTENT

Students will learn and put into practice the professional communication skills, both oral and written, useful to engineers. In the process, they will develop those of the following skill sets determined to best further their linguistic abilities and respond to their particular needs. The activities, associated with socializing skills in the world of work, may include:

- Holding a conversation on familiar topics
- Telling a story in a conversational context
- Initiating and respond to emails
- Hold a phone conversation on factual topics
- Giving instructions/ Explain a procedure
- Making a dynamic oral presentation using notes if needed on a topic of current events, travel, culture, other
- Preparing a personal or professional time abroad
- Organising after-work activities with colleagues
- Hosting clients and colleagues at lunch
- Learning and explaining professional rules of conduct as well as the company culture

BIBLIOGRAPHY

The Common European Framework of Reference for Languages, Council of Europe

PRE-REQUISITES

A2/B1 level

IDENTIFICATION

CODE : HU-4-S1-EC-L-ARA
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

-- understand and apply social codes from different countries

--have a conversation appropriate de the context

--confront communicational challenges of increasing difficulty

--produce diverse types of written expression

--understand authentic documents (written and audiovisual)

--talk about a society or a social phenomenon

--apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

--language learning methodology (RC, OC, WP, vocabulary learning and grammar training)

--oral presentation skills

--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-4-S1-EC-L-ANG
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation consists in continuous assessment of the student's preparation and oral work in class, as well as written work and oral presentations'

TEACHING AIDS

Teaching documents are chosen by the instructor according to the target level:
-Various instructional and authentic English-language materials
-Audio-visual documents

TEACHING LANGUAGE

English

CONTACT

Mme JOUFFROY Jeannie :
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AIMS

This EC contributes to the development of the following transversal skills :

CT3: INTERACT WITH OTHER PERSONS & WORK ON A TEAM
3.2 Put original discourse into context using explicit references.
3.3 Master non-verbal communication: posture and gesture.

CT7: WORK IN AN INTERNATIONAL, INTERCULTURAL CONTEXT
7.1 Communicate and interact in a foreign language.

The student-engineer will practice and be evaluated on the following learning outcomes:

--knowing how to communicate in person, by email and/or by telephone on factual topics
--knowing how to give instructions, explain a procedure, make a dynamic oral presentation
--mastery of the vocabulary and syntax necessary to the successful accomplishment of the above tasks

The course is taught in level groups; themes and linguistic skills vary between levels.

CONTENT

Students will learn and put into practice the professional communication skills, both oral and written, useful to engineers. In the process, they will develop those of the following skill sets determined to best further their linguistic abilities and respond to their particular needs. The activities, associated with socializing skills in the world of work, may include:

--Holding a conversation on familiar topics
--Telling a story in a conversational context
--Initiating and respond to emails
--Hold a phone conversation on factual topics
--Giving instructions/ Explain a procedure
--Making a dynamic oral presentation using notes if needed on a topic of current events, travel, culture, other
--Preparing a personal or professional time abroad
--Organising after-work activities with colleagues
--Hosting clients and colleagues at lunch
--Learning and explaining professional rules of conduct as well as the company culture

BIBLIOGRAPHY

The Common European Framework of Reference for Languages, Council of Europe

PRE-REQUISITES

A2/B1 level in English

IDENTIFICATION

CODE : HU-4-S1-EC-L-RUS
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional
documents, linked with chosen
themes

TEACHING LANGUAGE

CONTACT

M. Iakovlev Maxime :
maxime.iakovlev@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (B1-B2+) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE : HU-4-S1-EC-L-JAP
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

M. Mihara Norio :
norio.mihara@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

- understand and apply social codes from different countries
- have a conversation appropriate de the context
- confront communicational challenges of increasing difficulty
- produce diverse types of written expression
- understand authentic documents (written and audiovisual)
- talk about a society or a social phenomenon
- apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

- language learning methodology (RC, OC, WP, vocabulary learning and grammar training)
- oral presentation skills
- a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

None

IDENTIFICATION

CODE HU-0-S1-EC-L-FLE-
CIVB2C1

ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Project based evaluation.

TEACHING AIDS

Various authentic materials

TEACHING LANGUAGE

French

CONTACT

Mme Aumeunier Lise :
elisabeth.aumeunier@insa-lyon.fr

Mme Fradois Delphine :
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AIMS

Humanities Skills:

2. Work, learn, and develop independently
3. Interact with others and work in a team
4. Demonstrate creativity, innovation, and initiative
7. Work in an international and intercultural context

Language Activities: Written Expression, Written Expression, Oral Expression, Listening Comprehension, Oral Expression

An intercultural approach for a better understanding of French civilization and culture with the aim of fostering student autonomy and facilitating their integration into their studies, student life, and social and professional life.

CONTENT

The work will be organized around various activities: debates, presentations, projects, global simulations, forums, etc. on various past and present themes (society, culture, history, etc.) and will integrate an intercultural dimension.

BIBLIOGRAPHY

You will find a selection of resources available online at the following address <https://fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B2 level

IDENTIFICATION

CODE HU-0-S1-EC-L-FLE-
COMPB2

ECTS : 2

HOURS

Cours :	0h
TD :	26h
TP :	0h
Projet :	0h
Evaluation :	0h
Face à face pédagogique :	26h
Travail personnel :	0h
Total :	26h

ASSESSMENT METHOD

- Continuous assessment: 50%
- Final evaluation: 50%
- Attendance required

TEACHING AIDS

Various authentic documents
(paper, audio, video and
multimedia)

TEACHING LANGUAGE

French

CONTACT

Mme Aumeunier Lise :
elisabeth.aumeunier@insa-lyon.fr

Mme Fradois Delphine :
delphine.fradois@insa-lyon.fr

AIMS

Depending on the level:

- Review level B1 of CEFR and improve your knowledge of B2 level
- or
- Improve your knowledge of B2 level and begin learning level C1

CONTENT

Systematic practice of oral and written comprehension techniques, as well as narrative, summary, synthesis, and argumentation techniques. The program is consistent with the exercises required for the TCF.

Level B1 to review

Level B2

- Write a simple and coherent text on familiar topics
- Write personal letters to describe experiences and impressions.
- Write clear and detailed texts on a wide range of topics related to one's interests.
- Write an essay or report, conveying information or presenting reasons for or against a given opinion.
- Write a clear, well-structured text and develop a point of view.
- Write about complex topics in an essay or report.
- Synthesize ideas.
- Adopt a style appropriate to the audience.

BIBLIOGRAPHY

- La documentation française <https://www.vie-publique.fr/publications-de-la-documentation-francaise>
- Production écrite B1/B2 (Didier)
- Ecrire pour convaincre (Hachette)
- Vocabulaire expliqué du Français(CI)
- Francoscopie (Larousse)

Online resources :

<https://fle.satellite.insa-lyon.fr/content/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B1 Level

IDENTIFICATION

CODE : HU-4-S1-EC-L-ESP
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

The average is made up of half (50%) continuous assessment (assessments of the 5 skills carried out in class) and a common exam (50% of the average) which assesses Written Comprehension, Oral Comprehension, Grammatical and Lexical Skills and Written Expression. This common exam does not have the value of level validation.

TEACHING AIDS

Authentic and/or didactic documents related to the selected topics.
- "Pink" grammar and exercise booklets
- "Yellow" booklet: conjugation guides
- The CRL

TEACHING LANGUAGE

French

CONTACT

M. Suarez Lopez Gonzalo :
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AIMS

The targeted and mobilized skills are those of both the INSA Humanities Skills Framework (specifically skills 3 and 7) and the CEFR.
Consistent description of the CEFR skills.
<http://www.sciencespo-lille.eu/sites/default/files/cecrl.pdf>

At the end of the semester, the student will be able to...

- Communicate and interact in Spanish, both orally and in writing, adapting to the context and the interlocutors, from the transmission of simple information to mediation in more complex communication situations. (COMMUNICATIONAL COMPETENCY)
- Identify, understand, and utilize the cultural codes associated with the Spanish language and its speakers in order to adopt appropriate behavior and communication in diverse intercultural contexts. (INTERCULTURAL COMPETENCY)
- Understand spoken and written messages in Spanish in communication situations corresponding to the targeted proficiency level, in accordance with the Common European Framework of Reference for Languages (CEFR).
- Produce spoken and written messages in Spanish, effectively utilizing the linguistic resources specific to the targeted proficiency level (CEFR).

CONTENT

The five skills recognized by the Common European Framework of Reference for Languages (CEFR) are practiced and assessed several times throughout the semester/year, ensuring regular practice of the various skills and knowledge acquired.

The themes, grammar, and vocabulary covered in class are adapted to the target level (indicated in the group code) and are at the discretion of the teacher.

BIBLIOGRAPHY

Web resources:

- for listening comprehension, grammar, and vocabulary training (all levels): <http://www.ver-taal.com/index.htm>
- grammar and vocabulary exercises (all levels): <https://www.espanolfacile.com/>
- <https://moodle.insa-lyon.fr/course/index.php?categoryid=353>

PRE-REQUISITES

None. Courses range from beginner to advanced level. Each student will be placed in a group corresponding to their level, either through a test at the beginning of the year (for new students) or based on their level from the previous year for students already attending INSA.

IDENTIFICATION

CODE HU-0-S1-EC-L-FLE-
TTCB1B2

ECTS : 2

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

-Continuous assessment: 50%
-Final evaluation: 50%
-Attendance required

TEACHING AIDS

Various multimedia documents

TEACHING LANGUAGE

French

CONTACT

Mme Aumeunier Lise :
elisabeth.aumeunier@insa-lyon.fr

Mme Fradois Delphine :
delphine.fradois@insa-lyon.fr

AIMS

Depending on the level:

- Review level B1 of CEFR and improve your knowledge of B2 level
- or
- Improve your knowledge of B2 level and begin learning level C1

CONTENT

Systematic practice of oral and written comprehension techniques, as well as narrative, summary, synthesis, and argumentation techniques. The program is consistent with the exercises required for the TCF.

Level B1 to review

Level B2

- Write a simple and coherent text on familiar topics
- Write personal letters to describe experiences and impressions.
- Write clear and detailed texts on a wide range of topics related to one's interests.
- Write an essay or report, conveying information or presenting reasons for or against a given opinion.
- Write a clear, well-structured text and develop a point of view.
- Write about complex topics in an essay or report.
- Synthesize ideas.
- Adopt a style appropriate to the audience.

BIBLIOGRAPHY

Online resources :

<https://fle.insa-lyon.fr/fr/rubrique/pourquoi-travailler-en-autonomie>

PRE-REQUISITES

B1 level

IDENTIFICATION

CODE : HU-4-S1-EC-L-ALL
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

-2 written evaluations
-2 oral evaluations

TEACHING AIDS

Authentic and/or instructional documents, linked with chosen themes

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills :

CT7: Work in an international, intercultural context

--CT7.1: Communicate in a foreign language

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

The student CAN:

-- understand and apply social codes from different countries

--have a conversation appropriate de the context

--confront communicational challenges of increasing difficulty

--produce diverse types of written expression

--understand authentic documents (written and audiovisual)

--talk about a society or a social phenomenon

--apply grammar and vocabulary covered in class

The target level of acquisition (A1-B2) is dependent on the level of group.

CONTENT

Instructors use CEFR methodology to design lessons toward the completion of complex tasks that require the students to engage in the 5 linguistic activities, at a level and with linguistic input that are appropriate for the group. In-class and/or guided independent study of the forms and functions of the language is regular and adapted to the level of the group.

In the first semester, themes covered include:

--language learning methodology (RC, OC, WP, vocabulary learning and grammar training)

--oral presentation skills

--a topic about a society or a social phenomenon

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

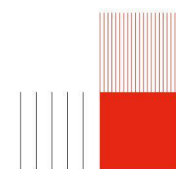
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membre de



IDENTIFICATION

CODE : HU-4-S1-EC-L-TAN
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 26h
Travail personnel : 0h
Total : 26h

ASSESSMENT METHOD

Evaluation in TANDEM consists in continuous assessment of their preparation and oral work in class, as well as written work and an oral presentation.

TEACHING AIDS

-Diverse authentic documents, both written and audiovisual

TEACHING LANGUAGE

French

CONTACT

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Mme Vincensini Catherine :
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Mme Strelow Isabel :
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Mme Fradois Delphine :
delphine.fradois@insa-lyon.fr

AIMS

This EC belongs to the UE: HUMANITIES

It contributes to the development of the following transversal skills valued by INSA:

CT7: Work in an international, intercultural context

7.1 Communicate in a foreign language

7.2 Decode cultural references in speech, attitudes and behaviors.

7.3 Put one's own values, beliefs and behaviors into a cultural perspective.

by allowing the student-engineer to develop and be evaluated on the following learning outcomes:

THE STUDENT CAN:

--talk about themselves and their country

--talk about a social phenomenon OR

--talk about a cultural phenomenon OR

--talk about a scientific phenomenon OR

--have a conversation appropriate to the context; communicate in an appropriate manner

--give an oral presentation AND/OR write clear expository prose

--learn and apply usage and vocabulary discovered in class

--understand and apply diverse linguistic and social codes

By the end of the semester, the student will be able to...

- Communicate and interact in one or more foreign languages, both orally and in writing, adapting to the context and the interlocutors, from the transmission of simple information to mediation in more complex communication situations. (Communication skills)

- Identify, understand, and utilize the cultural codes associated with their own language and those of their interlocutors in order to adopt appropriate behavior and communication in diverse intercultural contexts. (Intercultural skills)

- Produce written or oral messages in their native language or in the target language. (Linguistic skills)

CONTENT

The TANDEM modules offer linguistic and cultural exchanges between French-speaking students and German-speaking, English-speaking, Spanish-speaking and Portuguese-speaking students in each of the languages.

The INTERCOMPREHENSION module is all about learning to understand native speakers of a Romance language different to one's own. Each student speaks their native language and learns to understand the others.

The work is carried out by organizing a debate around a controversial subject, by writing a story, by presenting an element belonging to the specific culture, by playing board games ... in short, by interacting permanently and in any form possible with other students fluent in the different working languages.

Of course, oral comprehension is also at the heart of this course. It is a living, authentic learning in the spirit of sharing. Everyone has something to learn and to pass on.

BIBLIOGRAPHY

PRE-REQUISITES

FOR THE TANDEM

For francophone students:

.B2 in German, English, Spanish

.B1 in Portuguese

For anglophone students: B1 minimum in French

FOR INTERCOMPREHENSION

Must be native speaker of a romance language (French, Spanish, Catalan, Italian, Portuguese, Romanian)

IDENTIFICATION

CODE : CDS-4-S1-EC-EPS
ECTS : 1

HOURS

Cours : 0h
TD : 20h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 0h
Total : 20h

ASSESSMENT METHOD

Assessment in Physical Education concerns the teaching of Sports and Artistic Physical Activities (APSA), and will take the form of continuous assessment with half-yearly marking.

The mark depends on the degree of acquisition of the skills expected in each of the activities, and the progress made over all the sessions in the cycle. The mark takes into account :

Individual and/or team performance
mastery of execution
Progress in the sports project
Responsibility and autonomy

TEACHING AIDS

All physical, sporting, artistic and competitive activities

TEACHING LANGUAGE

French

CONTACT

Mme JAUSSAUD Marie :
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AIMS

This EC is part of the Teaching Unit: SHS and contributes to the development of the School's transversal competences

1* Auto-evaluating one's own performance

Knowledges :

- Fundamentals, principles of action and terminology of sports activities
- Criteria for observation, achievement and success.

Abilities :

- Assess your level of practice
- Build up a warm-up
- Set goals for progress
- Manage physical and mental potential

2* Work, learn and develop independently

Knowledge :

- PSAA rules
- Observation criteria
- Principles of warm-up and cool-down

Abilities :

- Mobilise resources
- Analyse, observe, question
- Take on different roles (referee, choreographer)

3* Interact with others, work as part of a team

Knowledges :

- Roles and functions in each sports activity

Abilities :

- Communicate appropriately: verbal, non-verbal and postural communication.
- Integrate into a group
- Take part in and develop a group project
- Take the initiative
- Be a good listener

4* Be creative, innovative and enterprising

Knowledge :

- Artistic disciplines

Abilities :

- Draw on knowledge and resources from different artistic fields to produce an original work.
- Mobilise the imagination and sensibility and make them visible through dance movement
- Access the symbolism of the body

5* Act responsibly in a complex world

Knowledge :

- Safety and operating rules

Abilities :

- Identify uncertainties and risks and act to reduce them
- Integrate a responsible dimension into their actions
- Show respect and fair play in a power struggle

6* Working in an international context

Knowledge :

- Socio-cultural differences

Abilities :

- Integrate cultural diversity into group work
- Act with respect for self and others

CONTENT

Physical Education and Sport lessons are organised around traditional Physical Education lessons, or advanced lessons, or appropriate practices (EPSA), or competitive practices within the framework of the Section Sportive Haut Niveau.

1. Physical Education lessons :

Students choose one or two physical and sporting activities per year from among the activities offered by the sports centre (individual, group, dual).

2. Appropriate Physical Education lessons: For all students who are exempt from physical activity for at least 2 months:

Swimming, Body-building, Nordic Walking, Somatic Exercise, Sophrology, Wheelchair Basketball, Pilates, Table Tennis, etc.

3. Advanced Physical Education courses :

Specialisation in a sporting activity, University training and competitions

4. SSHN (High-Level Athlete section)

BIBLIOGRAPHY

PRE-REQUISITES

- EPS: none
 - Appropriate Physical Education: subject to medical advice
 - Advanced courses and competitive practice: previous practice required subject to specific selection according to each activity
 - SHN: ministerial list
- Levels 1 and 2: Physical Education, Appropriate physical education
Level 3: Advanced courses and competitive practice, SHN

INSA LYON

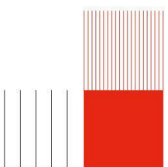
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membre de



IDENTIFICATIONCODE : HU-0-S1-EC-S-SERIE2
ECTS : 2**HOURS**

Cours :	0h
TD :	20h
TP :	0h
Projet :	0h
Evaluation :	0h
Face à face pédagogique :	20h
Travail personnel :	0h
Total :	20h

ASSESSMENT METHOD

Assessment will be conducted through continuous evaluation. The assessment methods will be presented at the beginning of the semester by the teaching team.

TEACHING AIDS

Materials are chosen by the instructor based on the module:

- Didactic documents related to the module
- Audiovisual materials
- Recommended readings

TEACHING LANGUAGE

French

CONTACTMme JOUISHOMME Delphine :
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A series of elective courses in Humanities and Social Sciences (HSS) offers several options for students to choose from, allowing them to develop and deepen specific skills. This course aims to develop one or more transversal skills among the following:

- CT1: Self-awareness and self-management
- CT2: Working, learning, and evolving independently
- CT3: Interacting with others, working in a team
- CT4: Demonstrating creativity
- CT5: Acting responsibly in a complex world
- CT6: Navigating and evolving within an organization
- CT7: Working in an international and intercultural context

The list of options available in Series 1 and the specific competencies for each option are detailed in the catalog on the IntranetHumas:
<https://intranethumas.insa-lyon.fr/sciences-humaines-sociales/offre-de-formation/cours-la-carte-0>

CONTENT

Each module is designed to encourage interaction and active student participation. The content is structured around the following key aspects:

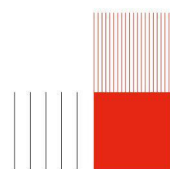
- Theoretical deepening related to the theme
- Reflection on the topic
- Practical exercises and activities
- Assessments and presentation of work

BIBLIOGRAPHY

The bibliography is selected by the instructor based on the module.

PRE-REQUISITES

French





IDENTIFICATION

CODE : GI-4-S1-EC-EIE
ECTS : 2

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h

ASSESSMENT METHOD

Written examination
Report

TEACHING AIDS

Powerpoint slides
Sustainable development and
societal responsibility serious
game (CIPE)

TEACHING LANGUAGE

French
English

CONTACT

M. ARBAOUI Taha :
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AIMS

This course belongs to teaching unit Continuous improvement and innovation(GI-4-S1-UE-AMCI) and contributes to the following skills :

A1 Analyze a system (real or virtual) or a problem (Level 2)

A2 Operate a model of a real or virtual system (Level 2)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)

C3 Evaluating, prototyping and simulating a system (Level 2)

C10 Developing and implementing an action plan as part of a quality and continuous improvement approach (Level 1)

C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 2)

C12 Changing organizations to meet new constraints or opportunities (Level 2)

C20 Implementing Corporate Social Responsibility (CSR) (Level 1)

B5 act responsibly in a complex world (Level 2)

B6 position oneself, work, and evolve within a company or a socio-productive organization (Level 1)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Definitions of industrial ecology, circular economy and associated terminology (B5, B6)

- Material Flow Analysis (A1, A2, C1, C2, C3)

- Integrated Assessment Model (A1, A2, C1, C2, C3)

- Waste definition, associated waste management rules, legal aspects (A1, C10)

- Challenges of circular economy and reverse logistics from the industrial engineering point of view (C20, B5)

- Eco-parks, industrial symbiosis, synergy and mutualization (C11, C12)

To be able to :

- Propose solutions limiting the consumption and waste of resources and the production of waste (C10, C20)

- Diagnose sustainable development issues within an industrial company (C10, C11)

CONTENT

- Definitions

- Sustainable development and societal responsibility serious game

- Material Flow Analysis

- Best Available Techniques

- Waste

- Reverse logistics

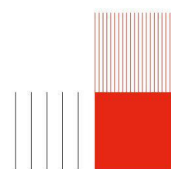
- Ecoparks

- Disassembly problems

BIBLIOGRAPHY

PRE-REQUISITES

Notions of life cycle analysis



IDENTIFICATION

CODE : GI-4-S1-EC-DTH
ECTS : 2

HOURS

Cours : 2h
TD : 24h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h

ASSESSMENT METHOD

project in group and evaluation
(2h)

TEACHING AIDS

Powerpoint slides

TEACHING LANGUAGE

French

CONTACT

MME SERNA Audrey :
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AIMS

Interactive systems design:
- Methods, tools and concepts for designing interactive systems
- User centered approach
- Ergonomic criterias

CONTENT

- HCI introduction and interaction paradigms
- Methods and models for designing UI (User Interfaces)
- Analysis of users requirements
- Sketching and prototyping
- Ergonomic criteria
- UI evaluation

BIBLIOGRAPHY

PRE-REQUISITES

IDENTIFICATION

CODE : GI-4-S1-EC-QLT
ECTS : 1

HOURS

Cours : 0h
TD : 12h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 14h
Travail personnel : 6h
Total : 20h

ASSESSMENT METHOD

1 two-hour exam

TEACHING AIDS

Handout in English, exercises in English, Minitab software and an online tutorial in French

TEACHING LANGUAGE

French
English

CONTACT

M. MOYAUX Thierry :
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AIMS

GI-4-S1-UE-AMCI
GI-4-S1-EC-QLT
Quality - Maintenance

Understand the issues and tools of quality (quality process vs. quality tools) and maintenance (impact of the organisation of quality on the other functions of companies).

Understand the usefulness and the operation of some tools (control charts, Gage R&R, reliability, etc.).

This EC contributes to:

- * School skills in Engineering Sciences / Level (1 for sensitization, 2 for appropriation, 3 for mastery) :
 - . Solve questions asked by a scientific problem by being aware of limits / 3
 - . Develop an experimental approach / 3
 - . Implement a scientific approach / 2
- * School skills specific to GI:
 - . Observing, measuring, analyzing and interpreting an activity or a system from data / 3
 - . Managing a production system and react to malfunctions / 3
 - . Developing and implementing an action plan as part of a quality and continuous improvement approach / 3
 - . Collectively managing a project: organization, communication, group coordination / 1

CONTENT

FR or EN - possibility to integrate a group with lecture/practical work in english (but online tutorial in French only)

ONLINE TUTORIAL about Six Sigma to do without the assistance of the teacher

QUALITY

- * General introduction: relations of quality and maintenance
- * Introduction to quality: definitions, goals, voice of the client, overview of tools
- * Six sigma : Introduction, Etapes du DMAIC, Organisation.
- * Overview of ISO9001:2008
- * Control charts
- * Gage R&R

MAINTENANCE

- * Introduction: Definitions, Types of maintenance
- * TPM (Total Productive Maintenance), Overview, The 8 pillars with details on Autonomous maintenance in Pillar 2 and overview of RCM (Reliability-Centered Maintenance) in Pillar 3 (Planned maintenance), Lean Six Sigma
- * Types of maintenance, Impact of OEE (Overall Equipment Effectiveness), Frequency of change calculated from reliability function

BIBLIOGRAPHY

Bufferne J. (www.jean-bufferne.com), Le guide de la TPM, Éditions d'Organisation, Eyrolles, 2006.

Duret D., Pillet M., Qualité en production; De l'ISO 9000 à Six Sigma, 3e édition, Editions d'Organisation, 2005.

Pillet M., Six Sigma; Comment l'appliquer, Editions d'Organisation, 2004.

PRE-REQUISITES

Probability - Statistics

IDENTIFICATIONCODE : GI-4-S1-EC-OEA
ECTS : 2**HOURS**Cours : 0h
TD : 0h
TP : 28h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h**ASSESSMENT METHOD**

Lab: 100 %

TEACHING AIDS

Slides et Polycopié

TEACHING LANGUAGE

French

CONTACTM. VERCRAENE Samuel :
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M. MONTEIRO Thibaud :
thibaud.monteiro@insa-lyon.fr**AIMS**

This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

Provide the basis of operational research (concepts, methods, models) for industrial engineers.

Enable the students to master linear programming, integer linear programming, graph theory, to model and solve industrial and logistic problems, using optimization softwares.

CONTENT

The first half of the course alternates between 2-hour lectures and practical sessions, introducing the main concepts of exact optimization and key metaheuristics. It covers a review of linear programming and the simplex method, as well as complexity, integer linear programming, path-based heuristics, and population-based heuristics.

The second half of the course is dedicated to a project. In 2024-25, it focused on a two-tier urban logistics problem, where a truck tour supplies depots, and bicycles then collect goods from the depots to deliver them to customers' homes. A code implementing an algorithmic solution to the problem is provided. Students are then required to replace the solution components with more advanced exact or heuristic methods.

BIBLIOGRAPHY**PRE-REQUISITES**Probability and statistics
Algorithmics
Programming
Integer Linear programming

IDENTIFICATIONCODE : GI-4-S1-EC-DDD
ECTS : 2**HOURS**Cours : 0h
TD : 0h
TP : 28h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h**ASSESSMENT METHOD**Continuous control
Lab report and oral presentation.**TEACHING AIDS**

PDF ressources on moodle

TEACHING LANGUAGE

English

CONTACTMME ROBARDET Celine :
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emil.dumitrescu@insa-lyon.fr**AIMS**

This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

CONTENT

- Building a Data Warehouse from raw data
- Introduction to data modeling for machine learning
- Methodology for the application of machine learning
- Application: predictive maintenance, building a model for failure prediction
- Application: Construction and operation of a model for failure prediction.

BIBLIOGRAPHY**PRE-REQUISITES**Notions of relational databases
PYTHON programming
Concepts of Statistics

IDENTIFICATION

CODE : GI-4-S1-EC-INR
ECTS : 2

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 2h
Travail personnel : 26h
Total : 50h

ASSESSMENT METHOD

written report and defense

TEACHING AIDS

Depending on the subject

TEACHING LANGUAGE

French
English

CONTACT

MME BOTTA-GENOULAZ Valerie
:
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AIMS

This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

A6 Communicate a scientific analysis or approach with scenarios adapted to their speciality (level 1)

B4 Be creative, innovative, entrepreneurial (Level 1)

C13 Take into account technological and methodological innovation (level 1)

In addition, it requires the mobilization of the following skills:

B2 Working, learning, evolving autonomously

B3 Interact with others, work in teams

B7 Working in an International and Intercultural Context

C1 Observe, measure, analyze and interpret an activity or system from data

C14 Leading a project collectively: organization, communication, animation, group coordination

C16 Identify, analyze and control the risks inherent in a project

By allowing the student to work and be evaluated on the following knowledge:

- Devices and actors of research in France and the place of engineers PhD (A6, B4, C13)

By allowing the student to work and be evaluated on the following abilities :

- Developing a scientific approach of questioning on a research problem (A6, B4, C13)

CONTENT

Research project, in pairs, allowing the student to learn and experiment a research/innovation approach, on a subject proposed and supervised by a teacher-researcher, according to the following steps:

- appropriation of the subject
- formalization of the research problem and proposal of possible solutions
- development of solutions and experiments
- analysis of results, proposal of perspectives

BIBLIOGRAPHY

Depending on the subject

PRE-REQUISITES

Depending on the subject

IDENTIFICATIONCODE : GI-4-S1-EC-SDF
ECTS : 2**HOURS**Cours : 0h
TD : 16h
TP : 12h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h**ASSESSMENT METHOD****TEACHING AIDS****TEACHING LANGUAGE**

French

CONTACTM. DUMITRESCU Emil :
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eric.zamai@insa-lyon.fr**AIMS**

This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

Knowledge :

- Defining a rationale for dimensioning
- MTBF MTTF MTTR
- State graph, Markov chain
- Fault Tree, Reliability Diagram
- Decision support

Capacities :

- Apprehend the phenomena of malfunctioning
- Preliminary Risk Analysis
- Lifetime, Reliability, Availability, Maintainability of material and organisational structures
- Contract, commitment and penalties
- Negotiate solutions to increase fault tolerance
- Performance indicators
- Dimensioning

CONTENT

Dependability assesment methods

BIBLIOGRAPHYSûreté de Fonctionnement des systèmes matériels, A. Villemeur, Edt Eyrolles
Reliability Eng., Elsevier**PRE-REQUISITES**Stochastic process
Markov Chains
Probability

IDENTIFICATION

CODE : GI-4-S1-EC-ORD
ECTS : 2

HOURS

Cours : 0h
TD : 8h
TP : 14h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 24h
Travail personnel : 12h
Total : 40h

ASSESSMENT METHOD

IE (written test)
ES (situational assessment)

TEACHING AIDS

Handout for the lecture-tutorials part with exercises.
Project description.
Access to Opcenter APS software package.

TEACHING LANGUAGE

French
English

CONTACT

M. FONDREVELLE Julien :
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AIMS

This course belongs to teaching unit Pilotage des opérations industrielles (GI-4-S1-UE-POPI) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 3)
- A2 Operate a model of a real or virtual system (Level 3)
- A3 Implement an experimental approach (Level 2)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 3)
- C5 Managing a production system and react to malfunctions (Level 3)
- C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Production scheduling techniques and methods: single machine, parallel machines, flowshop, jobshop, hybrid configurations. (A1, A2, A3, C1, C2, C3, C5, C6)

To be able to :

- Characterizing production master data; (A1, A2, C1, C2, C6)
- Specifying a scheduling problem; (A1, A2, C1, C2, C3)
- Choosing or proposing a method to solve a production scheduling problem that is best suited to a given context, and applying it; (A2, A3, C2, C3, C5, C6)
- Analyzing the impacts of scheduling methods. (A1, A2, A3, C1, C3, C5, C6)

CONTENT

1- Lecture-tutorials part :

Introduction: Positioning scheduling within the overall production management process, characteristics of scheduling problems, problem classification.

Single-machine problems.

Parallel machine problems.

Flowshop problems.

Jobshop problems.

Hybrid problems.

2- Project: using a scheduling software package (Opcenter APS) to solve a case study

Study of production master data.

Analysis of software functionalities.

Consideration of additional constraints.

Software evaluation and critical analysis.

BIBLIOGRAPHY

BAGLIN G., BRUEL O., GARREAU A., GREIF M., Van DELF C., Management Industriel et Logistique, Economica, 1996.

GIARD V. Gestion de la production et des flux, 3e édition, Economica, 2003.

JAVEL G. Organisation et gestion de la production, Masson, 1997.

PINEDO M., Scheduling: Theory, Algorithms, and Systems, Prentice Hall, 2001.

RINNOOY KAN A.H.G., Machine scheduling problems: Classification, complexity and computations, Nijhoff, The Hague (1976).Hall, 2001

PRE-REQUISITES

Basics of Production Management (production organization, master data, production planning)

Basic knowledge about algorithms.

IDENTIFICATIONCODE : GI-4-S1-EC-PCS
ECTS : 3**HOURS**Cours : 0h
TD : 16h
TP : 24h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 42h
Travail personnel : 18h
Total : 60h**ASSESSMENT METHOD**IE/COP ES (évaluation en situation)
P/COP ES (évaluation en situation)
IE/SCADA/MES ES (évaluation en situation)**TEACHING AIDS**

- printed and online book
- complementary docs
- Moodle page
- oral testimony by an industrialist

TEACHING LANGUAGE

French

CONTACTM. ZAMAI Eric :
eric.zamai@insa-lyon.fr**AIMS**

This course belongs to teaching unit Management of industrial operations (GI-4-S1-UE-POPI) and contributes to the following skills :

- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
- C3 Evaluating, prototyping and simulating a system (Level 2)
- C4 Sizing the hardware and / or software of a system (Level 2)
- C5 Managing a production system and react to malfunctions (Level 2)
- C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 2)
- C13 Considering technological and methodological innovation (Level 2)
- C15 Managing a project based on a master plan (planning, budgeting, definition of monitoring indicators) (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- (C4, C6, C13)
- (C15)
- (C1, C2, C3)

To be able to :

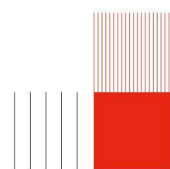
- (C1, C2, C3, C4, C5, C6? C13)

CONTENT

- Introduction to automation
- Automated systems of production (functions, uses)
- Conducting a project of automation (tools and methods for design and building)

BIBLIOGRAPHY**PRE-REQUISITES**

- Industrial Machine environment: sensors, actuators
- basic PLC programming
- System Engineering basics



IDENTIFICATION

CODE : GI-4-S2-EC-LEA
ECTS : 3

HOURS

Cours : 0h
TD : 20h
TP : 12h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 36h
Travail personnel : 24h
Total : 60h

ASSESSMENT METHOD

Assessment: participation (10%),
homework case study (30%),
practical work report (30%), poster
on a subject of your choice related
to the course (30%).
Plus 1 ECTS for exchange
students

TEACHING AIDS

Slides of the course, simulation
game, tutorial

TEACHING LANGUAGE

French
English

CONTACT

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AIMS

This course belongs to teaching unit Management and improvement of business performance (GI-4-S2-UE-GAPE) and contributes to the following skills :

- Appreciate lean as a way of management
- Use VSM to identify and eliminate the wastes,
- Develop the skills necessary to see the flow of a value stream and to create a future state map

CONTENT

- 2 sessions of 2 hours of lectures (CM):
 - Presentation of the history of Lean
 - Basic principles of Lean
 - 7 types of wastes, temple Toyota, Just In Time, Jidoka principles, standardization, management.
- 3 sessions of 2 hours of tutorials (TD):
 - Presentation of the Value Stream Mapping method,
 - Development of an action plan.
- A case study
 - 2 sessions of 4 hours of practice (TP): a simulation game of a production workshop of jacks (Synchro Lean Simulation on hardware FESTO).
 - 3 sessions of 2-hour seminars with industrial invited speakers: workstation organisation, problem solving and escalation, Lean Leadership

BIBLIOGRAPHY

- Learning to see. Mike Rother et John Shook. Avant propos: Jim Womack et Dan Jones .Traduction en Français : Line Perron. N° ISBN: 978-2-9529806-1-6
- Gold Mine. Freddy Ballé et Michael Ballé. N°ISBN: 0-9743225-6-3
- Le Modèle TOYOTA. Jeffrey Liker. Traduction en Français: Monique Sperry. Avant propos: Didier Leroy (président de Toyota Motor Manufacturing France. Préface: Michael Ballé et Godefroy Beauvallet. N° ISBN: 978-2-7440-7390-8
- Lexique Lean. Chet Marchwinski et John Shook. Traduction en Français: Institut Lean France. N° ISBN: 978-2-9529806-0-9
- Seeing the whole. James P. Womack and Daniel T. Jones. N° ISBN: 0-9667843-5-9
- Système Lean. Penser l'entreprise au plus juste 2ème édition. Jim Womack, Dan Jones. Préface: Michael Ballé et Godefroy Beauvallet. Traduction en français: Monique Sperry. N° ISBN: 978-2-7440-7391-5

PRE-REQUISITES

Production management
Problem solving methods

IDENTIFICATION

CODE : GI-4-S2-EC-BCG
ECTS : 2

HOURS

Cours : 8h
TD : 14h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 24h
Travail personnel : 16h
Total : 40h

ASSESSMENT METHOD

Continuous control + Final
examination (2h)

TEACHING AIDS

Textbook, exercices, case studies

TEACHING LANGUAGE

French

CONTACT

M. DUMITRESCU Emil :
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AIMS

This course belongs to teaching unit Management and improvement of business performance (GI-4-S2-UE-GAPE) and contributes to the following skills :

CONTENT

FINANCIAL FLOWS AND MANAGEMENT CONTROL

Financial mechanisms: cycles of a company, financial flows and formation of the needs of financial exploitation. These aspects are taught using a case study of a company creation. Indicators of activity, results and performance: Interpretation analysis, financial dashboard

BIBLIOGRAPHY

- * Gestion financière, Gérard Charreaux, ITEC.
- * Henri Amblard, Ph. Bernoux, G. Herreros, Y-F. Livian, Les nouvelles approches sociologiques des organisations, Seuil, 1996.
- * Philippe Bernoux, La sociologie des organisations, Le Seuil, coll. Points, 1985.
- * Reynald Bourque, Christian Thuderoz, Sociologie de la négociation, La Découverte, coll. Repères, 2002.
- * Philippe Cabin (coll.), Les Organisations. Etat des savoirs, Ed. Sciences Humaines, 1999.
- * Pierre Morin, Eric Delavallée, Le manager à l'écoute du sociologue, Ed. d'organisation, 2000.
- * Christian Thuderoz, Sociologie des entreprises, La Découverte, coll. Repères, 1997

PRE-REQUISITES

Basic knowledge of the operation of the industrial enterprise and its interactions with its environment.
The principles of financial accounting.

IDENTIFICATION

CODE : GI-4-S2-EC-ERP
ECTS : 3

HOURS

Cours : 0h
TD : 8h
TP : 24h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 32h
Travail personnel : 24h
Total : 60h

ASSESSMENT METHOD

Individual assessment
Summary of case studies,
screenshots

TEACHING AIDS

Case study student booklet
Other documents on Moodle
Software package SAP ERP Help
online

TEACHING LANGUAGE

French

CONTACT

MME BOTTA-GENOULAZ Valerie
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AIMS

This course belongs to teaching unit Information systems projects (GI-4-S2-UE-PRSI) and contributes to the following skills :

- Mastery of an Enterprise Resource Planning (ERP) system for production planning and management in a discrete production context
- Consciousness raising to business process integration issues

CONTENT

Objective: Master the challenges of an integrated management system (ERP) to support operational and decision-making processes (production, sales, planning processes) by conducting case studies on a leading ERP system.

- Introduction and discovery of SAP S/4HANA (structure, concepts, applications)
- Master data for production planning and sales processes: customer, material, BOM, routing, workcenter, cost center,...
- Case Study : Implementation of several cross-functional business processes : production planning and execution scenarii (make to stock, assemble to order), and a sales management scenario
- discovery of the FIORI environment on an external supply scenario

BIBLIOGRAPHY

Tomas J.L. (2000) : ERP et progiciels intégrés. Dunod 2ème édition.
Reix R. (2002) : Systèmes d'information et management des organisations. Vuibert 2ème édition
Dunleavy J., Hurley J.R., Gibson A., Norris G., Wrigt I. (1999) : SAP/R3 Le guide du décideur. Editions Eyrolles

PRE-REQUISITES

Industrial Management, Introduction to Supply Chain, Business Information Systems
Basics and methods of production management and planning;
Forecasting methods, Sales & Operation Planning
Knowledge of ERP systems: content, challenges and ERP project management

IDENTIFICATIONCODE : GI-4-S2-EC-PSM
ECTS : 3**HOURS**Cours : 0h
TD : 0h
TP : 32h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 32h
Travail personnel : 24h
Total : 60h**ASSESSMENT METHOD**

For a group, evaluation on the report and the application and for each student, evaluation on the individual work in the group.

TEACHING AIDS

Lecture notes

TEACHING LANGUAGE

French

CONTACTM. ESCUDERO Cedric :
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This course belongs to teaching unit Information systems projects (GI-4-S2-UE-PRSI) and contributes to the following skills :

- Specify the needs for control systems and MES;
- Use an analysis and design method for a control and information system;
- Implement automation and MES systems;
- Analyze the results and improve physical and information systems.

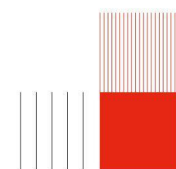
CONTENT

Application of the course on a real system, implemented with the AIP (local shared industrial workshop). Process and control analysis, study of the existing information system, design and implementation of MES new functions.

FR - practical work in french

BIBLIOGRAPHY

Heiko Meyer, Franz Fuchs and Klaus Thiel, Manufacturing Execution Systems (MES): Optimal Design, Planning, and Deployment: Optimal Design, Planning, and Deployment, McGraw-Hill Education, 2009, 273 p., ISBN 0071623833

PRE-REQUISITESAutomated Manufacturing Systems: role, architecture, components (sensors, activators), continuous et batch process control;
Information systems: role, architecture, design.
Production Management: role, methods.

IDENTIFICATION

CODE : CDS-4-S2-EC-EPS
ECTS : 1

HOURS

Cours : 0h
TD : 20h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 0h
Total : 20h

ASSESSMENT METHOD

Assessment in Physical Education concerns the teaching of Sports and Artistic Physical Activities (APSA), and will take the form of continuous assessment with half-yearly marking.

The mark depends on the degree of acquisition of the skills expected in each of the activities, and the progress made over all the sessions in the cycle. The mark takes into account :

Individual and/or team performance
mastery of execution
Progress in the sports project
Responsibility and autonomy

TEACHING AIDS

All physical, sporting, artistic and competitive activities

TEACHING LANGUAGE

French

CONTACT

Mme JAUSSAUD Marie :
marie.jaussaud@insa-lyon.fr

AIMS

This EC is part of the Teaching Unit: SHS and contributes to the development of the School's transversal competences

1* Auto-evaluating one's own performance

Knowledges :

- Fundamentals, principles of action and terminology of sports activities
- Criteria for observation, achievement and success.

Abilities :

- Assess your level of practice
- Build up a warm-up
- Set goals for progress
- Manage physical and mental potential

2* Work, learn and develop independently

Knowledge :

- PSAA rules
- Observation criteria
- Principles of warm-up and cool-down

Abilities :

- Mobilise resources
- Analyse, observe, question
- Take on different roles (referee, choreographer)

3* Interact with others, work as part of a team

Knowledges :

- Roles and functions in each sports activity

Abilities :

- Communicate appropriately: verbal, non-verbal and postural communication.
- Integrate into a group
- Take part in and develop a group project
- Take the initiative
- Be a good listener

4* Be creative, innovative and enterprising

Knowledge :

- Artistic disciplines

Abilities :

- Draw on knowledge and resources from different artistic fields to produce an original work.
- Mobilise the imagination and sensibility and make them visible through dance movement
- Access the symbolism of the body

5* Act responsibly in a complex world

Knowledge :

- Safety and operating rules

Abilities :

- Identify uncertainties and risks and act to reduce them
- Integrate a responsible dimension into their actions
- Show respect and fair play in a power struggle

6* Working in an international context

Knowledge :

- Socio-cultural differences

Abilities :

- Integrate cultural diversity into group work
- Act with respect for self and others

CONTENT

Physical Education and Sport lessons are organised around traditional Physical Education lessons, or advanced lessons, or appropriate practices (EPSA), or competitive practices within the framework of the Section Sportive Haut Niveau.

1. Physical Education lessons :

Students choose one or two physical and sporting activities per year from among the activities offered by the sports centre (individual, group, dual).

2. Appropriate Physical Education lessons: For all students who are exempt from physical activity for at least 2 months:

Swimming, Body-building, Nordic Walking, Somatic Exercise, Sophrology, Wheelchair Basketball, Pilates, Table Tennis, etc.

3. Advanced Physical Education courses :

Specialisation in a sporting activity, University training and competitions

4. SSHN (High-Level Athlete section)

BIBLIOGRAPHY

PRE-REQUISITES

- EPS: none
 - Appropriate Physical Education: subject to medical advice
 - Advanced courses and competitive practice: previous practice required subject to specific selection according to each activity
 - SHN: ministerial list
- Levels 1 and 2: Physical Education, Appropriate physical education
Level 3: Advanced courses and competitive practice, SHN

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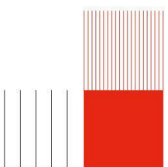
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membre de



IDENTIFICATIONCODE : GI-4-S2-EC-STI
ECTS : 14**HOURS**Cours : 0h
TD : 1h
TP : 0h
Projet : 229h
Evaluation : 0h
Face à face pédagogique : 1h
Travail personnel : 0h
Total : 230h**ASSESSMENT METHOD****TEACHING AIDS****TEACHING LANGUAGE**

French

CONTACTM. BERNARD-TREMOLET
Stéphane :
stephane.bernard@insa-lyon.fr
M. MONTEIRO Thibaud :
thibaud.monteiro@insa-lyon.fr**AIMS****CONTENT**

Developing observation and analysis skills.

Integrating technical and socio-economical aspects of a project / process.

Improving communication skills.

The industrial internship is an opportunity to experience the role of an engineer in a real industrial environment, where the student is likely to work in the future. It leads not only to technical achievements, but also to a detailed observation of the company, with its history, organization, activities, objectives, social dynamics ...

BIBLIOGRAPHY**PRE-REQUISITES**

IDENTIFICATION

CODE : GI-4-S2-EC-ASO-HU
ECTS : 1

HOURS

Cours : 0h
TD : 12h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 12h
Travail personnel : 8h
Total : 20h

ASSESSMENT METHOD

1 oral/written (collective)
assessment on a case study.

TEACHING AIDS

Textbook, exercices, case studies,
online ressources (website
"sociologos")

TEACHING LANGUAGE

French

CONTACT

M. MARIANO José-Pedro :
pedro.mariano@insa-lyon.fr

AIMS

CONTENT

STRATEGIC ANALYSIS OF AN ORGANISATION

Observation and internal strategic analysis, Integration of the technical, economical and socio-organizational aspects

Personal integration reinforced by contacting, necessary for targeted location

Identification of human and social aspects for the future engineering missions

BIBLIOGRAPHY

Henri AMBLARD, Ph. BERNOUX, G. HERREROS, Y-F. LIVIAN, Les nouvelles approches sociologiques des organisations, Seuil, 1996.

Howard BECKER, Les ficelles du métier, La Découverte, 2002.

Philippe BERNOUX, La sociologie des organisations, Seuil, coll. Points, 1985. Sociologie du changement, Seuil, 2004.

R. Bourque, Ch. Thuderoz, Sociologie de la négociation, La Découverte, coll° Repères, Paris 2002.

Philippe CABIN (coll.), Les Organisations. Etat des savoirs, Ed. Sciences Humaines, 1999.

Pierre MORIN, Eric DELAVALLEE, Le manager à l'écoute du sociologue, Ed. d'Organisation, 2000.

R. Reitter, F. Chevalier, H. Laroche, C. Mendoza, P. Pulicani, Cultures d'entreprises, Vuibert, 1991.

M. Vilette, Guide du stage en entreprise, La Découverte, 2006.

Christian THUDEROZ, Histoire et sociologie du management, PPUR, 2006. Clefs de sociologie pour ingénieur(e)s, PPUR, 2011. Sociologie des entreprises, La Découverte, coll. Repères, 1997, rééd° 2010.

PRE-REQUISITES

Fluent in French (speaking and writing)

IDENTIFICATIONCODE : GI-4-S2-EC-PCO
ECTS : 3**HOURS**Cours : 0h
TD : 0.016666666666666666h
TP : 0h
Projet : 0h
Evaluation : 4h
Face à face 4.016666666666667h
pédagogique :
Travail personnel : 48h
Total : 52.01666666666666h**ASSESSMENT METHOD**

Continuous assessment:
evaluation of processes:
- Final design
- Final realization
- Closing
and associated deliverables.
Final collective project defense.
Individual assessment of project
team members.

TEACHING AIDS

Numerous resources available on
the course's Moodle page
(presentation materials, process
notes, etc.).

TEACHING LANGUAGEFrench
English**CONTACT**M. FONDREVELLE Julien :
julien.fondreville@insa-lyon.fr**AIMS**

This course belongs to teaching unit Collectiv project (GI-4-S1-UE-PROJ) and contributes to the following skills :Project management methodes and tools (following a project, steeps of a project, risk management, AMDEC, formalization of the customers objectives , redaction of specifications, redaction of plan realisation.
Place the project and its stakes in the organization ; organize and manage a project and work in team; meetings management ; manage relevant information; communicate, negotiate and defend a project

CONTENT

Steps covered during this second part of the project: following delivery of the prototype and the preliminary acceptance phase, corrections and improvements to the product (final design, production and acceptance). Project assessment and feedback.

BIBLIOGRAPHY**PRE-REQUISITES**

This course is a direct continuation of the group project started in 4GI-S1: GI-4-S1-EC-PCO

IDENTIFICATION

CODE : GI-4-S2-EC-SIE
ECTS : 1

HOURS

Cours : 0h
TD : 12h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 12h
Travail personnel : 0h
Total : 12h

ASSESSMENT METHOD

Situational assessment; Group score

TEACHING AIDS

Project Subject, Company Data
Readings to be done before
certain sessions (theoretical
complements)
Methodological guides presented
during the sessions

TEACHING LANGUAGE

French

CONTACT

MME GZARA Lilia :
lilia.gzara@insa-lyon.fr

AIMS

This course belongs to the Management and Exploitation of Company Data (GI-4-S2-UE-RNIV) teaching unit and contributes to the following skills:

By allowing the student to work and be evaluated on the following knowledge:

- BPMN modeling
- Process Value Analysis
- Evaluation of the performance of a process
- Process mining
- The various industrial information systems

By allowing the student to work and be evaluated on the following abilities:

- Model a complex business process
- Diagnose a business process in terms of inefficiencies and non-value-added tasks/flows
- Identify and evaluate business process improvement solutions
- Specify the information system supporting a business process

CONTENT

This course is conducted as a team project and is based on an industrial case. The objective of the project is to carry out a preliminary study for the modernization of a manufacturing company's information system. This preliminary study includes several phases: (1) Analysis of the existing situation (AS-IS) through the modeling of current business processes, the diagnosis of these processes and the identification of improvement solutions; (2) Design of an improvement solution (TO-BE) through the modeling of new processes and the specification of a software/hardware solution to support the new organization (without however developing it).

BIBLIOGRAPHY

1. Marlon Dumas, Marcello La Rosa, Jan Mendling, Hajo A. Reijers. "Fundamentals of Business Process Management". Editions Springer, ISBN :9783662565094 / 3662565099, 2018, 527 pages
2. Heru Susanto, Fang-Yie Leu, Chin Kang Chen. "Business Process Reengineering: An ICT Approach". Editions Apple Academic Press, ISBN :9780429949302 / 0429949308, 2019, 248 pages
3. W.M.P. van der Aalst. "Process Mining: Data Science in Action". Editions Springer-Verlag, 2016. <http://www.springer.com/978-3-662-49850-7>

PRE-REQUISITES

- Process Modeling (BPMN)
- Object-oriented modeling (UML)

IDENTIFICATIONCODE : GI-4-S2-EC-PCS
ECTS : 1**HOURS**Cours : 0h
TD : 12h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 12h
Travail personnel : 0h
Total : 12h**ASSESSMENT METHOD**

For a group, evaluation on the report and the application and for each student, evaluation on the individual work in the group.

TEACHING AIDS

Lecture notes

TEACHING LANGUAGE

French

CONTACTM. ESCUDERO Cedric :
cedric.escudero@insa-lyon.fr**AIMS**

This course belongs to teaching unit Refresher (GI-4-S2-UE-RNIV) and contributes to the following skills :

- Specify the needs for control systems and MES;
- Use an analysis and design method for a control and information system;
- Implement automation and MES systems;
- Analyze the results and improve physical and information systems.

CONTENT

Application of the course on a real system, implemented with the AIP (local shared industrial workshop). Process and control analysis, study of the existing information system, design and implementation of MES new functions.

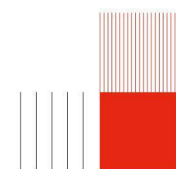
FR - practical work in french

BIBLIOGRAPHY

Heiko Meyer, Franz Fuchs and Klaus Thiel, Manufacturing Execution Systems (MES): Optimal Design, Planning, and Deployment: Optimal Design, Planning, and Deployment, McGraw-Hill Education, 2009, 273 p., ISBN 0071623833

PRE-REQUISITES

Automated Manufacturing Systems: role, architecture, components (sensors, activators), continuous et batch process control;
Information systems: role, architecture, design.
Production Management: role, methods.



IDENTIFICATION

CODE : GI-4-S2-EC-ISD
ECTS : 1

HOURS

Cours : 0h
TD : 12h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 12h
Travail personnel : 0h
Total : 12h

ASSESSMENT METHOD

TP/ADD TPS (Travaux pratiques surveillés)

Plus 1 ECTS for exchange students

TEACHING AIDS

TEACHING LANGUAGE

English

CONTACT

MME ROBARDET Celine :
celine.robardet@insa-lyon.fr

AIMS

This course belongs to teaching unit Refresher (GI-4-S2-UE-RNIV) and contributes to the following skills :

A3 Implement an experimental approach (Level 2)
A5 Process data (Level 3)
A6 Report on an analysis or on a scientific approach (Level 3)
C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)
C4 Sizing the hardware and / or software of a system (Level 2)
C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 1)
B2 work, learn, progress autonomously (Level 2)
B3 interact with people and work in a team (Level 3)

By allowing the engineering students to work and be evaluated on the following knowledge :

x
To be able to :
- (C4)
- (C1)
- (A3, A5, A6, C11, B2, B3)

CONTENT

DATA ANALYSIS

1.Data

Characteristics: origin, quality, coding
Preparation: restoration, filtering, synchronization, construction of classes
Dependence: distances, ultra metric, correlation

2.Tools :

Descriptive methods: principal components analysis, correspondences, preferences
Explanatory methods: regression, segmentation, canonical analysis, discriminated analysis

Structural methods: ultra metric and hierarchical classification, dynamic swarms.

3.Analysis :

Analysis methodologies
Interpretation rules
Case studies

BIBLIOGRAPHY

****Cours Analyse de données : ****

1. E.Diday, J.Lemaire, J.Pouget, F.Testu, Eléments d'analyse de données, Dunod 1982
2. G.Saporta, Probabilités, Analyse de données et Statistiques, Technip 1990
3. Jean-Pierre , Analyse interactive des données (ACP, AFC) avec Excel 2000 : théorie et pratique / Georgin. Presses universitaires de Rennes , 2002
4. Alain Morineau et Yves-Marie Chatelin , L'analyse statistique des données [Texte imprimé] : apprendre, comprendre et réaliser avec Excel, Ellipses , 2005 , 407 p

****Cours Entrepôts de données : ****

1. Entrepôts de données. Guide pratique de modélisation dimensionnelle, 2ème édition, R. Kimball, M. Ross
2. Piloter l'entreprise grâce au data Warehouse, J.-M. Franco, S de Lignerolles

PRE-REQUISITES

Statistics, data base

IDENTIFICATION

CODE : GI-0-S2-EC-VAC
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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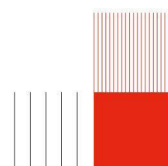
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IDENTIFICATION

CODE : GI-0-S2-EC-CESURE

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

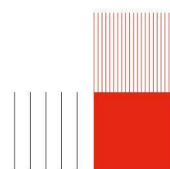
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IDENTIFICATIONCODE : GI-0-S2-EC-LABECH15
ECTS : 15**HOURS**Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 300h
Total : 300h**ASSESSMENT METHOD****TEACHING AIDS****TEACHING LANGUAGE**French
English**CONTACT**M. DUMITRESCU Emil :
emil.dumitrescu@insa-lyon.fr**AIMS****CONTENT**

The reception and supervision of students takes place in a scientific laboratory. The student works on a well-defined scientific problem, following a project approach, which requires managing objectives, planning, monitoring and deliverables. The host laboratories are DISP, AMPERE, LIRIS and LAMCOS. Available on application and selection, and depending on subject availability.

BIBLIOGRAPHY**PRE-REQUISITES**

IDENTIFICATION

CODE : GI-0-S2-EC-STAGECH30

ECTS : 30

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h

pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French
English

CONTACT

M. DUMITRESCU Emil :
emil.dumitrescu@insa-lyon.fr

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

IDENTIFICATION

CODE : GI-0-S2-EC-STAGECH15

ECTS : 15

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h

pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French
English

CONTACT

M. Dumitrescu Emil :
emil.dumitrescu@insa-lyon.fr

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

IDENTIFICATION

CODE : GI-0-S2-EC-LABECH30

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 600h

Total : 600h

ASSESSMENT METHOD**AIMS****CONTENT**

The reception and supervision of students takes place in a scientific laboratory. The student works on a well-defined scientific problem, following a project approach, which requires managing objectives, planning, monitoring and deliverables. The host laboratories are DISP and AMPERE. Available on application and selection, and depending on subject availability.

BIBLIOGRAPHY**PRE-REQUISITES****TEACHING AIDS****TEACHING LANGUAGE**French
English**CONTACT**M. DUMITRESCU Emil :
emil.dumitrescu@insa-lyon.fr

IDENTIFICATION

CODE : GI-5-S1-EC-CYB
ECTS : 2

HOURS

Cours : 0h
TD : 6h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

M. ZAMAI Eric :
eric.zamai@insa-lyon.fr

AIMS

CONTENT

Objectives: Understand the challenges related to cybersecurity in industrial systems and the specific features of this field. Understand the basic elements for identifying the vulnerabilities of these systems, as well as the recommendations and methodology for strengthening the cybersecurity level of existing systems and the key points for designing new systems. Understand the cybersecurity stakeholders in companies and support the implementation of the approach. Understand how to communicate with IT and OT to unite teams around these new challenges.

Main concepts of cybersecurity for information systems in general:
Definitions of cybersecurity and main concepts
Challenges, current status, history;
Classical attacks (MITM, spoofing, social engineering, denial of service, session hijacking, DDOS, APT, worms);
Specific attacks (malicious organizations): PLC code hijacking, takeover of the SCADA/MES information system, takeover of PLCs and industrial equipment
Duality of operational safety and cybersecurity; Examples of attacks (study, practical work on platform attacks);
Classic vulnerabilities and attack vectors;
Main principles for implementing a cybersecurity project (risk analysis, DEP, PSSI, etc.)
Best practices and recommendations;
Overview of norms and standards (2700X, product certification, etc.);
In France, the LPM;
Introduction to cryptography (symmetric/asymmetric encryption, hash functions, signatures, etc.).
IDS modeling (SED, ECC link)
Configuring industrial firewalls and understanding bad ideas.
The industrial system cybersecurity project in a real environment (transfer room practical work); Vulnerability analysis
Security strengthening
In the shoes of a cyber attacker

BIBLIOGRAPHY

PRE-REQUISITES

Basics of automation and different types of industrial systems;
Composition of an industrial system (ECC);
PLC programming languages (ECC, COP, TP API);
Industrial protocols and fieldbuses (ECC);
Classic network architectures (ECC, COP);
Operational safety (CPS/SDF to be extended with SIL standards and consolidate FMEA + EBIOS);
Overview of norms and standards (ECC, API labs, COP, SCADA/MES)
Computer Basics:
Concepts of classic hardware architectures (microprocessor, memory, bus, peripherals, firmware)
Operating systems (main systems used)
Concepts of software architectures, databases
Network Basics:
Main concepts and vocabulary used in this field:
o 7-layer OSI model,
o Data units (frame, packet, segment, etc.),
o MAC address, IP address, port concept (UDP/TCP), o Network discovery and diagnostics (ARP, ICMP),
o Connected (TCP) and offline (UDP) modes.
Services that may be present in industrial equipment (PLC, MES, ERP, HMI, VFD, wireless terminal, etc.)
o Web (HTTP) and secure Web (HTTPS),
o Equipment management (SNMP, SYSLOG, etc.). VPN using a VPN gateway, router, gateway, firewall.
Information systems fundamentals
Data warehouses and data lakes

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IDENTIFICATIONCODE : GI-5-S1-EC-IFU
ECTS : 2**HOURS**Cours : 0h
TD : 0h
TP : 20h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h**ASSESSMENT METHOD**

Report and oral presentation.

TEACHING AIDS

To be determined.

TEACHING LANGUAGE

French

CONTACTM. HADJ HAMOU Khaled :
khaled.hadj-hamou@insa-lyon.frM. MONCLA Ludovic :
ludovic.moncla@insa-lyon.fr**AIMS**

"This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

A1 Analyze a system (real or virtual) or a problem (Level 3)

A3 Implement an experimental approach (Level 3)

A5 Process data (Level 3)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)

C5 Managing a production system and react to malfunctions (Level 3)

C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 1)

C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 1)

C13 Considering technological and methodological innovation (Level 3)

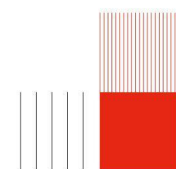
B3 interact with people and work in a team (Level 3)

CONTENT**BIBLIOGRAPHY****PRE-REQUISITES**

Python (Jupyter) programming.

Pulp language (CPLEX/GUROBI solver mathematical programming MIP language).

Courses: GI-4-S1-EC-ISD and GI-5-S1-EC-OTL



IDENTIFICATION

CODE : GI-5-S1-EC-KNM
ECTS : 2

HOURS

Cours : 0h
TD : 18h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

M. LOISEAU Mathieu :
mathieu.loiseau@insa-lyon.fr

AIMS

This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

C2 Modeling and designing an information, decision and production system, of goods and services (Level 1)
C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 1)
C10 Developing and implementing an action plan as part of a quality and continuous improvement approach (Level 1)
C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 1)
C12 Changing organizations to meet new constraints or opportunities (Level 1)
C18 Identifying the critical skills and knowledge of an organization and implement tools and methods to sustain the (Level 1)
C19 Conducting a socio-organizational analysis to better understand the effects of change and to adapt strategies (Level 1)
B5 act responsibly in a complex world (Level 1)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Industrial knowledge-related assets
 - Information, knowledge, and expertise
 - Knowledge management methods in industry, collaborative knowledge management tools
 - Digital knowledge management tools
 - Knowledge management strategies and continuous improvement
 - Technological, human, and organizational knowledge and structure
- To be able to :
- Distinguishing the type of knowledge created by the use of information and Communication
 - Identify industrial capital (human, technological, organizational, informational)
 - Assess the limitations of knowledge management systems

CONTENT

BIBLIOGRAPHY

- Ackoff, Russell Lincoln. 1989. « From Data to Wisdom ». Journal of Applied Systems Analysis 16: 3–9.
- Dejoux, Cécile. 2013. Gestion des compétences et GPEC. 2e édition. Dunod.
- Giezendanner, François Daniel. 2006. « Philosophie et caractéristiques des wikis ». Blog. Ici et là. 2 juin 2006. <http://icietla-ge.ch/voir/spip.php?article124>.
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PRE-REQUISITES

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membre de



IDENTIFICATION

CODE : GI-5-S1-EC-OTL
ECTS : 3

HOURS

Cours :	0h
TD :	24h
TP :	0h
Projet :	0h
Evaluation :	2h
Face à face pédagogique :	26h
Travail personnel :	30h
Total :	60h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

M. HADJ HAMOU Khaled :
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AIMS

This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

A2 Operate a model of a real or virtual system (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)

C5 Managing a production system and react to malfunctions (Level 2)

C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 1)

C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 1)

C13 Considering technological and methodological innovation (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

To be able to :

CONTENT

The engineering student will work and be assessed on the following knowledge and skills :

- (10h) Know the "classical" transportation problems : Traveling Salesman Problem (TSP), Vehicle Routing Problem (VRP), etc.

- (2h) Know the basic methods and tools to design logistic networks.

- (8h) Know the principles and tool for distributed decision-making: game theory, multi-agent systems and agent-based simulation coupled with flow simulation (for instance with Anylogic)

- (4h) Know the tools (especially decomposition methods) to perform joint optimisation.

- Be able to formulate and solve a supply chain optimisation problem, from the viewpoint of transportation and flow management.

BIBLIOGRAPHY

Ilya Grigoryev. AnyLogic 7 in Three Days : A Quick Course in Simulation Modeling. 3rd edition.

Jacques Teghem. Recherche opérationnelle. Tome 1, Tome 1, OCLC : 819196866. Paris : Ellipses, 2012. isbn : 978-2-7298-7509-1.

Paolo Toth et Daniele Vigo, éd.s. Vehicle routing : problems, methods, and applications. Second edition. MOS-SIAM series on optimization. Philadelphia : Society for Industrial et Applied Mathematics : Mathematical Optimization Society, 2014. 463 p. isbn : 978-1-61197-358-7

Michael Wooldridge. An Introduction to Multi-Agent Systems. 2d edition. New York, NY, USA: John Wiley & Sons, Inc., 2009.

Murat Yildizoglu. Introduction à la Théorie des Jeux. Dunod, Paris (France), 2003.

PRE-REQUISITES

Inventory management : EOQ, basic replenishment methods

Flow management

Production management : S&OP, MPS & MRP

Scheduling: models, standard scheduling notation, heuristic solving methods for scheduling problems

Combinatorial optimisation: graph theory

Logistic

IDENTIFICATION

CODE : GI-5-S1-EC-DSC
ECTS : 3

HOURS

Cours : 0h
TD : 24h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 26h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

Final exam 1h50.

TEACHING AIDS

Slides (lectures) + Python Jupyter notebooks (practicals)

TEACHING LANGUAGE

English

CONTACT

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AIMS

This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

A5 Process data (Level 2)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 1)

C13 Considering technological and methodological innovation (Level 2)

CONTENT

The Data Science course is structured into two core parts:

1. Introduction to Artificial Intelligence – This section focuses on Deep Learning and Reinforcement Learning, covering key concepts such as deep neural networks (RNN, LSTM, Transformers), activation functions, optimization techniques, and loss functions. It also explores real-world applications, including self-driving car simulations and AI-driven decision-making.

2. Introduction to Natural Language Processing (NLP) – This section covers the foundations and applications of Large Language Models (LLMs). Topics range from the history of NLP techniques to modern industrial applications in Customer Support, Legal Tech, and Industry 4.0. The course covers essential techniques such as text classification, word/document representation, and Retrieval-Augmented Generation (RAG).

The course includes hands-on labs using Python and Jupyter notebooks, providing practical experience in AI and NLP model development.

BIBLIOGRAPHY

Charu C. Aggarwal. (2023). Neural networks and deep learning. A Textbook. Switzerland: Springer International Publishing.

Nithin Buduma, Nikhil Buduma, Joe Papa (2022) Fundamentals of Deep Learning, 2nd Edition.

Laura Graesser and Wah Loon Keng. (2019) Foundations of Deep Reinforcement Learning Theory and Practice in Python. Addison-Wesley Professional.

Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition.

PRE-REQUISITES

Python programming

Probability theory

Introduction to Data Science

IDENTIFICATION

CODE : GI-5-S1-EC-ADM
ECTS : 2

HOURS

Cours : 0h
TD : 6h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

continuous assessment

TEACHING AIDS

Lecture book and exercises

TEACHING LANGUAGE

French

CONTACT

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M. VERCRAENE Samuel :
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AIMS

The Multicriteria Decision Aid course covers methods and tools that help a decision-maker make a complex decision involving multiple criteria, often contradictory or in conflict.

Three approaches will be covered in this course:

- The AHP method (Analytic Hierarchy Process);
- The TOPSIS method (Technique for Order Preference by Similarity to Ideal Solution);
- Multicriteria optimization.

CONTENT

TD1 2h introduction ADM + AHP
TP1 4h TP AHP CC1
TD2 2h OPTIM + TOPSIS
TP2 4h TP TOPSIS CC2
TD3 2h TD Lexico + Pareto
TP3 4h TP Pareto CC3

BIBLIOGRAPHY

Bouyssou D. et Roy B. : Aide multicritère à la décision : méthode et cas, Economica, 1993.
Carlier J. et Chrétienne P. : Problèmes d'ordonnancement : modélisation / complexité / algorithmes, Masson, 1988.
Garfinkel R.S. et Nemhauser : Integer programming, Wiley Interscience, 1972.
Roy B. : Méthodologie multicritère d'aide à la décision, Economica, 1985.

PRE-REQUISITES

Monocriterion decision support (Linear Programming), basic and advanced production management

IDENTIFICATIONCODE : GI-5-S1-EC-APF
ECTS : 3**HOURS**Cours : 0h
TD : 24h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 26h
Travail personnel : 30h
Total : 60h**ASSESSMENT METHOD**Continuous evaluation + final exam
Exam in English possible / Data
bilingual given**TEACHING AIDS**No support in English : Work and
project are accepted in English
Verbal exchange possible in
English**TEACHING LANGUAGE**

French

CONTACTM. BERNARD-TREMOLET
Stéphane :
stephane.bernard@insa-lyon.fr**AIMS**Obtain a clear understanding about the ideal cycle for sourcing, described in a step by
step method.
Apply management project's tools to sourcing
Put in place a supplier's monitoring
Be able to communicate, reproduce and transfert those competencies.**CONTENT**

N/A

BIBLIOGRAPHY

N/A

PRE-REQUISITES

IDENTIFICATION

CODE : GI-5-S1-EC-MTC
ECTS : 2

HOURS

Cours : 0h
TD : 6h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD**TEACHING AIDS**

Course and tutorials handouts
Softwares Carl Source (CMMS),
Minitab (reliability) and Anylogic
(simulation)

TEACHING LANGUAGE

French

CONTACT

M. MOYAUX Thierry :
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AIMS

This course belongs to teaching unit Excellence opérationnelle (GI-5-S1-UE-EXOP) and contributes to the following skills :

A1 Analyze a system (real or virtual) or a problem (Level 2)

A2 Operate a model of a real or virtual system (Level 2)

A3 Implement an experimental approach (Level 1)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 1)

C3 Evaluating, prototyping and simulating a system (Level 1)

C5 Managing a production system and react to malfunctions (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- CMMS Functions

- Maintenance Processes

- Maintenance Optimization

- Maintenance Tools: Manufacturer Documentation, Reliability, etc.

To be able to :

- Understanding the challenges of purchasing a CMMS (C2, C5)

- Defining a maintenance policy (A1, A2, A3, C1, C3, C5)

CONTENT

Computerized Maintenance Management System

FR - lecture/tutorial/practical work in french

* Overview of the functions of a CMMS

* Maintenance process

* Optimisation of maintenance

* Maintenance tools (manufacturer's documentation, reliability, etc.)

BIBLIOGRAPHY**PRE-REQUISITES**

IDENTIFICATION

CODE : GI-5-S1-EC-EVP
ECTS : 2

HOURS

Cours : 0h
TD : 16h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 16h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

Individual Assessment
Collective case study

TEACHING AIDS

Course and TD support
Exercices + Case studies

TEACHING LANGUAGE

English

CONTACT

Mme TRILLING Lorraine :
lorraine.trilling@insa-lyon.fr

AIMS

At the end of this teaching module, students should be able to:

- Define, characterise and formalise the needs of a decision-maker (5 and H)
- Propose a clear and legible format that is easy to understand, straightforward and unique.
- Take into account the maintainability and robustness of indicators when constructing a dashboard.
- Analyse indicator results, draw up a diagnosis and identify levers for action.
- Identify statistical biases and minimise them when proposing solutions.
- Demonstrate critical thinking when assessing results
- Demonstrate objectivity and neutrality in the choice of indicator and its presentation.

CONTENT

Teaching programme:

lecture/tutorial, 4 x 2h sessions: theoretical input on performance evaluation, with examples and exercises

Project, 4 hours supervised + 4 hours independent + 4 hours feedback: practical case study

BIBLIOGRAPHY

Setiawan, Indra & Purba, Humiras. (2020). A Systematic Literature Review of Key Performance Indicators (KPIs) Implementation. Journal of Industrial Engineering and Management. 1. 200-208. 10.7777/jiemar.v1i3.79.

Berrah Lamia, « Indicateur de performance : concepts et application », Edition Eyrolles, 2002

PRE-REQUISITES

Introduction to Supply Chain



IDENTIFICATION

CODE : GI-5-S1-EC-IFD
ECTS : 3

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 28h
Travail personnel : 30h
Total : 58h

ASSESSMENT METHOD

TEACHING AIDS

James A. Tompkins, 2010, Facilities Planning, John Wiley & Sons Inc
Mikell P. Groover, 2016, Work Systems: The Methods, Measurement & Management of Work, PEARSON edition

TEACHING LANGUAGE

English

CONTACT

M. BABOLI Armand :
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AIMS

CONTENT

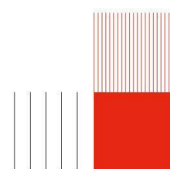
- 1- Introduction to facility design
 - a. Relationship between the product, process and scheduling design and the facility design
 - b. Flow system (Flow shop, Job Shop, cellular manufacturing, Fixed-position)
 - c. Process modeling and material flow analysis (Process Flowchart, IDEF0)
 - d. Activity relationship
- 2- Requirements of Manufacturing Facility (plant) design
 - a. Motion and Time study
 - b. Capacity determination (machine, human resource)
 - c. Performance measurement
- 3- Material Handling system, Storage and Warehouse system design
 - a. Type of material handling (receiving and shipping, dock equipment, moving equipment)
 - b. Type of Storage and Warehousing (Picking cart, Picking truck, vertical Warehouse, Automated Warehouse)
 - c. Transfer Line (Straight Line Flow, Circular Flow, U-Flow, Spine-Flow)
 - d. Equipment selection and Capacity determination
- 4- Cellular Manufacturing System (CMS) design
 - a. Clustering approach (Rank ordering, clustering, similarity coefficient, mathematical programming)
 - b. Designing and planning issues of CMS (capacity, material handling, layout)
- 5- Facility Layout
 - a. Basic layout types
 - b. Multi-floor facility layout,
 - c. Service layout
 - d. Flow shop and line balancing
 - e. Job shop and department location problem
 - f. Development alternatives
 - g. Mathematical modeling and Algorithms
- 6- Flexible Manufacturing System (FMS) design and optimization: (types of flexibilities, flexibilities classification, flexibility modeling and measurement, decision-making and optimization of the level of flexibility)
- 7- Industry 4.0 (Cyber Physical system, IIoT, Data Analytics and prescription of decision)
- 8- Configuration and experimentation of a flexible manufacturing system: Practical work on the Smart Factory platform (cyber-physical system).

BIBLIOGRAPHY

S. Heragu, 2008, Facilities Design, CRC Press Inc; Third Edition,
Ralph M. Barnes, Motion and Time Study: Design and Measurement of Work, 1980
Alberto Garcia-Diaz & J. MacGregor Smith, 2024, Facilities Planning and Design, Springer

PRE-REQUISITES

Mathematical modeling and optimization, Statistic and Probability, Linear programming, Inventory management, Lean manufacturing



IDENTIFICATION

CODE : GI-5-S1-EC-ETH-HU
ECTS : 2

HOURS

Cours : 0h
TD : 20h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

Press review (collective oral presentation)
Ethical analysis of professional practice (individual written report)

TEACHING AIDS

Case Studies
Scientific and Theoretical Articles in Humanities and Social Sciences
Press Articles
Slideshow Presentations

TEACHING LANGUAGE

French

CONTACT

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AIMS

The "Engineer Ethics" course aims to meet the following objectives. The details of each are indicated in the course syllabus.

- 1) Explain, using concrete examples, the types (ethical, moral, or legal), the natures (political, social, economic, and environmental), and the scales (micro, meso, macro) of responsibilities of the engineer.
- 2) Discern the causes and stakes of an ethical problem in an engineering context, as well as on the scale of a societal debate.
- 3) Justify a stance (point of view and/or a decision) by mobilizing ethical reasoning according to one's role in the specified context.

CONTENT

Session 1: Positioning of Engineers
Session 2: Socio-history of Engineers
Session 3: Engineering Ethics
Session 4: Session Project 1
Session 5: Traditions of Ethics
Session 6: Ethics of Science and Technology
Session 7: Session Project 2
Session 8: Gender and Work: Raising Awareness of VSS in Businesses
Session 9: Environmental Ethics
Session 10: Session Project 3
Session 11: Democratic Technical Approach
Session 12: Presentations and Wrap-up

BIBLIOGRAPHY

Beck Ulrich, La société du risque, Paris, Flammarion, 1995.
Boltanski Luc et Chiapello Ève, Le Nouvel esprit du capitalisme, Paris, Gallimard, 1999.
Bonneuil Christophe et Joly Pierre-Benoît, Sciences, techniques et société, Paris, La Découverte, 2013.
Bonneuil Christophe, Jean-Baptiste Fressoz, L'événement anthropocène, Paris, Seuil, 2016.
Callon Michel & al, Agir dans un monde incertain, Paris, Seuil, 2001.
Capron Michel et Quairel-Lanoizelée Françoise, La responsabilité sociale d'entreprise, Paris, La Découverte, 2016.
Capron Michel et Quairel-Lanoizelée Françoise, L'entreprise dans la société, Paris, La Découverte, 2015.
Didier Christelle, "Éthique de l'ingénierie. Un champ émergent pour l'éthique professionnelle", Techniques de l'Ingénieur, 2015.
Dubar Claude & al., Sociologie des professions, Paris, Armand Colin, 2015.
Jonas Hans, Le Principe de responsabilité, Paris, Flammarion, 1979.
Latour Bruno, Petites leçons de sociologie de sciences, Paris, La Découverte, 1993.
Renouard Cécile, Éthique et entreprise, Paris, L'Atelier, 2015.
Roby Catherine, Évolution du rôle social à la responsabilité sociétale des ingénieurs, Techniques de l'ingénieur, 2017.
Van de Poel Ibo and Royackers Lamber, Ethics, Technology and Engineering. An introduction, Wiley-Blackwell, 2011.
Vinck Dominique et Sainsaulieu Ivan, Ingénieur aujourd'hui, Lausanne, PPUR, 2015.
Vinck Dominique, Science et société, Paris, Armand Colin, 2007.

PRE-REQUISITES

Sociological Analysis of Organizations (ASO-4GI)
Professional internship

IDENTIFICATION

CODE : GI-5-S1-EC-RSE
ECTS : 3

HOURS

Cours : 0h
TD : 20h
TP : 8h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 30h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

Mme SUBAI Corinne :
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AIMS

CONTENT

- Sustainable development, corporate ethics, CSR
- Challenges of sustainable development for business

BIBLIOGRAPHY

PRE-REQUISITES

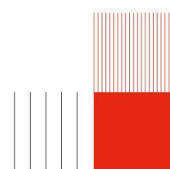
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membre de



IDENTIFICATION

CODE : GI-5-S1-EC-SSC
ECTS : 3

HOURS

Cours : 0h
TD : 28h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 30h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

TEACHING AIDS

Sunil Chopra, Supply Chain Management: Strategy, Planning and Operations, (6th Edition) 2016

TEACHING LANGUAGE

English

CONTACT

M. BABOLI Armand :
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AIMS

CONTENT

- 1- Supply chain Strategic Framework
 - a. understanding the supply chain,
 - b. Drivers and metrics
 - c. KPI and performance measurement,
- 2-) Designing the Supply Chain Network
 - a. Distribution networks,
 - b. Facility location and capacity allocation,
 - c. Risk management in global supply chains
- 3- Planning and Coordinating Demand and Supply in a Supply Chain
 - a. Demand forecasting, aggregate planning,
 - b. Sales and operations planning,
 - c. Coordination in a supply chain,
- 4- Planning and Managing Inventories
 - a. Managing economies of scale in a supply chain,
 - b. Managing uncertainty in a supply chain,
 - c. Determining the optimal level of product availability
- 5- Designing and Planning Transportation Networks
 - a. Challenges about Transportation Networks
 - b. Multimodal transportation
- 6- Managing Cross-Functional Drivers in a Supply Chain
 - a. Sourcing decisions,
 - b. Pricing and revenue management,
 - c. IT in a supply chain

BIBLIOGRAPHY

- 1- Sunil Chopra, 2018, Supply Chain Management: Strategy, Planning and Operations, (7th Edition), Pearson Education
- 2- F. Robert Jacobs & Richard Chase, 2023, Operations and Supply Chain Management ISE, McGraw-Hill Education; 17edition
- 3- John J. Coyle et al., 2020, Supply Chain Management: A Logistics Perspective, South-Western College Publishing
- 4- Roberta R. Russell, 2016, Operations Management: Creating Value Along the Supply Chain, John Wiley & Sons

PRE-REQUISITES

Production Planning and Scheduling, ERP, Mathematical modeling and optimization, Statistic and Probability, Linear programming, Inventory Control

IDENTIFICATION

CODE : GI-5-S1-EC-MGI-HU
ECTS : 3

HOURS

Cours : 0h
TD : 28h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 30h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

Continuous and/or final evaluation

TEACHING AIDS

Tutorials, case studies and projects, role plays
Online PowerPoint files
Exercises, case studies, and document work

TEACHING LANGUAGE

French

CONTACT

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Stéphane :
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AIMS

CONTENT

Management of human resources
- Introduction to the management of human resources and social relations.
- Reinforcement of one's listening abilities and of the knowledge in interpersonal communication collective intelligence, also in new cultural contexts.
- Putting it into practice with real collective cases
- Team work and information-communication; management roles and styles; principles and tools of HR (Vision, recruitment, evaluation and competence management, remuneration, career management, etc.) ; motivation and change management.
- System thinking and change management
- Social relations and employment contracts: social dialogue, IRP, trade-union organizations and collective bargaining; collective agreements; elements of labor law.

BIBLIOGRAPHY

- N.Alter, Sociologie de l'entreprise et de l'innovation, PUF, Paris 1996H.
- R.Blake & J.Mouton, Troisième dimension du management, Ed° d'organisation, Paris 1987
- O.Devillard, La dynamique des équipes, Ed° d'Organisation, Paris 2000
- J-L.Emery & E.Albert, Le manager est un psy, Ed° d'organisation, Paris 1998
- Y.Enrègle, Du conflit à la motivation, Ed° d'Organisation, Paris 1985.
- P.Fabart, Révélez le manager qui est en vous, Ed° d'organisation, Paris 2002
- E.Friedberg, Le pouvoir et la règle : dynamiques de l'action organisée, Le Seuil, Paris 1993
- B.Galambaud, L'initiative contrôlée ou le nouvel art du manager, EME ,Ed° ESF, Paris 1988
- V.Guibert, Comment manager, Ed° d'Organisation, Paris 2009
- B.Jarrosson, 100 ans de management, Ed° Dunod, Paris 2000
- R-V.Joule & J-L.Beauvois, Petit traité de manipulation des honnêtes gens, PU Grenoble, 2002
- M.Lallement, Sociologie des relations professionnelles, La Découverte, coll° Repères, Paris 1996.
- T.Levitt, Réflexions sur le management (trad. M.Speny), Dunod, Paris 1991
- P.Morin, L'art du manager: de Babylone à l'Internet, Ed° d'Organisation, Paris 1997
- Pichault, F. et Nizet, J., Les pratiques de gestion des ressources humaines. Approches contingente et politique, Seuil, coll. Points/Essais Sciences humaines, Paris 2000
- J-D.Reynaud, Les règles du jeu. L'action collective et la régulation sociale, Ed. Armand Colin, Paris 1997
- D.Segrestin, Les chantiers du manager. L'innovation en entreprise : où en sommes-nous ? Comment piloter les changements et les maîtriser ?, Armand Colin, Paris 2004
- C.Thuderoz & J.Vandewattyne, Un siècle d'histoire du management des hommes et des organisations, Presses Polytechniques Romandes, 2006
- Frédéric Laooux, Reinventing Organizations, Diateino, 2015
- Tom Nixon, Work with Source, 2020
- Et les films : « 12 hommes en colère » (Sidney Lumet, USA 1957), « Riens du tout » (Cédric Klapisch, France 1992), « Ressources humaines » (Laurent Cantet, France 1999), « El metodo » (Marcelo Piñeyro, Espagne et Argentine 2005), « Rien de personnel » (Mathias Gokalp, France 2009), "La loi du marché" (Stéphane Brizé, France 2015) etc.
- + ANACT (Agence Nationale pour les conditions de travail) publications

PRE-REQUISITES

Experience of several months in business in an engineering student internship.
Concepts of management control and industrial management.
General knowledge about the industrial enterprise
Fluent in French (speaking and writing)

IDENTIFICATION

CODE : GI-5-S1-EC-PCO
ECTS : 3

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 2h
Travail personnel : 58h
Total : 60h

ASSESSMENT METHOD

Contrôle continu en situation d'activité :
- Validation de cahier des charges client (CDC), de réponse à appel d'offre (RAO), d'un plan directeur de projet (PDP) incluant un tableau de bord
- Validation d'un ensemble de livrables jalonnant le projet
- Validation finale cohérente en une procédure de recette avec le maître d'ouvrage
- Défense collective finale du projet devant l'ensemble de la promotion
- Évaluation d'un rapport technique par le "tuteur technique"
- Évaluation d'un rapport management par le "tuteur MPC"

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

project management methodes and tools (following a project, steeps of a project, risk management, AMDEC, formalization of the customers objectives , redaction of specifications, redaction of plan realisation.

place the project and its stakes in the organization ; organize and manage a project and work in team; meetingsmanagement ; manage relevant information; communicate, negotiate and defend a project

CONTENT

Theoretical course of projet management (following a project, etapes of a project, a tool and methodes (risk management, AMDEC, formalization of the customers objectives , redaction of specifications, redaction of plan realisation)

Project: realisation of a project over 6 months by group of 8 students followed by two tutors

BIBLIOGRAPHY

- GIDEL T, ZONGHERO W. « Management de Projet 1 et 2 », Hermès Sciences, 2006
- BOURGEOIS J-P. « Gestion de Projet » Technique de l'Ingénieur T 7700
- CHVIDCHENKO I., CHEVALLIER J. « Conduite et gest. de proj. Princ. et prat. pour pts et grds projets » Cépaduès Edition, 1997
- AFITEP « Le Management de Projet, Principes et Pratique » Afnor Gestion, 1991
- GIARD V. « La Gestion de Projet » Economica, 1991
- PMI Project management Body of Knowledge; Project Management Institute; www.pmi.org
- R. MARCINIAK , M. PAGERIE Gestion de projet . Guide pratique de tous vos proj. et prod. indust., Editions WEKA
- J. MAISONNEUVE La dynamique des groupes, Paris Que-sais je ?, PUF, 1964
- B. KAYE Pédagogie de groupe sciences de l'éducation, Dunod, 1975
- R. MUCHIELLI Les méth. act. dans la pédago des adult., form. perm. en sci. hum., 1991 Gestion de Projets , Tome 1 et 2 , Edition WEKAP7

PRE-REQUISITES

- Méthodologie de résolution de problème
- Outils de la qualité

IDENTIFICATION

CODE : GI-5-S1-EC-PRI
ECTS : 3

HOURS

Cours : 0h
TD : 0h
TP : 20h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 24h
Travail personnel : 36h
Total : 60h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Group Project (GI-5-S1-UE-PROJ) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 3)
- A2 Operate a model of a real or virtual system (Level 1)
- A4 Design a system that meets a set of specifications (Level 3)
- A5 Process data (Level 2)
- A6 Report on an analysis or on a scientific approach (Level 1)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 3)
- C4 Sizing the hardware and / or software of a system (Level 3)
- C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 3)
- C7 Developing and implement a sourcing strategy (Level 3)
- C12 Changing organizations to meet new constraints or opportunities (Level 3)
- C13 Considering technological and methodological innovation (Level 3)
- C14 Collectively managing a project: organization, communication, group coordination (Level 3)
- C15 Managing a project based on a master plan (planning, budgeting, definition of monitoring indicators) (Level 3)
- C16 Identifying, analyzing and controlling the risks inherent to a project (Level 3)
- C17 Identifying, formalizing and contractualising the requirements of a client, following their evolution and validating their compliance (requirements traceability) (Level 3)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Industrial Organization (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)
- Continuous Improvement (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)
- Production Data Analysis and System Design (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)
- Material Sourcing and Replenishment (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)

To be able to :

- Develop an organization's observation and strategic analysis capabilities (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)

CONTENT

Learn how to develop a response to a call for tenders, taking into account the technical, economic, legal, and human aspects of the project.

Each project is supervised and led by an industry project manager for six weeks. Engineering students are grouped into teams of six.

Each team, as a potential project manager, is responsible for developing solutions in response to specifications.

At the end of the project, each team proposes and defends its technical, organizational, economic, and time-related solutions in a competitive environment to secure the project owner's contract.

These projects most often address the following aspects:

- production organization, logistics,
- operational safety, quality, and maintenance.

BIBLIOGRAPHY

PRE-REQUISITES

IDENTIFICATION

CODE : GI-5-S1-EC-RGI
ECTS : 3

HOURS

Cours : 0h
TD : 2h
TP : 6h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 12h
Travail personnel : 48h
Total : 75h

ASSESSMENT METHOD

Written report + Oral presentation

TEACHING AIDS

Depending on the topic of the project.

TEACHING LANGUAGE

French
English

CONTACT

MME BOTTA-GENOULAZ Valerie
:
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AIMS

This teaching unit (EC) contributes to the following skills:

A3 Implement an experimental approach or a production approach
A5 Process data
A6 Communicate an analysis, a scientific approach, a proof or a solution in a reasoned and logical way
B2 Work, learn, evolve independently
B4 Show creativity, innovate, undertake
C1 Observing, measuring, analyzing and interpreting an activity or a system from data
C2 Modeling and designing an information, decision and production system, of goods and services
C3 Evaluating, prototyping and simulating a system
C13 Considering technological and methodological innovation

Skills mobilized :

A1 Analyze a system (real or virtual) or a problem
A2 Exploit a model of a real or virtual system
C8 Managing supply in connection with the planning and inventory management policy
C14 Collectively managing a project: organization, communication, group coordination
C16 Identifying, analyzing and controlling the risks inherent to a project
B3 Interact with others, team work
B7 Work in an international and intercultural context

CONTENT

Research project, in pairs, allowing the student to learn and experiment a research/innovation approach, on a subject proposed and supervised by a teacher-researcher, according to the following steps:

- appropriation of the subject
- formalization of the research problem and proposal of possible solutions
- development of solutions and experiments
- analysis of results, proposal of perspectives
- writing of a document respecting the editorial standards of a research article

BIBLIOGRAPHY

Depending on the topic of the project.

PRE-REQUISITES

Depending on the topic of the project.

IDENTIFICATION

CODE : CDS-5-S1-EC-EPS
ECTS : 1

HOURS

Cours : 0h
TD : 20h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 0h
Total : 20h

ASSESSMENT METHOD

Assessment in Physical Education concerns the teaching of Sports and Artistic Physical Activities (APSA), and will take the form of continuous assessment with half-yearly marking.

The mark depends on the degree of acquisition of the skills expected in each of the activities, and the progress made over all the sessions in the cycle. The mark takes into account :

Individual and/or team performance
mastery of execution
Progress in the sports project
Responsibility and autonomy

TEACHING AIDS

All physical, sporting, artistic and competitive activities

TEACHING LANGUAGE

French

CONTACT

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AIMS

This EC is part of the Teaching Unit: SHS and contributes to the development of the School's transversal competences

1* Auto-evaluating one's own performance

Knowledges :

- Fundamentals, principles of action and terminology of sports activities
- Criteria for observation, achievement and success.

Abilities :

- Assess your level of practice
- Build up a warm-up
- Set goals for progress
- Manage physical and mental potential

2* Work, learn and develop independently

Knowledge :

- PSAA rules
- Observation criteria
- Principles of warm-up and cool-down

Abilities :

- Mobilise resources
- Analyse, observe, question
- Take on different roles (referee, choreographer)

3* Interact with others, work as part of a team

Knowledges :

- Roles and functions in each sports activity
- Communicate appropriately: verbal, non-verbal and postural communication.
- Integrate into a group
- Take part in and develop a group project
- Take the initiative
- Be a good listener

4* Be creative, innovative and enterprising

Knowledge :

- Artistic disciplines

Abilities :

- Draw on knowledge and resources from different artistic fields to produce an original work.
- Mobilise the imagination and sensibility and make them visible through dance movement
- Access the symbolism of the body

5* Act responsibly in a complex world

Knowledge :

- Safety and operating rules

Abilities :

- Identify uncertainties and risks and act to reduce them
- Integrate a responsible dimension into their actions
- Show respect and fair play in a power struggle

6* Working in an international context

Knowledge :

- Socio-cultural differences

Abilities :

- Integrate cultural diversity into group work
- Act with respect for self and others

CONTENT

Physical Education and Sport lessons are organised around traditional Physical Education lessons, or advanced lessons, or appropriate practices (EPSA), or competitive practices within the framework of the Section Sportive Haut Niveau.

1. Physical Education lessons :

Students choose one or two physical and sporting activities per year from among the activities offered by the sports centre (individual, group, dual).

2. Appropriate Physical Education lessons: For all students who are exempt from physical activity for at least 2 months:

Swimming, Body-building, Nordic Walking, Somatic Exercise, Sophrology, Wheelchair Basketball, Pilates, Table Tennis, etc.

3. Advanced Physical Education courses :

Specialisation in a sporting activity, University training and competitions

4. SSHN (High-Level Athlete section)

University training and competitions

EPS5 GMPP OYONNAX :Group cohesion project
Autonomy Lessons at S1 on Wednesday afternoons
Hiking outing

BIBLIOGRAPHY

OTTAWA Charter (1986): 'health is seen as a resource for everyday life; it is a positive concept that highlights social and individual resources, as well as physical abilities'.

PRE-REQUISITES

- EPS: none
 - Appropriate Physical Education: subject to medical advice
 - Advanced courses and competitive practice: previous practice required subject to specific selection according to each activity
 - SHN: ministerial list
- Levels 1 and 2: Physical Education, Appropriate physical education
Level 3: Advanced courses and competitive practice, SHN

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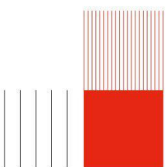
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membre de



IDENTIFICATION

CODE : GI-5-S1-EC-REN
ECTS : 0

HOURS

Cours : 2h
TD : 2h
TP : 2h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 6h
Travail personnel : 0h
Total : 6h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

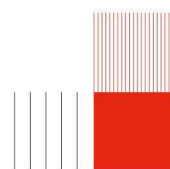
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IDENTIFICATION

CODE : HU-0-S1-EC-S-PPH
ECTS : 1

HOURS

Cours : 0h
TD : 20h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 0h
Total : 20h

ASSESSMENT METHOD

Written report (10 pages minimum)
and oral defence (in presence of
tutor and guest).

TEACHING AIDS

Présentation du PPH sur Moodle :
<http://moodle.insa-lyon.fr>

TEACHING LANGUAGE

French

CONTACT

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AIMS

The intended learning outcomes are:

- Formulate a societal issue
- Define a methodological framework for data collection to address it
- Organize the elements of the response and provide a personal analysis
- Demonstrate critical thinking, both in the selection of sources and in their analysis
- Present the results of this analysis in an educational manner, both in writing and orally.

CONTENT

Work on a particular theme with a tutor chosen by the student.
Filling in of a project sheet (elaboration of the question, definition of the personal
approach, bibliography, etc),
Step by step meetings with the tutor (plan, analysis, etc),
Report writing and oral presentation.

BIBLIOGRAPHY**PRE-REQUISITES**

IDENTIFICATIONCODE : HU-0-S1-EC-S-SERIE5
ECTS : 2**HOURS**

Cours :	20h
TD :	20h
TP :	0h
Projet :	0h
Evaluation :	0h
Face à face pédagogique :	40h
Travail personnel :	0h
Total :	40h

ASSESSMENT METHOD

Assessment will be conducted through continuous evaluation. The assessment methods will be presented at the beginning of the semester by the teaching team.

TEACHING AIDS

Materials are chosen by the instructor based on the module:

- Didactic documents related to the module
- Audiovisual materials
- Recommended readings

TEACHING LANGUAGE

French

CONTACTMme JOUISHOMME Delphine :
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carine.goutaland@insa-lyon.fr**AIMS**

A series of elective courses in Humanities and Social Sciences (HSS) offers several options for students to choose from, allowing them to develop and deepen specific skills. This course aims to develop one or more transversal skills among the following:

- CT1: Self-awareness and self-management
- CT2: Working, learning, and evolving independently
- CT3: Interacting with others, working in a team
- CT4: Demonstrating creativity
- CT5: Acting responsibly in a complex world
- CT6: Navigating and evolving within an organization
- CT7: Working in an international and intercultural context

The list of options available in Series 1 and the specific competencies for each option are detailed in the catalog on the IntranetHumas:
<https://intranethumas.insa-lyon.fr/sciences-humaines-sociales/offre-de-formation/cours-la-carte-0>

CONTENT

Each module is designed to encourage interaction and active student participation. The content is structured around the following key aspects:

- Theoretical deepening related to the theme
- Reflection on the topic
- Practical exercises and activities
- Assessments and presentation of work

BIBLIOGRAPHY

The bibliography is selected by the instructor based on the module.

PRE-REQUISITES

French

IDENTIFICATION

CODE : GI-5-S1-EC-MAS
ECTS : 22

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

45 min presentation
Project dissertation
Summary

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

IDENTIFICATION

CODE : GI-5-S1-EC-RSE
ECTS : 3

HOURS

Cours : 0h
TD : 20h
TP : 8h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 30h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

CONTENT

- Sustainable development, corporate ethics, CSR
- Challenges of sustainable development for business

BIBLIOGRAPHY

PRE-REQUISITES

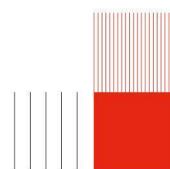
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membre de



IDENTIFICATION

CODE : GI-5-S1-EC-CYB
ECTS : 2

HOURS

Cours : 0h
TD : 6h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

CONTENT

Objectives: Understand the challenges related to cybersecurity in industrial systems and the specific features of this field. Understand the basic elements for identifying the vulnerabilities of these systems, as well as the recommendations and methodology for strengthening the cybersecurity level of existing systems and the key points for designing new systems. Understand the cybersecurity stakeholders in companies and support the implementation of the approach. Understand how to communicate with IT and OT to unite teams around these new challenges.

Main concepts of cybersecurity for information systems in general:
Definitions of cybersecurity and main concepts
Challenges, current status, history;
Classical attacks (MITM, spoofing, social engineering, denial of service, session hijacking, DDOS, APT, worms);
Specific attacks (malicious organizations): PLC code hijacking, takeover of the SCADA/MES information system, takeover of PLCs and industrial equipment
Duality of operational safety and cybersecurity; Examples of attacks (study, practical work on platform attacks);
Classic vulnerabilities and attack vectors;
Main principles for implementing a cybersecurity project (risk analysis, DEP, PSSI, etc.)
Best practices and recommendations;
Overview of norms and standards (2700X, product certification, etc.);
In France, the LPM;
Introduction to cryptography (symmetric/asymmetric encryption, hash functions, signatures, etc.).
IDS modeling (SED, ECC link)
Configuring industrial firewalls and understanding bad ideas.
The industrial system cybersecurity project in a real environment (transfer room practical work); Vulnerability analysis
Security strengthening
In the shoes of a cyber attacker

BIBLIOGRAPHY

PRE-REQUISITES

Basics of automation and different types of industrial systems;
Composition of an industrial system (ECC);
PLC programming languages (ECC, COP, TP API);
Industrial protocols and fieldbuses (ECC);
Classic network architectures (ECC, COP);
Operational safety (CPS/SDF to be extended with SIL standards and consolidate FMEA + EBIOS);
Overview of norms and standards (ECC, API labs, COP, SCADA/MES)
Computer Basics:
Concepts of classic hardware architectures (microprocessor, memory, bus, peripherals, firmware)
Operating systems (main systems used)
Concepts of software architectures, databases
Network Basics:
Main concepts and vocabulary used in this field:
o 7-layer OSI model,
o Data units (frame, packet, segment, etc.),
o MAC address, IP address, port concept (UDP/TCP), o Network discovery and diagnostics (ARP, ICMP),
o Connected (TCP) and offline (UDP) modes.
Services that may be present in industrial equipment (PLC, MES, ERP, HMI, VFD, wireless terminal, etc.)
o Web (HTTP) and secure Web (HTTPS),
o Equipment management (SNMP, SYSLOG, etc.). VPN using a VPN gateway, router, gateway, firewall.
Information systems fundamentals
Data warehouses and data lakes

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IDENTIFICATION

CODE : GI-5-S1-EC-IFD
ECTS : 3

HOURS

Cours : 0h
TD : 26h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 28h
Travail personnel : 30h
Total : 58h

ASSESSMENT METHOD

TEACHING AIDS

James A. Tompkins, 2010, Facilities Planning, John Wiley & Sons Inc
Mikell P. Groover, 2016, Work Systems: The Methods, Measurement & Management of Work, PEARSON edition

TEACHING LANGUAGE

English

CONTACT

M. BABOLI Armand :
armand.baboli@insa-lyon.fr

AIMS

CONTENT

- 1- Introduction to facility design
 - a. Relationship between the product, process and scheduling design and the facility design
 - b. Flow system (Flow shop, Job Shop, cellular manufacturing, Fixed-position)
 - c. Process modeling and material flow analysis (Process Flowchart, IDEF0)
 - d. Activity relationship
- 2- Requirements of Manufacturing Facility (plant) design
 - a. Motion and Time study
 - b. Capacity determination (machine, human resource)
 - c. Performance measurement
- 3- Material Handling system, Storage and Warehouse system design
 - a. Type of material handling (receiving and shipping, dock equipment, moving equipment)
 - b. Type of Storage and Warehousing (Picking cart, Picking truck, vertical Warehouse, Automated Warehouse)
 - c. Transfer Line (Straight Line Flow, Circular Flow, U-Flow, Spine-Flow)
 - d. Equipment selection and Capacity determination
- 4- Cellular Manufacturing System (CMS) design
 - a. Clustering approach (Rank ordering, clustering, similarity coefficient, mathematical programming)
 - b. Designing and planning issues of CMS (capacity, material handling, layout)
- 5- Facility Layout
 - a. Basic layout types
 - b. Multi-floor facility layout,
 - c. Service layout
 - d. Flow shop and line balancing
 - e. Job shop and department location problem
 - f. Development alternatives
 - g. Mathematical modeling and Algorithms
- 6- Flexible Manufacturing System (FMS) design and optimization: (types of flexibilities, flexibilities classification, flexibility modeling and measurement, decision-making and optimization of the level of flexibility)
- 7- Industry 4.0 (Cyber Physical system, IIoT, Data Analytics and prescription of decision)
- 8- Configuration and experimentation of a flexible manufacturing system: Practical work on the Smart Factory platform (cyber-physical system).

BIBLIOGRAPHY

S. Heragu, 2008, Facilities Design, CRC Press Inc; Third Edition,
Ralph M. Barnes, Motion and Time Study: Design and Measurement of Work, 1980
Alberto Garcia-Diaz & J. MacGregor Smith, 2024, Facilities Planning and Design, Springer

PRE-REQUISITES

Mathematical modeling and optimization, Statistic and Probability, Linear programming, Inventory management, Lean manufacturing

IDENTIFICATION

CODE : GI-5-S1-EC-EVP
ECTS : 2

HOURS

Cours : 0h
TD : 16h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 16h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

Individual Assessment
Collective case study

TEACHING AIDS

Course and TD support
Exercices + Case studies

TEACHING LANGUAGE

English

CONTACT

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AIMS

At the end of this teaching module, students should be able to:

- Define, characterise and formalise the needs of a decision-maker (5 and H)
- Propose a clear and legible format that is easy to understand, straightforward and unique.
- Take into account the maintainability and robustness of indicators when constructing a dashboard.
- Analyse indicator results, draw up a diagnosis and identify levers for action.
- Identify statistical biases and minimise them when proposing solutions.
- Demonstrate critical thinking when assessing results
- Demonstrate objectivity and neutrality in the choice of indicator and its presentation.

CONTENT

Teaching programme:

lecture/tutorial, 4 x 2h sessions: theoretical input on performance evaluation, with examples and exercises

Project, 4 hours supervised + 4 hours independent + 4 hours feedback: practical case study

BIBLIOGRAPHY

Setiawan, Indra & Purba, Humiras. (2020). A Systematic Literature Review of Key Performance Indicators (KPIs) Implementation. Journal of Industrial Engineering and Management. 1. 200-208. 10.7777/jiemar.v1i3.79.

Berrah Lamia, « Indicateur de performance : concepts et application », Edition Eyrolles, 2002

PRE-REQUISITES

Introduction to Supply Chain

IDENTIFICATION

CODE : GI-5-S1-EC-MGI-HU
ECTS : 3

HOURS

Cours : 0h
TD : 28h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 30h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

Continuous and/or final evaluation

TEACHING AIDS

Tutorials, case studies and projects, role plays
Online PowerPoint files
Exercises, case studies, and document work

TEACHING LANGUAGE

French

CONTACT

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AIMS

CONTENT

Management of human resources
- Introduction to the management of human resources and social relations.
- Reinforcement of one's listening abilities and of the knowledge in interpersonal communication collective intelligence, also in new cultural contexts.
- Putting it into practice with real collective cases
- Team work and information-communication; management roles and styles; principles and tools of HR (Vision, recruitment, evaluation and competence management, remuneration, career management, etc.) ; motivation and change management.
- System thinking and change management
- Social relations and employment contracts: social dialogue, IRP, trade-union organizations and collective bargaining; collective agreements; elements of labor law.

BIBLIOGRAPHY

- N.Alter, Sociologie de l'entreprise et de l'innovation, PUF, Paris 1996H.
- R.Blake & J.Mouton, Troisième dimension du management, Ed° d'organisation, Paris 1987
- O.Devillard, La dynamique des équipes, Ed° d'Organisation, Paris 2000
- J-L.Emery & E.Albert, Le manager est un psy, Ed° d'organisation, Paris 1998
- Y.Enrègle, Du conflit à la motivation, Ed° d'Organisation, Paris 1985.
- P.Fabart, Révélez le manager qui est en vous, Ed° d'organisation, Paris 2002
- E.Friedberg, Le pouvoir et la règle : dynamiques de l'action organisée, Le Seuil, Paris 1993
- B.Galambaud, L'initiative contrôlée ou le nouvel art du manager, EME ,Ed° ESF, Paris 1988
- V.Guibert, Comment manager, Ed° d'Organisation, Paris 2009
- B.Jarrosson, 100 ans de management, Ed° Dunod, Paris 2000
- R-V.Joule & J-L.Beauvois, Petit traité de manipulation des honnêtes gens, PU Grenoble, 2002
- M.Lallement, Sociologie des relations professionnelles, La Découverte, coll° Repères, Paris 1996.
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- P.Morin, L'art du manager: de Babylone à l'Internet, Ed° d'Organisation, Paris 1997
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- J-D.Reynaud, Les règles du jeu. L'action collective et la régulation sociale, Ed. Armand Colin, Paris 1997
- D.Segrestin, Les chantiers du manager. L'innovation en entreprise : où en sommes-nous ? Comment piloter les changements et les maîtriser ?, Armand Colin, Paris 2004
- C.Thuderoz & J.Vandewattyne, Un siècle d'histoire du management des hommes et des organisations, Presses Polytechniques Romandes, 2006
- Frédéric Laooux, Reinventing Organizations, Diateino, 2015
- Tom Nixon, Work with Source, 2020
- Et les films : « 12 hommes en colère » (Sidney Lumet, USA 1957), « Riens du tout » (Cédric Klapisch, France 1992), « Ressources humaines » (Laurent Cantet, France 1999), « El metodo » (Marcelo Piñeyro, Espagne et Argentine 2005), « Rien de personnel » (Mathias Gokalp, France 2009), "La loi du marché" (Stéphane Brizé, France 2015) etc.
- + ANACT (Agence Nationale pour les conditions de travail) publications

PRE-REQUISITES

Experience of several months in business in an engineering student internship.
Concepts of management control and industrial management.
General knowledge about the industrial enterprise
Fluent in French (speaking and writing)



IDENTIFICATION

CODE : GI-5-S1-EC-OTL
ECTS : 3

HOURS

Cours : 0h
TD : 24h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 26h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

- A2 Operate a model of a real or virtual system (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
- C5 Managing a production system and react to malfunctions (Level 2)
- C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 1)
- C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 1)
- C13 Considering technological and methodological innovation (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

To be able to :

CONTENT

The engineering student will work and be assessed on the following knowledge and skills :

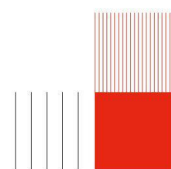
- (10h) Know the "classical" transportation problems : Traveling Salesman Problem (TSP), Vehicle Routing Problem (VRP), etc.
- (2h) Know the basic methods and tools to design logistic networks.
- (8h) Know the principles and tools for distributed decision-making: game theory, multi-agent systems and agent-based simulation coupled with flow simulation (for instance with Anylogic)
- (4h) Know the tools (especially decomposition methods) to perform joint optimisation.
- Be able to formulate and solve a supply chain optimisation problem, from the viewpoint of transportation and flow management.

BIBLIOGRAPHY

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- Jacques Teghem. Recherche opérationnelle. Tome 1, Tome 1, OCLC : 819196866. Paris : Ellipses, 2012. isbn : 978-2-7298-7509-1.
- Paolo Toth et Daniele Vigo, eds. Vehicle routing : problems, methods, and applications. Second edition. MOS-SIAM series on optimization. Philadelphia : Society for Industrial et Applied Mathematics : Mathematical Optimization Society, 2014. 463 p. isbn : 978-1-61197-358-7.
- Michael Wooldridge. An Introduction to Multi-Agent Systems. 2d edition. New York, NY, USA: John Wiley & Sons, Inc., 2009.
- Murat Yildizoglu. Introduction à la Théorie des Jeux. Dunod, Paris (France), 2003.

PRE-REQUISITES

Inventory management : EOQ, basic replenishment methods
Flow management
Production management : S&OP, MPS & MRP
Scheduling: models, standard scheduling notation, heuristic solving methods for scheduling problems
Combinatorial optimisation: graph theory
Logistic



IDENTIFICATIONCODE : GI-5-S1-EC-APF
ECTS : 3**HOURS**Cours : 0h
TD : 24h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 26h
Travail personnel : 30h
Total : 60h**ASSESSMENT METHOD**Continuous evaluation + final exam
Exam in English possible / Data
bilingual given**TEACHING AIDS**No support in English : Work and
project are accepted in English
Verbal exchange possible in
English**TEACHING LANGUAGE**

French

CONTACTM. BERNARD-TREMOLET
Stéphane :
stephane.bernard@insa-lyon.fr**AIMS**Obtain a clear understanding about the ideal cycle for sourcing, described in a step by
step method.
Apply management project's tools to sourcing
Put in place a supplier's monitoring
Be able to communicate, reproduce and transfert those competencies.**CONTENT**

N/A

BIBLIOGRAPHY

N/A

PRE-REQUISITES

IDENTIFICATION

CODE : GI-5-S1-EC-DSC
ECTS : 3

HOURS

Cours : 0h
TD : 24h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 26h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

Final exam 1h50.

TEACHING AIDS

Slides (lectures) + Python Jupyter notebooks (practicals)

TEACHING LANGUAGE

English

CONTACT

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AIMS

This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

A5 Process data (Level 2)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 1)

C13 Considering technological and methodological innovation (Level 2)

CONTENT

The Data Science course is structured into two core parts:

1. Introduction to Artificial Intelligence – This section focuses on Deep Learning and Reinforcement Learning, covering key concepts such as deep neural networks (RNN, LSTM, Transformers), activation functions, optimization techniques, and loss functions. It also explores real-world applications, including self-driving car simulations and AI-driven decision-making.

2. Introduction to Natural Language Processing (NLP) – This section covers the foundations and applications of Large Language Models (LLMs). Topics range from the history of NLP techniques to modern industrial applications in Customer Support, Legal Tech, and Industry 4.0. The course covers essential techniques such as text classification, word/document representation, and Retrieval-Augmented Generation (RAG).

The course includes hands-on labs using Python and Jupyter notebooks, providing practical experience in AI and NLP model development.

BIBLIOGRAPHY

Charu C. Aggarwal. (2023). Neural networks and deep learning. A Textbook. Switzerland: Springer International Publishing.

Nithin Buduma, Nikhil Buduma, Joe Papa (2022) Fundamentals of Deep Learning, 2nd Edition.

Laura Graesser and Wah Loon Keng. (2019) Foundations of Deep Reinforcement Learning Theory and Practice in Python. Addison-Wesley Professional.

Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition.

PRE-REQUISITES

Python programming

Probability theory

Introduction to Data Science

IDENTIFICATION

CODE : GI-5-S1-EC-PRI
ECTS : 3

HOURS

Cours : 0h
TD : 0h
TP : 20h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 24h
Travail personnel : 36h
Total : 60h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Group Project (GI-5-S1-UE-PROJ) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 3)
- A2 Operate a model of a real or virtual system (Level 1)
- A4 Design a system that meets a set of specifications (Level 3)
- A5 Process data (Level 2)
- A6 Report on an analysis or on a scientific approach (Level 1)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 3)
- C4 Sizing the hardware and / or software of a system (Level 3)
- C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 3)
- C7 Developing and implement a sourcing strategy (Level 3)
- C12 Changing organizations to meet new constraints or opportunities (Level 3)
- C13 Considering technological and methodological innovation (Level 3)
- C14 Collectively managing a project: organization, communication, group coordination (Level 3)
- C15 Managing a project based on a master plan (planning, budgeting, definition of monitoring indicators) (Level 3)
- C16 Identifying, analyzing and controlling the risks inherent to a project (Level 3)
- C17 Identifying, formalizing and contractualising the requirements of a client, following their evolution and validating their compliance (requirements traceability) (Level 3)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Industrial Organization (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)
- Continuous Improvement (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)
- Production Data Analysis and System Design (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)
- Material Sourcing and Replenishment (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)

To be able to :

- Develop an organization's observation and strategic analysis capabilities (A1, A2, A4, A5, A6, C1, C2, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17)

CONTENT

Learn how to develop a response to a call for tenders, taking into account the technical, economic, legal, and human aspects of the project.

Each project is supervised and led by an industry project manager for six weeks. Engineering students are grouped into teams of six.

Each team, as a potential project manager, is responsible for developing solutions in response to specifications.

At the end of the project, each team proposes and defends its technical, organizational, economic, and time-related solutions in a competitive environment to secure the project owner's contract.

These projects most often address the following aspects:

- production organization, logistics,
- operational safety, quality, and maintenance.

BIBLIOGRAPHY

PRE-REQUISITES

IDENTIFICATION

CODE : GI-5-S1-EC-KNM
ECTS : 2

HOURS

Cours : 0h
TD : 18h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

C2 Modeling and designing an information, decision and production system, of goods and services (Level 1)
C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 1)
C10 Developing and implementing an action plan as part of a quality and continuous improvement approach (Level 1)
C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 1)
C12 Changing organizations to meet new constraints or opportunities (Level 1)
C18 Identifying the critical skills and knowledge of an organization and implement tools and methods to sustain the (Level 1)
C19 Conducting a socio-organizational analysis to better understand the effects of change and to adapt strategies (Level 1)
B5 act responsibly in a complex world (Level 1)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Industrial knowledge-related assets
 - Information, knowledge, and expertise
 - Knowledge management methods in industry, collaborative knowledge management tools
 - Digital knowledge management tools
 - Knowledge management strategies and continuous improvement
 - Technological, human, and organizational knowledge and structure
- To be able to :
- Distinguishing the type of knowledge created by the use of information and Communication
 - Identify industrial capital (human, technological, organizational, informational)
 - Assess the limitations of knowledge management systems

CONTENT

BIBLIOGRAPHY

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- Dejoux, Cécile. 2013. Gestion des compétences et GPEC. 2e édition. Dunod.
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- Nonaka, Ikujiro, et Hirotaka Takeuchi. 1995. The Knowledge-Creating Company: How Japanese companies create the dynamics of innovation. Oxford University Press.
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- Reagle, J.M. 2010. Good Faith Collaboration: The Culture of Wikipedia. Édité par Michael Buckland et Jonathan Furner. History and foundations of information science. Cambridge, MA: MIT Press. <http://mitpress.mit.edu/books/good-faith-collaboration>.
- Zins, Chaim. 2007. « Conceptual approaches for defining data, information, and knowledge ». Journal of the American Society for Information Science and Technology 58 (4): 479–493. <https://doi.org/10.1002/asi.20508>.

PRE-REQUISITES

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membre de



IDENTIFICATION

CODE : GI-5-S1-EC-IFU
ECTS : 2

HOURS

Cours : 0h
TD : 0h
TP : 20h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

Report and oral presentation.

TEACHING AIDS

To be determined.

TEACHING LANGUAGE

French

CONTACT

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AIMS

"This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

A1 Analyze a system (real or virtual) or a problem (Level 3)

A3 Implement an experimental approach (Level 3)

A5 Process data (Level 3)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)

C5 Managing a production system and react to malfunctions (Level 3)

C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 1)

C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 1)

C13 Considering technological and methodological innovation (Level 3)

B3 interact with people and work in a team (Level 3)

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

Python (Jupyter) programming.

Pulp language (CPLEX/GUROBI solver mathematical programming MIP language).

Courses: GI-4-S1-EC-ISD and GI-5-S1-EC-OTL

IDENTIFICATION

CODE : GI-5-S1-EC-SSC

ECTS : 3

HOURS

Cours : 0h

TD : 28h

TP : 0h

Projet : 0h

Evaluation : 2h

Face à face pédagogique : 30h

Travail personnel : 30h

Total : 60h

ASSESSMENT METHOD

TEACHING AIDS

Sunil Chopra, Supply Chain Management: Strategy, Planning and Operations, (6th Edition) 2016

TEACHING LANGUAGE

English

CONTACT

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AIMS

CONTENT

- 1- Supply chain Strategic Framework
 - a. understanding the supply chain,
 - b. Drivers and metrics
 - c. KPI and performance measurement,
- 2-) Designing the Supply Chain Network
 - a. Distribution networks,
 - b. Facility location and capacity allocation,
 - c. Risk management in global supply chains
- 3- Planning and Coordinating Demand and Supply in a Supply Chain
 - a. Demand forecasting, aggregate planning,
 - b. Sales and operations planning,
 - c. Coordination in a supply chain,
- 4- Planning and Managing Inventories
 - a. Managing economies of scale in a supply chain,
 - b. Managing uncertainty in a supply chain,
 - c. Determining the optimal level of product availability
- 5- Designing and Planning Transportation Networks
 - a. Challenges about Transportation Networks
 - b. Multimodal transportation
- 6- Managing Cross-Functional Drivers in a Supply Chain
 - a. Sourcing decisions,
 - b. Pricing and revenue management,
 - c. IT in a supply chain

BIBLIOGRAPHY

- 1- Sunil Chopra, 2018, Supply Chain Management: Strategy, Planning and Operations, (7th Edition), Pearson Education
- 2- F. Robert Jacobs & Richard Chase, 2023, Operations and Supply Chain Management ISE, McGraw-Hill Education; 17edition
- 3- John J. Coyle et al., 2020, Supply Chain Management: A Logistics Perspective, South-Western College Publishing
- 4- Roberta R. Russell, 2016, Operations Management: Creating Value Along the Supply Chain, John Wiley & Sons

PRE-REQUISITES

Production Planning and Scheduling, ERP, Mathematical modeling and optimization, Statistic and Probability, Linear programming, Inventory Control

IDENTIFICATION

CODE : GI-5-S1-EC-ADM
ECTS : 2

HOURS

Cours : 0h
TD : 6h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

continuous assessment

TEACHING AIDS

Lecture book and exercises

TEACHING LANGUAGE

French

CONTACT

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M. VERCRAENE Samuel :
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AIMS

The Multicriteria Decision Aid course covers methods and tools that help a decision-maker make a complex decision involving multiple criteria, often contradictory or in conflict.

Three approaches will be covered in this course:

- The AHP method (Analytic Hierarchy Process);
- The TOPSIS method (Technique for Order Preference by Similarity to Ideal Solution);
- Multicriteria optimization.

CONTENT

TD1 2h introduction ADM + AHP
TP1 4h TP AHP CC1
TD2 2h OPTIM + TOPSIS
TP2 4h TP TOPSIS CC2
TD3 2h TD Lexico + Pareto
TP3 4h TP Pareto CC3

BIBLIOGRAPHY

Bouyssou D. et Roy B. : Aide multicritère à la décision : méthode et cas, Economica, 1993.
Carlier J. et Chrétienne P. : Problèmes d'ordonnancement : modélisation / complexité / algorithmes, Masson, 1988.
Garfinkel R.S. et Nemhauser : Integer programming, Wiley Interscience, 1972.
Roy B. : Méthodologie multicritère d'aide à la décision, Economica, 1985.

PRE-REQUISITES

Monocriterion decision support (Linear Programming), basic and advanced production management

IDENTIFICATION

CODE : GI-5-S1-EC-MTC
ECTS : 2

HOURS

Cours : 0h
TD : 6h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

TEACHING AIDS

Course and tutorials handouts
Softwares Carl Source (CMMS),
Minitab (reliability) and Anylogic
(simulation)

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Excellence opérationnelle (GI-5-S1-UE-EXOP) and contributes to the following skills :

A1 Analyze a system (real or virtual) or a problem (Level 2)

A2 Operate a model of a real or virtual system (Level 2)

A3 Implement an experimental approach (Level 1)

C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)

C2 Modeling and designing an information, decision and production system, of goods and services (Level 1)

C3 Evaluating, prototyping and simulating a system (Level 1)

C5 Managing a production system and react to malfunctions (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- CMMS Functions

- Maintenance Processes

- Maintenance Optimization

- Maintenance Tools: Manufacturer Documentation, Reliability, etc.

To be able to :

- Understanding the challenges of purchasing a CMMS (C2, C5)

- Defining a maintenance policy (A1, A2, A3, C1, C3, C5)

CONTENT

Computerized Maintenance Management System

FR - lecture/tutorial/practical work in french

* Overview of the functions of a CMMS

* Maintenance process

* Optimisation of maintenance

* Maintenance tools (manufacturer's documentation, reliability, etc.)

BIBLIOGRAPHY

PRE-REQUISITES

IDENTIFICATION

CODE : GI-5-S1-EC-ETH-HU
ECTS : 2

HOURS

Cours : 0h
TD : 20h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 20h
Travail personnel : 20h
Total : 40h

ASSESSMENT METHOD

Press review (collective oral presentation)
Ethical analysis of professional practice (individual written report)

TEACHING AIDS

Case Studies
Scientific and Theoretical Articles in Humanities and Social Sciences
Press Articles
Slideshow Presentations

TEACHING LANGUAGE

French

CONTACT

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AIMS

The "Engineer Ethics" course aims to meet the following objectives. The details of each are indicated in the course syllabus.

- 1) Explain, using concrete examples, the types (ethical, moral, or legal), the natures (political, social, economic, and environmental), and the scales (micro, meso, macro) of responsibilities of the engineer.
- 2) Discern the causes and stakes of an ethical problem in an engineering context, as well as on the scale of a societal debate.
- 3) Justify a stance (point of view and/or a decision) by mobilizing ethical reasoning according to one's role in the specified context.

CONTENT

Session 1: Positioning of Engineers
Session 2: Socio-history of Engineers
Session 3: Engineering Ethics
Session 4: Session Project 1
Session 5: Traditions of Ethics
Session 6: Ethics of Science and Technology
Session 7: Session Project 2
Session 8: Gender and Work: Raising Awareness of VSS in Businesses
Session 9: Environmental Ethics
Session 10: Session Project 3
Session 11: Democratic Technical Approach
Session 12: Presentations and Wrap-up

BIBLIOGRAPHY

Beck Ulrich, La société du risque, Paris, Flammarion, 1995.
Boltanski Luc et Chiapello Ève, Le Nouvel esprit du capitalisme, Paris, Gallimard, 1999.
Bonneuil Christophe et Joly Pierre-Benoît, Sciences, techniques et société, Paris, La Découverte, 2013.
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Capron Michel et Quairel-Lanoizelée Françoise, La responsabilité sociale d'entreprise, Paris, La Découverte, 2016.
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Van de Poel Ibo and Royackers Lamber, Ethics, Technology and Engineering. An introduction, Wiley-Blackwell, 2011.
Vinck Dominique et Sainsaulieu Ivan, Ingénieur aujourd'hui, Lausanne, PPUR, 2015.
Vinck Dominique, Science et société, Paris, Armand Colin, 2007.

PRE-REQUISITES

Sociological Analysis of Organizations (ASO-4GI)
Professional internship

IDENTIFICATION

CODE : GI-0-S1-EC-CESURE
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

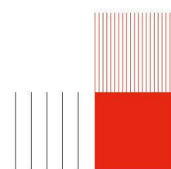
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IDENTIFICATION

CODE : GI-0-S1-EC-VAC
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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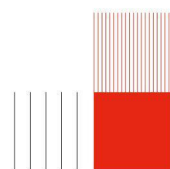
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IDENTIFICATION

CODE : GI-5-S1-EC-DDINSTN-S9

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

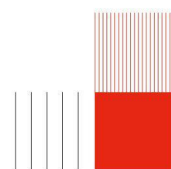
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IDENTIFICATION

CODE :GI-5-S1-EC-DDTOHOKU-S9

ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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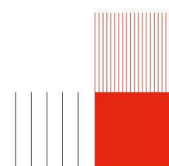
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IDENTIFICATION

CODE : GI-5-S1-EC-DDETS-S9

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

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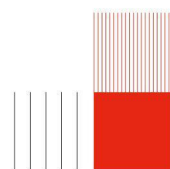
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IDENTIFICATION

CODE :GI-5-S1-EC-DDPOLITO-S9

ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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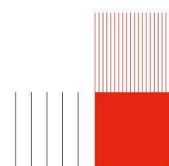
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IDENTIFICATION

CODE : GI-5-S1-EC-DDKAIST-S9

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-5-S1-EC-DDPUC-S9

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

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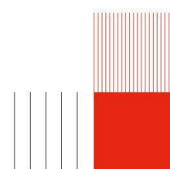
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IDENTIFICATION

CODE : GI-5-S1-EC-DDSHERBK-S9

ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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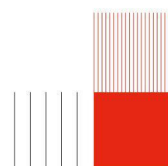
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IDENTIFICATION

CODE : GI-5-S1-EC-DDUPC-S9

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

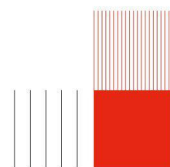
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IDENTIFICATION

CODE : GI-5-S1-EC-DDEPM-S9

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

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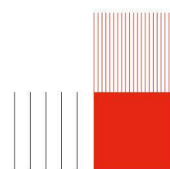
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IDENTIFICATION

CODE : GI-5-S1-EC-DDIIT-S9

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

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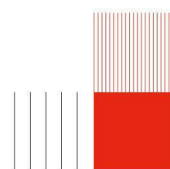
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IDENTIFICATION

CODE : GI-5-S1-EC-DDETS-S11

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

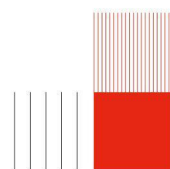
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IDENTIFICATION

CODE : GI-5-S1-EC-PFE00

ECTS : 0

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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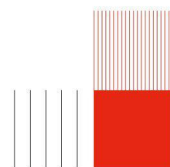
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IDENTIFICATION

CODE : GI-5-S1-EC-PFE30
ECTS : 30

HOURS

Cours : 0h
TD : 352h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 352h
Travail personnel : 0h
Total : 352h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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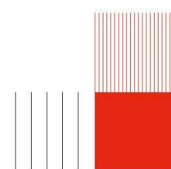
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IDENTIFICATION

CODE : GI-5-S1-EC-DDIIT-S11
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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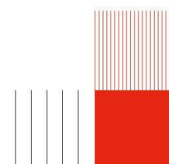
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IDENTIFICATION

CODE :GI-5-S1-EC-DDPOLITO-
S11

ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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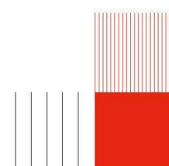
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IDENTIFICATION

CODE : GI-5-S1-EC-PFE30

ECTS : 30

HOURS

Cours : 0h

TD : 352h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 352h

Travail personnel : 0h

Total : 352h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-5-S1-EC-DDEPMC-S11

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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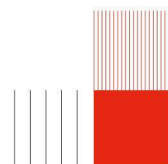
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membre de





IDENTIFICATION

CODE : GI-5-S1-EC-PFE00

ECTS : 0

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-5-S1-EC-PFE30

ECTS : 30

HOURS

Cours : 0h

TD : 352h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 352h

Travail personnel : 0h

Total : 352h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-5-S1-EC-DDKAIST-S11

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

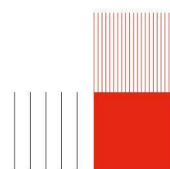
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IDENTIFICATION

CODE : GI-5-S1-EC-PFE00

ECTS : 0

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

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TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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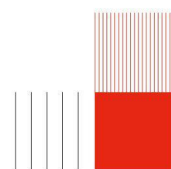
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IDENTIFICATION

CODE GI-5-S1-EC-DDSHERBK-S11

ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
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Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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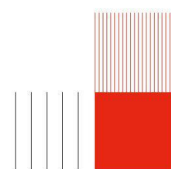
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IDENTIFICATION

CODE : GI-5-S1-EC-DDUPC-S11

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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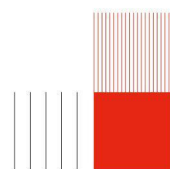
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IDENTIFICATION

CODE : GI-5-S1-EC-DDPUC-S11

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

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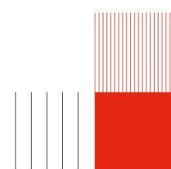
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IDENTIFICATION

CODE : DD-5-S1-EC-IFP
ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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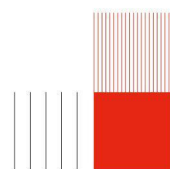
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IDENTIFICATION

CODE : GI-0-S1-EC-ATCFLE
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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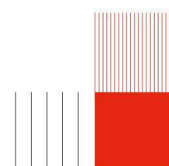
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IDENTIFICATION

CODE : GI-0-S1-EC-ATJAUME

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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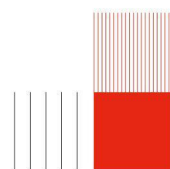
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IDENTIFICATION

CODE GI-5-S1-EC-DDTOHOKU-S11

ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

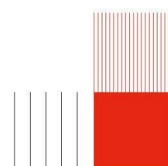
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IDENTIFICATION

CODE : GI-0-S1-EC-ATCANG
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

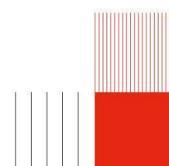
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IDENTIFICATION

CODE : GI-5-S1-EC-DDEPMR-S11

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

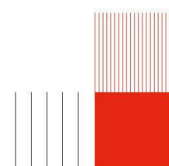
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IDENTIFICATION

CODE : GI-0-S1-EC-ATUPC
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-5-S1-EC-PFE00

ECTS : 0

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

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IDENTIFICATION

CODE GI-5-S1-EC-DDTOHOKU-S13

ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-5-S1-EC-ECH
ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

MME LADIER Anne-Laure :
anne-laure.ladier@insa-lyon.fr

MME PELLET Lorraine :
lorraine.trilling@insa-lyon.fr

IDENTIFICATION

CODE : GI-5-S1-EC-PFE30
ECTS : 30

HOURS

Cours : 0h
TD : 352h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 352h
Travail personnel : 0h
Total : 352h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

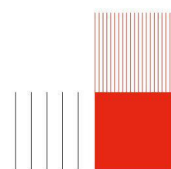
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IDENTIFICATION

CODE LEAN-5-S2-EC-CTLEAN-
PROJ

ECTS : 30

HOURS

Cours : 0h

TD : 0.01666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.01666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.01666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE LEAN-5-S2-EC-CTLEAN-
ENS

ECTS : 0

HOURS

Cours : 0h
TD : 320h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 320h
Travail personnel : 0h
Total : 320h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

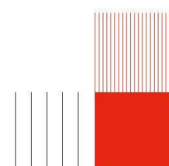
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IDENTIFICATION

CODE : GI-5-S2-EC-VIE
ECTS : 1

HOURS

Cours : 0h
TD : 2h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 2h
Travail personnel : 0h
Total : 2h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

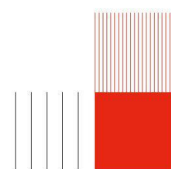
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IDENTIFICATIONCODE : GI-5-S2-EC-PFE30
ECTS : 29**HOURS**Cours : 0h
TD : 7h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 7h
Travail personnel : 0h
Total : 7h**ASSESSMENT METHOD**45 min presentation
Project dissertation
Summary**TEACHING AIDS****TEACHING LANGUAGE**

French

CONTACTMME BOTTA-GENOULAZ Valerie
:
valerie.botta@insa-lyon.fr**AIMS**

- To develop the capacities of observation and strategic analysis of an organization.
- To integrate the technical and socio-economic aspects of a project or a process (of production, management...).
- To improve the capacities of communication and analysis of situation, by an precise identification of professional environment, by the multiplication of the contacts and the working relationships in the company, etc.

CONTENT

During 18 consecutive weeks (minimum), the final project assignment aims to resolve an industrial problem in a company.
The project provides a mission similar to the one of an engineer. It is in general a question of adopting a project manager behaviour taking part in different phases like analysis and design of a product, implementation of the manufacturing process, organization of the production system...

BIBLIOGRAPHY**PRE-REQUISITES**

All the Industrial Engineering training.

IDENTIFICATION

CODE : GI-5-S2-EC-ECH
ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-0-S2-EC-VAC
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-0-S2-EC-ATJAUME
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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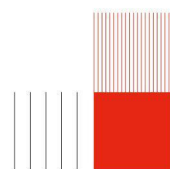
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IDENTIFICATION

CODE : GI-0-S2-EC-ATUPC
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-5-S2-EC-DDPUC-S10

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

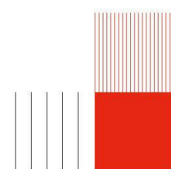
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IDENTIFICATION

CODE : GI-5-S2-EC-DDUPC-S10

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

IDENTIFICATION

CODE GI-5-S2-EC-DDTOHOKU-S10

ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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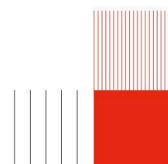
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IDENTIFICATION

CODE : GI-0-S2-EC-CESURE

ECTS : 0

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

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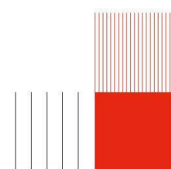
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IDENTIFICATION

CODE : GI-5-S2-EC-DDUPC-S12

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

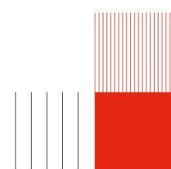
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IDENTIFICATION

CODE : GI-5-S2-EC-PFE00

ECTS : 0

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

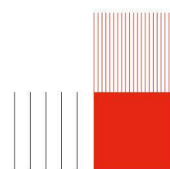
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IDENTIFICATION

CODE : GI-5-S2-EC-DDEPMR-S12

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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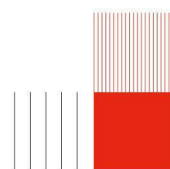
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IDENTIFICATION

CODE : GI-5-S2-EC-PFE00

ECTS : 0

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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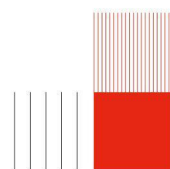
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IDENTIFICATION

CODE : GI-5-S2-EC-PFE00

ECTS : 0

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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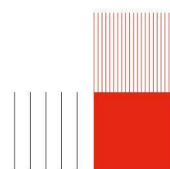
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IDENTIFICATION

CODE : GI-5-S2-EC-DDPUC-S12

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

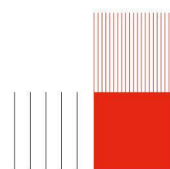
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IDENTIFICATION

CODE GI-5-S2-EC-DDTOHOKU-S12

ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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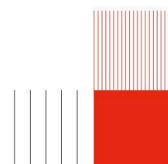
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IDENTIFICATION

CODE : GI-5-S2-EC-ECH
ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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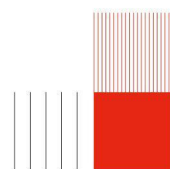
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IDENTIFICATION

CODE :GI-5-S2-EC-DDPOLITO-
S10

ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

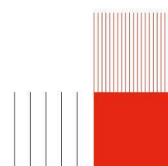
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IDENTIFICATION

CODE : GI-5-S2-EC-PFE00

ECTS : 0

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

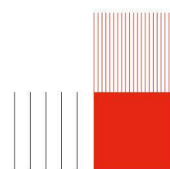
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IDENTIFICATION

CODE : DD-5-S2-EC-IFP
ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

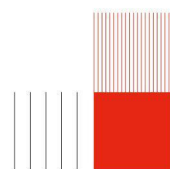
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IDENTIFICATION

CODE :GI-5-S2-EC-DDINSTN-S10

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

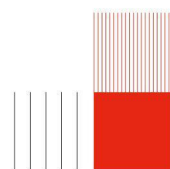
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IDENTIFICATION

CODE : GI-5-S2-EC-PFE00

ECTS : 0

HOURS

Cours : 0h

TD : 0.016666666666666666h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face 0.016666666666666666h
pédagogique :

Travail personnel : 0h

Total : 0.016666666666666666h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

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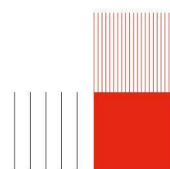
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IDENTIFICATION

CODE : GI-5-S2-EC-DDEPM-S10

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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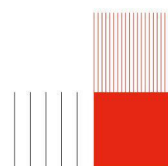
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IDENTIFICATION

CODE : GI-5-S2-EC-DDIIT-S10

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

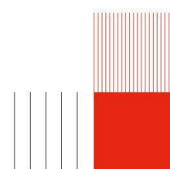
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IDENTIFICATION

CODE : GI-5-S2-EC-DDKAIST-S10

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

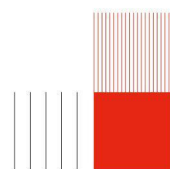
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IDENTIFICATION

CODE GI-5-S2-EC-DDSHERBK-S10

ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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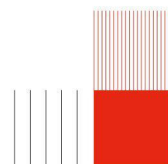
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IDENTIFICATION

CODE : GI-5-S2-EC-DDETS-S10

ECTS : 30

HOURS

Cours : 0h

TD : 0h

TP : 0h

Projet : 0h

Evaluation : 0h

Face à face pédagogique : 0h

Travail personnel : 0h

Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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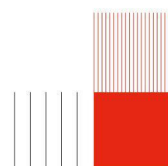
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IDENTIFICATION

CODE : GI-5-S2-EC-ECH
ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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IDENTIFICATION

CODE : GI-5-S2-EC-ECH
ECTS : 30

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
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ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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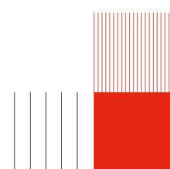
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IDENTIFICATION

CODE : GI-5-S2-EC-VALID
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
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Face à face pédagogique : 0h
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ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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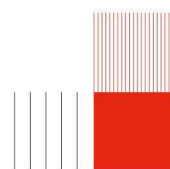
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IDENTIFICATION

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ECTS : 0

HOURS

Cours : 0h
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Projet : 0h
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Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

INSA LYON

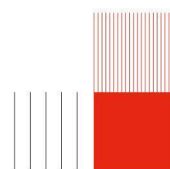
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IDENTIFICATION

CODE : GI-0-S2-EC-ATCANG
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

INSA LYON

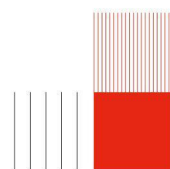
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IDENTIFICATION

CODE : GI-0-S2-EC-ATCFLE
ECTS : 0

HOURS

Cours : 0h
TD : 0h
TP : 0h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 0h
Travail personnel : 0h
Total : 0h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

CONTACT

AIMS

CONTENT

BIBLIOGRAPHY

PRE-REQUISITES

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